Prodesy

An LLM-maintained wiki on protein design — book edition

Compiled from the Prodesy wiki

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Introduction — the field at a glance

Evolving synthesis of the Prodesy wiki. Current scope: the deep-learning core (batch 1), the generative-design expansion (batch 2), the symmetric-assembly history (batch 3), repeat/parametric design plus relaxed-sequence hallucination (batch 4), binder design (batch 5), the Genie lineage (batch 6), and enzyme design (batch 7). Will broaden as more of `sources/` is ingested.

The through-line

A single pipeline now organizes the field:

> **generate a backbone → design its sequence → predict/filter before you build**

- **Generate** — Diffusion generative design: Watson 2023 (fine-tuned RoseTTAFold) and, independently, Ingraham 2023 (bespoke polymer-aware diffusion) sample novel, steerable backbones.
- **Sequence** — Sequence design (fixed-backbone): Dauparas 2022 assigns a foldable sequence to a backbone in seconds, displacing Rosetta.
- **Filter** — Structure prediction: Jumper 2021 confidence scores decide, *in silico*, which designs are worth synthesizing.

Bennett 2023 shows the payoff concretely: dropping ProteinMPNN + AF2 filtering into the older Rosetta binder workflow raised success ~10×. De novo binder design is the clearest application of the whole stack, and is now a chapter.

How the pipeline was built and where it's going (batch 2)

- **Precursors.** Before diffusion, the same goals were met by **network hallucination** and **inpainting**. Anishchenko 2021 established the method — invert a structure predictor by optimizing a random sequence until the network's own prediction sharpens — and Wang 2022 (Motif scaffolding (functional-site design)) and Wicky 2022 (Symmetric assembly design) are the same engine under two additional loss terms. RFdiffusion later absorbed all of it as conditional-diffusion tasks: what was a bespoke objective becomes a conditioning input.
- **Symmetric assemblies are now a chapter.** Symmetric oligomer design traces the longest continuous arc in the wiki, from Harbury 1993 and the fusion strategy of Yeates 2002, through the parametric era (Huang 2014, Thomson 2014, Boyken 2016, Courbet 2022) and its repeat-protein basis (Brunette 2015, Wood 2018), to hallucination, conditional diffusion, and back to gradient-based hallucination (Frank 2024).

- **Hallucination is not superseded.** The 2023 consensus that diffusion had displaced predictor inversion did not hold: Frank 2024 fixed hallucination's two weaknesses — slow random search, and gradient descent in a discrete sequence space — by optimizing in a **relaxed** continuous sequence space, and reaches 1000-residue single chains (median TM 0.89) where Watson 2023 struggles past ~600. The two lineages now look complementary rather than sequential.
- **All-atom is the current frontier**, pursued by ≥3 distinct routes: atomize-in-predictor (Krishna 2024), intrinsic side-chain **superposition** (Chu 2024, Huang lab), and efficient **token+atom** generation (Butcher 2025, ~10× cheaper, DNA binders). This mirrors, on the design side, AF3's all-atom move on the prediction side.
- **Synthetic structures as training data is now a settled bet.** Lin 2024 was "the first protein diffusion model to train on AFDB", replacing ~20,000 PDB structures of usable length with 588,570 FoldSeek-clustered AlphaFold predictions at pLDDT > 80 — moving diversity from 0.642 to 0.91 and PDB novelty from 0.041 to 0.41 relative to its predecessor, at designability matched with RFdiffusion. Lin 2023 had already observed that "increasing training set size markedly improves Genie's sampling quality", and Didi 2026 makes the same bet for *complexes* with its synthetic Teddymer dataset built from domain–domain interactions of predicted monomers. Three sources, one direction. The open question the wiki cannot answer is what the ceiling of predicted-structure training is, since Lin 2026 uses **no experimentally determined monomeric structures at all**.
- **Sequence design is converging on codesign.** Dauparas 2022 and the learned-potential Anand 2022 (Huang lab) both showed learned energies beat Rosetta; Protpardelle now folds sequence *into* generation.
- **De novo antibodies now exist, at a measurable cost.** Bennett 2026 designs epitope-specific VHHs, scFvs and full IgGs from a **fine-tuned RFdiffusion2**, supplying the framework global-frame-invariantly so the antibody's rigid-body placement against the epitope is designed rather than assumed. Cryo-EM confirms **all six CDR loops** for one scFv and a 0.8 Å CDR3 for a VHH. The price of the format is explicit: **0–2% success per target**, against 19% for hotspot-conditioned miniprotein design from the same lab (Watson 2023). The wiki's reading is that de novo design has displaced stage 1 of antibody discovery — epitope-specific generation, previously done by immunization followed by epitope mapping — and not stage 2, since OrthoRep maturation still supplies the final two orders of magnitude.
- **Applications reach harder targets.** Vasquez-Torres 2023 extends De novo binder design to floppy peptide hormones via **partial diffusion**, reaching picomolar affinity, and turns binders into biosensors and serum mass-spec reagents. Stark 2025 pushes the same frontier to intrinsically disordered regions and to small molecules.

Binder design is now a chapter (batch 5)

- **The progression is a story of deletions.** De novo binder design organises the field by what each generation stopped asking the designer to supply: the **binding mode** (grafting required a natural interaction to copy; hotspot conditioning removed that), then the **scaffold** (Watson 2023), then the **chemistry** (Krishna 2024, Butcher 2025), then the **modality** (Stark 2025). The **epitope** has survived every generation as a manual input.
- **Hot spots finally have a source.** Halperin 2004 supplies what the wiki had been using without defining: a hot spot is an alanine-scanning $\Delta\Delta G \geq 1.9$ kcal/mol residue, hot spots carry 4.2 cross-interface neighbours against 2.3 for other interface residues, and occupied volume around a residue correlates with its $\Delta\Delta G$ at $\mathbf{r^2 = 0.94}$. That is why hotspot conditioning is a coherent instruction: it points a generator at the densely packed region where binding energy concentrates.
- **There is an independent lineage, and it works.** Gainza 2020 and Gainza 2023 design binders by matching learned descriptors of molecular *surfaces* and grafting the retrieved

seed — the inverse of telling a generator where to bind — reaching nanomolar affinities on spike, PD-L1 and CTLA-4 with crystal and cryo-EM confirmation, and a 4.2 µM PD-1 binder with no experimental optimisation at all. Its evaluation splits are the strictest in the wiki. Until batch 5 the wiki's binder coverage was a single lab's account of the field presented as the field.

- **The generative/hallucination split has collapsed.** Didi 2026 argues the dichotomy is false — generative methods are pure training-time optimisation, hallucination pure inference-time optimisation with no prior — and supplies both halves, beating BindCraft under *normalised compute budgets*. Cho 2025 reaches the same place with no training at all, by letting a diffusion predictor's own hallucination be the prior. The wiki's reading is that hallucination's recurring adversarial-overfitting failure (see Wicky 2022) was never about the optimiser but about the absence of a prior to optimise inside — though no source runs the controlled experiment.
- **The architectural consensus is less settled than it looked.** Two moves the field made in 2024–25 — abandon SE(3)-equivariance to get all-atom detail, and replace generation with hallucination to get hit rates — are both reversed by Lin 2026, which keeps equivariance via a **branched-polymer** representation with partial atomization, runs 10× faster by moderating the reverse diffusion through a direction scale rather than a noise scale, and still beats the hallucination state of the art (BindCraft) on 7 of 10 benchmark binder problems, by a wider margin once runtime is normalised. Its blind-competition result is the strongest wet-lab datapoint in the batch: a **92 nM** binder to Nipah Glycoprotein G at a **1/8** hit rate, against aggregate 3/60, 1/100 and 2/288 for RFdiffusion, BindCraft and BoltzGen in the same challenge. The wiki's reading is that the consensus was adopted faster than it was tested.
- **The oracle's precision now has a number, and it is 12%.** Lin 2026 reports maximum 12% precision across its binder evaluation pipelines with complete failure on three of eleven targets, and says plainly that this "place[s] a low ceiling on the efficiency of binder design using *in silico* oracles as filters". It also caught designs passing an AlphaFold-Multimer benchmark while binding **off-target**, and repaired the filter by requiring ≥80% of specified hotspots within 5 Å — the published counterpart of the gate pathologies Ruijgrok 2026 measured operationally.
- **The filter, not the generator, sets the hit rate.** Three independent demonstrations: inserting the AF2 filter raised success ~10× with the generator unchanged (Bennett 2023); an earlier BoltzGen whose ranking was close to random produced **no hits at all** from a substantially identical generator (Stark 2025); and the generate-to-test ratio is **60,000:15** (Stark 2025). Every published binder success rate is a rate measured after a filter of roughly that severity.

Enzyme design is now a chapter (batch 7)

- **The field now has an independent enzyme-design lineage, and it reaches higher activities.** Braun 2026 (Oberdorfer lab) builds *on* Watson 2023 rather than retraining it, enumerating **artificial motif libraries** by rotamer inversion and enforcing a substrate pocket with an **α-helix placeholder that is deleted after diffusion** — because RFdiffusion's own substrate potential "failed to promote the formation of well-defined substrate binding pockets". From the same RA95.5-8F retroaldolase theozyme that Ahern 2025 used, it reaches **290 M^{-1} s^{-1} against an extensively evolved enzyme's 320**, with **99% enantiomeric excess**, from 36 designs rather than 96 — roughly **45× the catalytic efficiency**. The wiki's reading is that this reframes §5's activity ceiling as a property of particular pipelines rather than of de novo design.
- **Preorganization has now been filtered prospectively, once.** Kim 2026 selected on **PLACER** ensembles, ranked its best design (53,000 M^{-1} s^{-1}, against 3–60 for prior designed metallohydrolases) first — and then found that **seven designs from the same**

backbone family, rejected for suboptimal PLACER metrics, had very low or no activity. Same campaign, same backbone, separated by an ensemble score. Taken with the productive-conformation correlation in Choi 2026, the ensemble is now demonstrably the filterable quantity.

- **The amide barrier fell three times in one year.** Peptide bonds have half-lives of hundreds of years in neutral water (Choi 2026), which is why de novo enzyme design had been "largely limited to low-energy-barrier model reactions". Three 2026 campaigns cleared it by three chemistries: an activated **cysteine** nucleophile at 13/69 = 19% with rate enhancements to 3×10^7 (Choi 2026); a water-activating **zinc** centre at 36% of 135, reaching $>10^8$-fold and $>10^{10}$ after four substitutions (Chen 2026); and **free-atom diffusion** delivering zero-shot phosphotriesterases and transthyretin-cleaving proteases at 56/96 in cis (Butcher 2026).
- **The remaining gap is preorganization, and everyone says so.** Crystal structures confirm the active sites were built as designed — Cα r.m.s.d. < 1.2 Å across six structures, 0.82–1.35 Å elsewhere — while efficiencies stay **10^3–10^4-fold below native** (Chen 2026). The mechanism is ensemble, not geometry: PLACER ensembles across the catalytic cycle show a non-productive His rotamer suppressed by one substitution, with the **fraction of productive conformations tracking activity** across a design series (Choi 2026). The wiki's reading is that the apparatus optimises a single structure while catalysis depends on a distribution.
- **What you model is what you get.** The largest single effect on outcome anywhere in the wiki comes from modelling the **catalytic water** in the filtered enzyme–substrate complex: 3/89 versus 31/38 one-hour hit rate, same generator, same designers, **unchanged numerical thresholds** (Chen 2026, Chen 2026 (suppl.)). The converse is equally sharp — AlphaFold3 gave pLDDT **86.55** to a metal in a site with one coordinating ligand that binds no zinc, a design its authors note "would have been classified as a success" by standard filtering (Elnesr 2026).
- **A second hypothesis about where catalysis comes from.** Every Baker-lab route transplants a native catalytic motif. Elnesr 2026 argues motif geometry alone "does not guarantee catalytic competence" because second-shell residues mediate reactivity, and optimises the **coordination sphere** directly by multi-objective evolutionary search over LigandMPNN and Durr 2023. It arrived at a **binuclear zinc site nature also uses, without being shown one**, at a rate enhancement of 3×10^{13} — the highest reported for any de novo hydrolase.
- **Training-set artefacts surface only when a predictor is used generatively.** Metal3D's corpus contains zinc sites with fewer than three coordinating residues that likely reflect crystallisation additives; its own supplement stratifies nearly every metric by coordination number and occupancy (Durr 2023 (suppl.)). That was a reporting detail until a design pipeline started optimising against it (Elnesr 2026). The same pattern as the interface prior in Anishchenko 2021, in a different modality.

Cross-cutting observations

- **Prediction ⇄ design convergence.** Prediction models are reused as design oracles (AF2 filtering) and as generators (RoseTTAFold → RFdiffusion). Meanwhile prediction itself adopted diffusion in Abramson 2024. The two problems are merging onto shared machinery.
- **Independent replication of the paradigm, now three ways.** RFdiffusion (Baker lab), Chroma (Generate Biomedicines) and Genie (Lin 2023, AlQuraishi lab) arrived at diffusion-based design separately and by three different mechanisms — fine-tuning a predictor, a bespoke correlated-noise prior, and a forward/reverse representational asymmetry over oriented reference frames. Taken together the wiki reads this as a sign the approach is robust rather than a single group's trick. Genie is the only one of the three

that is SE(3)-equivariant end to end, and it did it at 4.1 M parameters against RFdiffusion's 59.8 M (Lin 2023).

- **Validation discipline is the real unlock.** The recurring lesson across sources is not just better generators but cheap, reliable *in silico* filtering that collapses wet-lab search space.
- **But the oracle is now the thing to be suspicious of.** Taken together, batch 5 gives four distinct ways an *in silico* filter misleads: **adversarial overfitting** when it becomes an objective rather than a gate (Wicky 2022); **circularity** when generator and scorer share weights, which is a structural consequence of the unify-design-and-prediction direction both Stark 2025 and Didi 2026 take, and which Ruijgrok 2026 handles by declaring a scoring tier circular and promoting an independent predictor; **gates that deplete the signal they select for**, measured in that report as an AF2 gate reducing epitope contact and a promotion threshold discarding on-epitope designs; and **plain disagreement with experiment**, where Gainza 2023 confirmed designed interfaces by competition and mutagenesis for several binders whose AlphaFold predictions did *not* match the design models.
- **Atomic representation per pipeline stage is a validity condition, not a software detail.** Ruijgrok 2026 audits which stages can see sidechains and finds interface numbers being read off a Cα-only file in one pipeline and a generator's sidechains discarded before scoring in another. Given that interface energetics are a packing phenomenon correlating with occupied volume at $r^2 = 0.94$ (Halperin 2004), a backbone-only stage cannot evaluate the quantity that matters.
- **Conditioning moved from the coordinates into the sampling process.** Symmetric design is the cleanest place to watch this happen: symmetry was first *built into* a coordinate generator (Huang 2014 (suppl.)), then *rewarded* in a loss against a predictor (Wicky 2022), then *projected* onto the state at every denoising step (Watson 2023 (suppl.)). The wiki's reading, developed in Symmetric oligomer design, is that this shift traded a geometry bottleneck for an interface-quality bottleneck — coverage improved far faster than yield.
- **The pre-deep-learning literature is doing real work.** Batch 3 showed that the wiki's synthesis was resting on Baker-lab-2022-onward sources alone for a problem with a thirty-year history. Where a chapter's claims trace only to recent papers, that is worth treating as a gap rather than a summary.
- **Generative models inherit priors nobody asked for.** Anishchenko 2021 modelled every design as a monomer and got homo-oligomers and soluble aggregates anyway, because trRosetta had learned the sequence signatures of native protein–protein interfaces and did not disfavour surface hydrophobics enough. The same bias shows up as adversarial surfaces in Wicky 2022, and again in Frank 2024, whose designs formed homo-oligomers whenever the explicit negative-design filter against them was omitted. Three independent sightings across three years. The wiki's reading is that a predictor-derived generator carries its training distribution's interface statistics into every design — self-association is the default output, and every symmetric-design method is a way of *aiming* a propensity that is already there. See Symmetric oligomer design §5.1 and §10.

Open threads to pursue (next batches / queries)

- **Prediction ecosystem** (batch 3): ColabFold, Foldseek, lDDT, Boltz/Boltzgen — the tooling around the predictors used as design oracles.
- ~~Surface-fingerprint design~~ and ~~remaining generative models (BoltzGen, Protein Hunter, test-time compute)~~ — **done in batch 5.**
- **Antibody-specific design** (IgLM; Bennett RFdiffusion antibodies) as a sub-branch of De novo binder design — **blocked**: `sources/` holds the Bennett 2025 supplement but not its main text, so §8.2 of the chapter is a placeholder. The main text must be acquired.

- **BindCraft** — the most widely used open binder pipeline in practice, and the baseline two ingested papers measure themselves against. BindCraft is entirely secondary-sourced. Highest-priority acquisition in the wiki.
- Add a `benchmarks/` thread (CASP, sequence recovery, binder success rates, the RFD3 PPI/DNA/small-molecule/AME suites) to make cross-paper numbers comparable. **Batch 5 raised the urgency and supplied the argument**: De novo binder design §12 shows the field's rates differ in assay, bar, denominator *and* target difficulty simultaneously, and Ruijgrok 2026 demonstrates the same within one project. Three benchmark and affinity-dataset PDFs sit un-ingested in `sources/Proteins/`.
- **Protein dynamics** (Kern lab papers already in folder) — a different lens (conformational ensembles) largely absent from the current design-centric synthesis.

Sequence and Interfaces

De novo binder design

A chapter on designing a protein that binds a chosen target, starting from the target's structure — or, latterly, from its sequence alone — without immunization and without library selection. The subject is worth a chapter rather than a concept page because it is the part of protein design where the field's methods, its oracles and its wet-lab discipline all meet, and because it has been *solved* repeatedly in a specific sense: each generation of method deleted one thing the designer previously had to supply. First the **binding mode** went — motif grafting required a natural interaction to copy, and hotspot-conditioned generation removed that requirement. Then the **scaffold** went — the designer supplies a target and a patch, and the backbone is invented. Then the **chemistry** went — all-atom models dropped the requirement that the target be a protein at all. One specification has survived every generation: somebody still has to decide **where on the target to bind**, and that decision is still largely made by hand.

Running underneath that progression is a second story, and it is the one this chapter treats as its main argument. As generation got easier, the binding constraint moved downstream — from generation, to *in silico* filtering, to experimental throughput — and the field's reported success rates became **mutually incommensurable** along the way. A yeast-display SC50 hit rate, a biolayer-interferometry single-point response, an AlphaFold2 self-consistency pass rate and a fraction of designs reaching nanomolar K_D are four different numbers with four different denominators, and papers quote whichever their assay produced. This chapter states each figure in its own units and declines to average them. The most direct evidence that this is not pedantry comes from outside the published literature: Ruijgrok 2026 measured six binder pipelines on one target class inside one project and found six different denominators, with a dedicated comparability ledger explaining why no two of them can be placed side by side.

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1. What de novo binder design means, and what it replaces

The object of the exercise is a protein that binds a chosen target with useful affinity and specificity, designed from structural information about the target rather than discovered by screening. Bennett 2023 characterises the state this capability had reached by 2023: design of high-affinity binders from target structure alone had become possible by Rosetta-based motif grafting, but success rates were low enough that large numbers of candidates had to be screened experimentally. The therapeutic and diagnostic motivation is stated across the sources — Watson 2023 calls high-affinity binder design a grand challenge with numerous therapeutic applications, and Vasquez-Torres 2023 carries its designs through to serum diagnostics.

Three things must still be specified, and it is worth separating them because the chapter's history is the history of the first two being automated:

- **The target**, as a structure, a sequence, or — for the all-atom methods of §7 — a small molecule or nucleic acid.
- **The epitope**: where on the target to bind. In practice this is a list of hotspot residues, and §2 explains why that is a physically coherent instruction. No method in this chapter chooses the epitope for the designer, with one partial exception: Gainza 2023 uses MaSIF-site to *predict* which surface patches have a high propensity to be engaged by a protein, which is epitope nomination rather than epitope selection.
- **The modality**: a 50–130-residue miniprotein, a peptide, a nanobody, an antibody. Until recently this was a property of the method; Stark 2025 is the first ingested source for which it is an input, and Bennett 2026 the first for which an **antibody framework** is a conditioning input with the rigid-body placement left to be designed.

The chapter excludes affinity maturation of existing binders, interface *prediction* as an end in itself, and the interfaces internal to symmetric homo-oligomers, which are treated in Symmetric oligomer design.

What this replaces is the weakest-sourced claim in the chapter. Binder discovery has conventionally proceeded by immunization or by display-library selection, and the de novo literature positions itself against that. No ingested source states this history directly; the closest available support is indirect — Gainza 2023 compares a designed SARS-CoV-2 binder's potency to the clinical antibody imdevimab and notes that its buried interface area of 1,452 $Å^2$ is smaller than the ~2,071 ± 456 $Å^2$ typical of antibodies, and Stark 2025 motivates nanobody design by that modality's developability advantages. A primer on display or antibody engineering is flagged in §14.

2. The biophysics: what an interface is, and where its energy lives

Every method after §3 is told where to bind by naming **hotspot residues**, so the chapter needs to say what a hotspot is. Halperin 2004 gives the operational definition: a hot spot is a residue whose mutation to alanine causes a significant drop in binding affinity, measured as the change in free energy of binding, with the working threshold set at **ΔΔG ≥ 1.9 kcal/mol**. That is a mutagenesis definition, not a geometric one, and the paper's contribution is to connect the two.

Its dataset is two-sided, which is the point. Structurally conserved residues were drawn from a non-redundant interface set that had grown from 1,629 interfaces in 351 clusters in 1996 to **21,686 interfaces in 3,799 clusters**; experimental hot spots came from the Alanine Scanning Energetics database, screened for complexes where *both* partners had been alanine-scanned and the pair had been crystallised together, yielding **8 complexes, 429 alanine mutations and 53 hot spots** (Halperin 2004).

The physical result is that hot spots are more densely packed than the rest of the interface, and the packing tracks the energetics. Hot spots have on average **4.2 neighbours across the interface** against **2.3** for non-hot-spot interface residues, a difference significant at p = 0.01 and 2.9 standard deviations across 53 hot spots and 551 non-hot-spot residues. The occupied volume in a residue's vicinity is **69.2 Å^3 (SD 42.5) around hot spots** against **38.1 Å^3 (SD 17.1)** elsewhere, significant at the 0.99 confidence level. And the occupied volume **correlates with ΔΔG at r^2 = 0.94** (Halperin 2004). The effect is not an artefact of residue size: amino-acid volume correlates with hot-spot propensity at r^2 = 0.01, and with neighbour count at r^2 = 0.03.

The synthesising claim in that paper is that a protein–protein interface "consists of regions of high and low packing density, with the hot spots organized in the former", and that the high local packing density in interfaces is **reminiscent of protein cores** (Halperin 2004). An interface is therefore not a uniform patch of surface: it has a core-like dense region carrying most of the binding energy and a looser periphery. Hot spots also **couple across** the interface more often than a random distribution predicts, with expected couple counts of 2.6 (SD 4.0) under a purely random null and 5.4 (SD 6.0) under a null that additionally conditions on neighbour count — so the elevated neighbour count accounts for much, though not all, of the coupling.

The wiki's reading is that this is what makes hotspot conditioning a coherent instruction rather than a convenient fiction. Telling a generative model to contact a named set of target residues is telling it to build into the region where binding energy is concentrated and where packing is densest — and because the property is regional rather than pointwise, *soft* conditioning that merely biases toward those residues can still work, which is how every method in §5 and §7 in fact implements it (Ruijgrok 2026).

Chemistry is the second axis, and it is not the same as packing. Halperin 2004 finds strong Cys–Cys conservation across interfaces; glycine preferentially coupled with aromatics, polar and charged residues and with small hydrophobics but disfavoured beside large ones, which the authors read as glycine's absent side chain letting it pack tightly while its backbone hydrogen-bonds across the interface; preferred aromatic–aromatic and hydrophobic–hydrophobic couplings, though not involving Ile or Leu; and, strikingly, **charged–charged couples underrepresented**, with polar–polar underrepresented less strongly. The complementary constraint is satisfaction of buried polar atoms: Boyken 2016 establishes that buried polar interactions are hard to design because a hydrogen bond is partially covalent and so restricts donors and acceptors to narrow ranges of orientation and distance, and because nearly all buried polar atoms must find a partner or incur a considerable desolvation penalty — and that a hydrogen-bond network is inherently *many-body*, which is why conventional design algorithms expressing energy as a sum of pairwise residue interactions handle it badly.

Taken together, these two sources give the chapter the distinction it needs between **affinity** and **specificity**. Packing density and buried area buy affinity; the identity and satisfaction of polar contacts buy discrimination between similar targets. Boyken 2016 demonstrated the point in the oligomer setting, where designs with hydrophobic cores each folded correctly yet cross-assembled promiscuously until hydrogen-bond networks were installed, after which the networks conferred interaction specificity in vivo. The binder literature's specificity evidence is thinner and is almost always a negative control rather than a designed property: no binding of a PTH binder to PTHrP at 34% identity, and no cross-reactivity of an NPY binder to PYY

(Vasquez-Torres 2023); no binding of a PD-1 binder to porcine PD-1 (Gainza 2023). §13 returns to whether specificity is ever designed or only observed.

What no ingested source supplies is the kinetic half. Every affinity in this chapter is an equilibrium constant; association and dissociation rates appear nowhere, and a source on binding kinetics is flagged in §14.

3. The pre-generative era — grafting a binding mode

The founding strategy was to transplant a binding mode that already existed. The designer's problem was then not *what interaction to make* but *what protein to put it on*: given a functional site — a binding motif, a catalytic constellation — build a scaffold that presents it in the right geometry.

Wang 2022 is the deep-learning generalisation of that idea and the wiki's only ingested source for the era, which is a real limitation discussed in §14. The concept page is Motif scaffolding (functional-site design). The structural point for this chapter is that grafting bounds the accessible space by the space of *known* interactions: you can only scaffold a motif you already possess, so targets with no characterised binding partner are out of reach, and the designed interface is the natural one plus whatever the scaffold adds around it.

Two traces of the era survive in the sources as quantitative baselines. Bennett 2023 describes inheriting a Rosetta motif-grafting pipeline whose success rates were low enough that campaigns were screened rather than shortlisted, and Watson 2023 later measured its own 19% hit rate as **roughly two orders of magnitude above the previous Rosetta-based method on the same five targets** — which places the grafting-era rate at a few tenths of a percent on those targets, by that assay. The era's throughput signature is equally stark: Watson 2023 reports testing fewer than 100 designs per target "compared to thousands in previous studies".

Grafting did not disappear. It is the backbone of §6: Gainza 2023 matches surface descriptors to find binding *seeds* and then grafts them onto scaffolds with Rosetta MotifGraft against a database of 1,300 monomeric scaffolds plus designed-miniprotein and AlphaFold-proteome databases. And it remains the highest-affinity route in the diffusion literature: the tightest binders in Watson 2023 are not free-generated at all but **scaffolded around the p53 helix** that binds MDM2, reaching 0.5 and 0.7 nM against a native affinity of 600 nM. The wiki's reading is that this is a standing result the field has not fully absorbed — when a natural binding mode exists, building around it still beats inventing one.

4. Deep learning enters as a filter, not a generator

The pivotal move in this chapter is not a generative model. It is the insertion of a structure predictor **in front of the ordering decision**, and it is the best-sourced section here.

Bennett 2023 keeps the Rosetta binder pipeline and makes two substitutions. ProteinMPNN replaces Rosetta for Sequence design (fixed-backbone), greatly cutting compute; and Jumper 2021 (with RoseTTAFold as a cross-check) is used to score two distinct probabilities before anything is synthesised — that the designed sequence **folds** to the intended monomer, and that it **binds** the target in the designed pose. The reported effect of the filter is a **~10× increase in experimental success rate** (Bennett 2023).

The supplement is where the discipline becomes operational, and three details from it are load-bearing for everything downstream (Bennett 2023 (suppl.)):

- **The two metrics.** Monomer **pLDDT** answers "does it fold as designed?"; **pAE_interaction** answers "does it dock as designed?". Figure S6's ROC and true-positive-rate analysis

quantifies the discriminative power of each, which is the empirical justification for using them as go/no-go gates rather than as soft rankings (Bennett 2023 (suppl.)).
- **The operational success criterion** is a **yeast-surface-display SC50 < 4 µM**. The ~10× improvement is defined against that bar and against no other (Bennett 2023, Bennett 2023 (suppl.)).
- **The practical trick that makes complex prediction reliable enough to filter on**: AlphaFold2 is run on the complex **with the target as a template and the Rosetta design model as an initial guess**. This is absent from the main text and is the difference between a usable and an unusable interface score (Bennett 2023 (suppl.)).

Figure S7 adds a prospective comparison across three targets showing ProteinMPNN is not merely cheaper than Rosetta FastDesign but at least comparable in experimental success once deep-learning filtering is applied (Bennett 2023 (suppl.)).

Two consequences deserve stating. First, the quantity that improved tenfold was the **yield of the shortlist**, not the quality of the generator — the generator was unchanged. Second, this is the moment the field acquired an oracle, and §10 is about what happens when a community optimises against one.

5. Hotspot-conditioned diffusion — the scaffold requirement dissolves

Watson 2023 removes the requirement that the binder resemble anything. RFdiffusion fine-tunes the RoseTTAFold structure-prediction network on a denoising task, so that noised residue frames are iteratively denoised into a coherent backbone, and conditioning inputs steer where it goes.

The conditioning machinery for binders is specific and is mapped in the supplement (Watson 2023 (suppl.)): **interface hotspots** — the target residues the binder should contact — plus **secondary-structure and block-adjacency conditioning**, plus a dedicated **PPI fine-tune** of the base model. That last item is easy to miss and worth flagging: binder design was not a free capability of unconditional RFdiffusion but a separately trained mode, which bears on how far the paper's "single general framework" claim extends. A further fine-tuned model adds coarse-grained control over binder scaffold **topology**, so a fold can be requested alongside the hotspots (Watson 2023).

The campaign, in its own units. Five targets chosen to match the prior Rosetta benchmark — influenza A H1 haemagglutinin, IL-7Rα, PD-L1, insulin receptor and TrkA — with designs generated both with and without fold conditioning, filtered by AlphaFold2 confidence in the interface and the monomer, and **95 designs selected per target** for experimental characterisation. Binding was assessed by single-point biolayer interferometry at 10 µM binder, with success defined as response at or above 50% of the positive control's maximum. The result: **19% overall**, called a conservative estimate because some designs that bound had too little material to screen at 10 µM; binders for **all five targets**; and full BLI titrations on a subset giving **nanomolar affinities with no experimental optimisation**, including haemagglutinin and IL-7Rα binders at roughly **30 nM** (Watson 2023).

The motif-scaffolded arm of the same paper is the stronger result and is discussed in §3: scaffolding the p53 helix against MDM2 gave **55 of 95 designs binding**, 32 of them monomeric, with **0.5 and 0.7 nM** affinities against a 600 nM native interaction and on average 31% more interface surface area than the native helix (Watson 2023).

The *in silico* criteria used to select those 95 designs per target are recorded as **pAE < 5, pLDDT > 90 and RMSD < ~1 Å** (Watson 2023 (suppl.)). They matter because they are not the criteria anyone else uses: Didi 2026 requires pLDDT > 90, interface pAE < 7.0 and binder

RMSD < 1.5 Å, and Cho 2025 requires interface PAE ≤ 1.5 and ipTM ≥ 0.8 for small-molecule work. The thresholds are close enough to look comparable and different enough that they are not.

Hotspot compliance separates generation from hallucination. Lin 2026 reports that for the H1 and IL-17A targets "a sizable proportion of otherwise successful BindCraft designs fail to bind the specified interface", and offers an architectural rather than incidental explanation: in hallucination, hotspot compliance is "an additional ad hoc loss term that is optimized during hallucination", whereas in a conditional generative model hotspots "are provided as inputs to the model during training and generation, resulting in a shift in the underlying distribution of the generative process towards compliant designs." The same paper shows the specification can be improved post hoc by two heuristics that expand it: **extended hotspots**, adding any target surface residue within 6 Å (Cβ–Cβ) of a user hotspot, where surface means relative solvent accessibility ≥ 0.25 and SASA ≥ 10 $Å^2$; and **convergent hotspots**, adding interface residues found in the intersection of all successful designs from previous rounds. Both raise success rates, convergent substantially more so on multimeric targets, and applied iteratively to TNFα — the hardest target in the set — the success rate climbs nearly monotonically with design round (Lin 2026).

How hotspots are actually enforced varies more than the papers suggest. Ruijgrok 2026 audits four generative pipelines and finds four mechanisms: Protpardelle-1c pins the target as a *motif to scaffold* via an atomistic contig and carries a placement diagnostic (`ca_motif_sample_rmsd`, ~0.030 Å on working runs); Proteína-Complexa supplies the target as *context* with `binder_gen_only: True` and — consequentially — **re-generates rather than copies** the target chains; RFdiffusion3 specifies hotspots at the level of **individual sidechain atoms**, mapping each residue to named atoms; and ProteinHunter supplies **no target structure at all**. In all four, hotspot conditioning is **soft guidance rather than a hard constraint** (Ruijgrok 2026). That report also records that Protpardelle's global secondary-structure proportion control (`binder_sse_prop`, quantised to 0.1) is the only confirmed fold lever among them, with no counterpart in the other three — a practical narrowing of the topology conditioning that Watson 2023 introduced.

6. Surface fingerprints — the independent lineage

Everything in §4, §5 and §7 descends from one laboratory's stack. This section is the chapter's control on whether that stack's success reflects its method or its choice of targets.

Gainza 2020 proposes that a protein's **molecular surface** is fingerprinted with patterns of chemical and geometric features that reveal what it will interact with, and that proteins **with no sequence homology** undergoing similar interactions display similar patterns. The surface is decomposed into overlapping radial patches of fixed **geodesic** radius (9 Å or 12 Å, application-dependent), each vertex is assigned precomputed geometric and chemical features — hydropathy index, continuum electrostatics among them — and geodesic convolutional layers learn to embed each patch into a vector descriptor, trained end-to-end per task so descriptors are task-specific rather than universal. The geodesic parameterisation is the load-bearing choice and the supplement shows why with a single figure: on trypsin, a 12 Å *Euclidean* patch centred in the interface pocket reaches points on an entirely different face of the protein, while the 12 Å *geodesic* patch stays on the interacting face (Gainza 2020 (suppl.)).

Three applications were demonstrated — pocket–ligand similarity, protein-binding **site** prediction, and ultrafast surface **search** for complex configurations (Gainza 2020) — with MaSIF-site reaching ROC AUC 0.86 with three convolutional layers against 0.77 with one, over 2,192,870 test-set surface points (Gainza 2020 (suppl.)).

Gainza 2023 turns this into **MaSIF-seed**: fingerprint a chosen target patch, then evaluate

fingerprint complementarity against a large library of candidate binding motifs — both α-helical and β-strand — harvested from the PDB, and graft the hits onto scaffolds. Patches are 12 Å radius capturing on average nearly 400 $Å^2$ of surface, chosen for consistency with the buried surface areas of real interfaces.

The conceptual inversion is worth stating precisely, because it is the section's reason for existing. Hotspot-conditioned diffusion tells a generator *where* to bind and asks it to invent a binder. Surface-fingerprint design **searches for what is complementary** to a given patch and then builds a scaffold around the hit. The epitope is an input in both cases; what differs is whether the binding chemistry is generated or retrieved.

Benchmark. Recovering the native binder in the correct conformation (interface RMSD < 3 Å) from co-crystal structures among 1,000 decoys, MaSIF-seed placed the correct answer top-1 in **18 of 31** helical cases and **41 of 83** non-helical cases, against **3/31 and 7/83** for ZDock and **6/31 and 21/83** for ZDock+ZRank2 — at average times of 15 and 118 minutes against 2,206–2,946 minutes (Gainza 2023). The speed comparison needs a caveat the main text omits: the supplement records that ZDock was run on a 96-vCPU cloud server with times measured as CPU user time (Gainza 2020 (suppl.)), so this is partly a GPU-versus-CPU comparison.

Experimental results, on four therapeutically serious targets. For the SARS-CoV-2 receptor-binding domain, MaSIF-site predicted a site distinct from the ACE2 footprint but overlapping enough to inhibit, and MaSIF-seed returned **7,713 binding seeds** for it; the initial design bound at **4.6 µM** and an optimised version at **80 nM**, with a **cryo-EM structure at 2.9 Å local resolution** agreeing with the AlphaFold prediction at **1.4 Å interface RMSD**, one binder bound per RBD on the trimeric spike. That design engages the RBD through small residues with a buried interface area of **1,452 $Å^2$** — smaller than the ~2,071 ± 456 $Å^2$ typical of antibodies — and as an Fc fusion neutralised most variants of concern except those carrying L452R, with potency comparable to the clinical antibody imdevimab; the binding mode was near-identical between wild-type and Omicron complexes at 1.4 Å backbone RMSD (Gainza 2023).

PD-L1 was attacked **because** its PD-1-engaged site is "extremely flat at the structural level" — the class of surface that resists small molecules. Designs progressed from **2 µM** to **256 nM**, **374 nM** and **120 nM** across library and site-saturation optimisation, with **crystal structures** matching the computational models at **0.8 Å RMSD** and 1.0–1.9 Å interface RMSD, and site-saturation mutagenesis confirming that the four hotspot residues MaSIF-seed had placed were **crucial** — any other residue at those positions abolished binding (Gainza 2023). That last result is the chapter's only direct experimental validation of placed hotspots, and it is the experimental counterpart of the packing-versus-ΔΔG argument in §2.

One-shot design. The paper names its own earlier limitation — designs still required in vitro evolution to reach characterisable affinity — and addresses it with a structurally diverse seed library and design biased for polar interface contacts. Screening the top **2,000** designed sequences per target by yeast display with deep sequencing against PD-L1, PD-1 and CTLA-4 gave binders for **all three**. The best PD-1 binder bound at **K_D 4.2 ± 2 µM with no experimental optimisation at all**, comparable to the native PD-L1–PD-1 interaction at 8.2 µM, improving to submicromolar with two mutations from the SSM data; a β-sheet-mediated PD-L1 binder reached an apparent **21.8 nM** on yeast while adopting a backbone conformation **different from the native** PD-1–PD-L1 interaction (Gainza 2023).

The homology-independence of the representation (Gainza 2020) and the strictness of its evaluation splits are what make this lineage a real control. The supplement reports that test sets were split by hierarchical clustering on TM-score matrices, and that a **stricter split removing every test structure with TM-score > 0.5 to any training member did not affect performance** — MaSIF-site from 359 proteins to 169, MaSIF-search from 957 to 635 (Gainza 2020 (suppl.)). For a wiki that repeatedly asks whether a design method has memorised the

PDB, this is the strongest such control in any ingested source.

7. Beyond protein targets — all-atom and ligand conditioning

A protein target can be represented adequately by its backbone. A small molecule cannot. The all-atom transition therefore matters for binder design specifically, and not merely as an accuracy improvement.

Krishna 2024 generalises the Baker-lab prediction and design stack to all-atom, ligand-aware territory, and its design arm **RFdiffusionAA** diffuses a binder backbone *around* a small molecule. Experimental validation by crystallography and binding measurements covers three ligands: the cardiac therapeutic **digoxigenin**, the cofactor **heme**, and the light-harvesting **bilin** (Krishna 2024). The supplement carries the featurisation — atomised groups, bonds, chirality, metals — and the per-ligand binder protocols (Krishna 2024 (suppl.)).

Butcher 2025 unifies the modalities in one all-atom token-and-atom model: protein–protein, protein–DNA and protein–small-molecule interfaces from a single generator, matching or beating prior methods on **protein-protein interaction, DNA-binder, small-molecule and atomic-motif-enzyme benchmarks at roughly one-tenth the computational cost** (Butcher 2025, Butcher 2025 (suppl.)). The cost reduction is best read not as a convenience but as an enabler of the search strategies in §11.

The all-atom turn is also where the generative and the predictive lineages converge: Abramson 2024 is the prediction-side counterpart, and Chu 2024 reaches all-atom generation by a different route, superposition-based rather than tokenised. Ingraham 2023 contributes the orthogonal idea that conditioning should be programmable rather than task-specific. The concept page is Diffusion generative design.

A fourth route disputes the premise. The three routes above all gave up SE(3)-equivariance to get atomistic detail, a move Lin 2026 describes as the field following "the evolution of AlphaFold2 to AlphaFold3" and eschewing "data-efficient SE(3)-equivariance". Genie 3 keeps the equivariance and gets all-atom detail from a **branched-polymer** treatment instead — sidechain frames built off the backbone frame in `atom14` ordering — with **partial atomization**, so only residues whose sidechains are actually known are represented atomistically. The equivariant frame representation it generalises dates to Lin 2023, whose 4.1 M-parameter model reached 0.790 designability against RFdiffusion's 0.951 at 59.8 M — the parameter-efficiency observation that became this lineage's thesis. That it then outperforms the non-equivariant models on binder design (§11) is the chapter's reason for treating the architectural consensus as unsettled rather than superseded.

One operational caveat on representation, from outside the literature. Ruijgrok 2026 records RFdiffusion3's output as **observed all-atom at 6.4 atoms per residue with full sidechains and a low-entropy co-designed sequence** — while noting that RFD3's own quality-control comments describe the same output as "backbone-only diffusion output", a discrepancy that report leaves unresolved. It also notes that in practice the distinction is moot for that pipeline, because the next stage discards the generator's sequence and re-derives it from backbone alone with ProteinMPNN. §9 takes up why that step is not as innocuous as it sounds.

8. Hard target classes

8.1 Flexible and disordered targets

Many peptide hormones adopt an α-helix only on binding their receptor; free in solution they are floppy. Vasquez-Torres 2023 states the difficulty as two compounding problems: the picomolar-affinity hyper-stable 50–65-residue minibinders that work against folded proteins have shapes suited to protein surfaces rather than to an extended helix, and peptides offer **fewer residues to interact with**.

The paper's three design generations are the cleanest affinity ladder in the chapter. **Parametric groove scaffolds** gave 8 of 12 designs binding for each of PTH, glucagon and NPY — but **micromolar only**. **RFjoint inpainting plus ProteinMPNN redesign**, extending the binder interface, gave 44 binders from 192 designs tested against PTH with the best at **6.1 nM**, about three orders of magnitude better than the parametric starting point; the control that matters is that **ProteinMPNN redesign of the original interface alone did not measurably increase affinity**, so the gain came from extending the interface rather than from better sequence design. The same route gave 231 nM for glucagon and 3.5 µM for NPY from 96 designs. **De novo RFdiffusion from noise** then gave **2.7 nM for secretin**, sub-100 nM for GLP1 and GIP, and **7 nM for the Bid BH3 peptide** — unstructured in isolation — a *higher* affinity than Bid's native partner Mcl-1 (Vasquez-Torres 2023).

Partial diffusion is the operator that closed the gap: noise an existing design partway, then re-denoise, diversifying the fold by an amount set by the noise level and "regularising" backbones so that ProteinMPNN sequence design on them is more designable. Applied to the 231 nM glucagon and 3.5 µM NPY binders it gave 25 of 96 designs binding glucagon and 20 of 96 binding NPY at 10 nM peptide, and final affinities of **5.6 nM for NPY** and, for glucagon, **below the limit of detection in the picomolar range**; a Bim binder reached **K_D < 500 pM**, described as the highest-affinity designed peptide binders then reported. Crystal structures at **1.95 Å and 1.81 Å** matched the design models at 0.72 Å and 0.6 Å RMSD (Vasquez-Torres 2023).

The applications close the loop to diagnostics: the 6.1 nM PTH binder was grafted into a luminescent caged reporter giving a **10 nM limit of detection with ~21-fold activation**, and the picomolar PTH and glucagon binders were used for immunoaffinity enrichment of peptide hormones **from human serum** followed by mass-spectrometric detection (Vasquez-Torres 2023).

Stark 2025 pushes the same frontier past peptides to **intrinsically disordered regions**. Twenty thousand peptide designs of 40–80 residues were generated against the disordered region of NPM1, specified to engage the disordered region while avoiding its structured β-sheet — which tasks the model with predicting how the disordered region will *become* structured in the presence of the designed binder. Five designs were tested, with reported evidence of binding **in live cells**; the paper is explicit that the in-cell experiment does not confirm that binding occurs specifically at the disordered region (Stark 2025).

8.2 Antibodies and CDR loops

Antibody design is the modality most therapeutic interest attaches to, and it is a genuinely different problem: the designable region is a set of **loops on a fixed framework** rather than a free backbone, and the interactions are loop-mediated rather than the secondary-structure-paired interfaces miniprotein design exploits.

What was missing, and why. Bennett 2026 opens on the observation that methods for the *first* stage of antibody development — discovering antibodies that bind a **specified epitope** — "do not exist", so discovery proceeds by immunization or library screening "**followed by**

subsequent epitope mapping". The epitope is normally found after the fact. Prior deep-learning work either sampled alternative *native* CDR loops to improve affinity, or used vanilla RFdiffusion, which was not built for loop-mediated interfaces.

How it is solved. A **fine-tuned RFdiffusion2** trained predominantly on antibody complexes, in which the **framework sequence and structure are supplied as conditioning** while the antibody structure is corrupted — and supplied **global-frame-invariantly**, through the template track as a 2D matrix, so framework and target templates carry no relative position (Bennett 2026). That is the architectural trick: the framework's internal structure stays fixed, inheriting a therapeutic scaffold, while the **rigid-body placement of the antibody against the epitope is designed rather than assumed**. CDR sequences come from Dauparas 2022. The filter is a **separately fine-tuned RoseTTAFold2** that must model the CDRs and find the pose given the target and epitope, and which outperformed the then-available models "especially at CDR H3" — the longest, most variable loop and the usual failure point. Its calibration is conditional rather than absolute: at **mean pAE < 10, 80.3% of structures are within 2 Å when 100% of hotspots are satisfied** (Bennett 2026 (suppl.)).

A detail from the supplement changes how the generalisation should be read: training used **SAbDab, a TCR/MHC-peptide set, and loop-mediated interactions in the PDB** (Bennett 2026 (suppl.)). The prior is over loop-mediated recognition broadly, not over antibodies — which is why the same network could be pointed at a peptide–MHC complex.

Results, in the chapter's units. Six targets (TcdB, influenza H1 haemagglutinin, RSV sites I and III, SARS-CoV-2 RBD, IL-7Rα), screened either at **9,000 designs per target by yeast display** or **95 per target** by *E. coli* expression and single-concentration SPR. **Success rates per target range from 0% to 2%** (Bennett 2026). Initial designs land at tens to hundreds of nanomolar: **78 nM** for haemagglutinin, **260 nM** for TcdB, **5.5 µM** for RBD. **OrthoRep** continuous hypermutation then buys roughly **two orders of magnitude**, giving single-digit nanomolar binders that retain epitope selectivity. scFvs assembled from designed heavy- and light-chain CDRs in **~10-million-member combinatorial libraries** reached **72 nM** (68 nM as full IgG1) against TcdB.

The specificity controls are the strongest in the chapter. The TcdB epitope was the **Frizzled interface, for which no antibody or VHH exists in the PDB**. Binding was abolished by competition with a structurally characterised de novo binder to the same site (FZD48, PDB 9CM5; AHB2, PDB 7UHB for RBD); there was **no binding to the 70%-homologous toxin TcsL**; and the VHH **neutralised TcdB toxicity in CSPG4-knockout cells at EC50 460 nM** (Bennett 2026). §9's open question about whether specificity is designed or observed is answered here more directly than anywhere else in the chapter.

Atomic accuracy, by cryo-EM. VHH_flu_01 on natively glycosylated trimeric haemagglutinin at **3.0 Å**: backbone RMSD **1.45 Å** to the design, **CDR3 RMSD 0.8 Å**, with named designed residues seen contacting the stem epitope as intended. Roughly **66% of particles** carried up to two VHHs per trimer, the partial occupancy attributed to the **N296 glycan** occluding the epitope in unbound subunits and reorienting on binding. For one TcdB scFv, high-resolution data verified the designed conformations of **all six CDR loops** (Bennett 2026).

What the number means. 0–2% against the **19%** hotspot-conditioned miniprotein rate from the same lab (Watson 2023) is the clearest measure in the chapter of what the antibody format costs — roughly an order of magnitude in yield per design. Set against Stark 2025, which obtained nanomolar nanobodies for 66% of targets from 15 tested but **60,000 generated**, the two are closer than they look: both reach useful binders by screening deeply rather than by high per-design success. The wiki's reading is that de novo design has displaced stage 1 of antibody discovery — epitope-specific generation — and has not displaced stage 2, since maturation still supplies the final two orders of magnitude.

Adjacent capabilities elsewhere: Cho 2025 demonstrates antibody CDR redesign, and Stark

2025 treats nanobodies as a first-class modality by fixing a framework and replacing its three CDRs with random-length loops from four default scaffolds.

8.3 Modality as an input

Stark 2025 is the first ingested source for which the binder's modality is specified rather than assumed. Design is controlled by a **specification language** over covalent bonds, structure constraints and binding sites, so a cyclic peptide is requested by declaring a covalent bond and a nanobody by fixing a framework. Design and structure prediction are unified in one model, which also reaches state-of-the-art folding performance (Stark 2025).

Its **novelty control** is the methodological reason this chapter leans on it. Nine targets were chosen for which **no protein with >30% sequence identity exists in a bound context anywhere in the PDB**. Against each, 60,000 nanobodies were generated *without specifying any binding site* and **15 tested**; separately 60,000 proteins of length 80–140, again 15 tested. The outcome was **nanomolar binders for 6 of 9 targets (66%) for nanobodies and the same 66% for protein binders** by SPR and BLI (Stark 2025). On five benchmark targets with known high-affinity binders, testing at most 20 proteins and 15 nanobodies gave nanomolar binders against **80%** of targets for both modalities.

The modality sweep and its affinity gradient are informative in themselves (Stark 2025): nanomolar against protein targets; **0.41 and 4.4 µM** against melittin and nanomolar or micromolar against three bioactive peptides from six designs each; **7 binders from 29** linear peptides against RagC with the best at 3.5 µM; disulfide-cyclised peptides against the RagA:RagC dimer giving binders but only **80 µM and 164 µM** as the two tightest resolved; **50–150 µM** against rucaparib and **30–250 µM** against a rhodamine derivative; and **19.5%** of antimicrobial peptide designs inhibiting cell growth more than fourfold against GyrA. The paper is candid that prior specialist work reached low-nanomolar on rucaparib through an expert-guided approach, and claims only that a general-purpose model achieves moderate affinity with simpler customisation. The wiki's reading of the gradient is that affinity still tracks how much surface the modality has to work with — which is the §2 packing argument reappearing as a constraint on modality rather than on epitope.

One internal inconsistency should be recorded rather than smoothed: the paper's summary states binders for **14 of 28** tested disulfide-bonded designs, while its §2.6 states 14 binders from **24** tested (Stark 2025).

9. Sequence design at an interface is not the monomer problem

An interface residue is solvent-exposed in the monomer and buried in the complex. A model conditioned on the monomer alone is therefore being asked the wrong question, and this section collects the evidence that the sequence step is a distinct problem rather than a detail.

Dauparas 2022 is the workhorse, and its relevance here is that it was built to be conditioned: sequence design on a fixed backbone with the option to tie or fix positions. Bennett 2023 (suppl.) establishes prospectively that it is at least comparable to Rosetta FastDesign for binders once filtering is applied, across three targets, while being far cheaper.

The **interface prior** is the subtlest issue in the chapter, and the evidence for it arrives from the opposite direction. Anishchenko 2021 designed monomers by hallucination and found that many came out as **oligomers in solution** — one design crystallising as a dimer — and diagnosed the cause precisely: trRosetta had been trained on native structures including many homo- and hetero-oligomers, so it never learned to disfavour surface hydrophobics to the degree required for soluble monomers, having absorbed **the sequence signatures of native**

protein–protein interfaces from the PDB. The rescue was manual substitution of exposed hydrophobics with polar residues, shown by paired before/after SEC traces (Anishchenko 2021 (suppl.)).

The wiki's reading is that this is a *resource* for binder design and a liability for monomer design, and that no ingested source settles which way it cuts when a generator fine-tuned from a structure predictor is asked for an interface. §13 records it as open.

Two operational findings sharpen the section beyond what the published sources establish. First, the step that discards a generator's own sequence is not free: Ruijgrok 2026 records that in the RFdiffusion3 pipeline the generator's co-designed sequence is **discarded** at `mpnn_redesign`, because ProteinMPNN is backbone-only by construction — so whatever sidechain information the all-atom generator produced is not used. Second, and more consequentially, that report finds that Proteína-Complexa's **co-designed sequence carries an antigen-copying defect**: 48 contiguous target residues reproduced inside a 79-residue binder, correlating with the search reward at ρ +0.54, severe enough that the project's headline metric is computed on MPNN redesigns because the co-designed sequence cannot be substituted (Ruijgrok 2026).

That finding stands in **direct contradiction** to the published claim about the same model. Didi 2026 reports that Complexa's self-generated sequences outperform all baselines including ProteinMPNN redesign, concluding that "such additional steps [are] unnecessary". The wiki's reading is that these are not two measurements of one quantity but measurements of two different things — an *in silico* benchmark success rate on standard targets, and an operated pipeline's behaviour on a hard real target — and that the gap between them is the single most instructive fact available to this chapter. It is taken up again in §12.

10. The oracle, its metrics, and its failure modes

The field runs on two numbers: a monomer confidence and an interface error estimate. §4 established their provenance (Bennett 2023 (suppl.)). This section is about what they cannot see and what happens when they are optimised against.

What the metrics measure. pLDDT is a local-confidence estimate from Jumper 2021; the interface predicted-aligned-error family answers whether the predictor places the two chains in the designed relative pose. Neither is an affinity. Didi 2026 states the empirical basis for using them at all — that interface confidence and alignment scores from structure predictors correlate with wet-lab success when evaluated on designed binder–target pairs — and treats them as standard *in silico* metrics on that basis.

Adversarial overfitting is the known failure, and it is severe. Wicky 2022 ran hundreds of Monte Carlo steps against AlphaFold2 with a symmetry objective, selected 150 designs with pLDDT > 0.7 and pTM > 0.7, and found that **virtually none expressed solubly** — median soluble yield 9 mg/L, with none of the marginally soluble ones showing both the expected oligomerisation state and a consistent circular-dichroism profile. The diagnosis was overfitting to the oracle: many optimisation steps against a confidence metric generate sequences the network confidently predicts as structured but which have aberrant biophysical properties. The fix was to discard the hallucinated sequences, keep the backbones, and redesign with Dauparas 2022 using **dozens rather than hundreds of evaluations** — after which 71 of 96 expressed at high levels with a median yield of 247 mg/L, a ~27-fold increase (Wicky 2022).

The structural lesson generalises past symmetry: **an oracle used as a gate is reliable in a way that an oracle used as an objective is not**, and the number of evaluations is the control variable. Frank 2024 pushes the objective route further with gradients in a relaxed

sequence space rather than MCMC, and its supplement records an interface filtering cascade (Frank 2024 (suppl.)).

Three independent demonstrations that the filter, not the generator, sets the hit rate.

- Bennett 2023: inserting the filter raised experimental success ~10× with the generator unchanged.
- Stark 2025: an earlier version of the model "suffered from a serious flaw resulting in close-to-random ranking and filtering", and for designs from that version **no hits were found** — the generator being substantially the same. This is the cleanest natural experiment in the chapter.
- Stark 2025 again, by ratio: **60,000 designs generated per target, 15 tested**. Every published binder success rate is a rate measured after a filter of roughly that severity — 4,000:1 — and this is the only source that states the magnitude.

The oracle's precision has now been measured, and it is low. Lin 2026 reports that "all evaluation pipelines exhibit low precision (maximum of 12%) and completely fail to identify binders against H3, TGFβ, and TIE2", and states the consequence directly: this "place[s] a low ceiling on the efficiency of binder design using *in silico* oracles as filters." That is a harder number than anything in Bennett 2023 (suppl.)'s ROC analysis, and it comes from a pipeline whose filtering power had been retrospectively calibrated against an experimentally validated dataset of both positives and negatives.

Standard oracles are also blind to *where* a binder binds. Lin 2026 found designs that passed an AlphaFold-Multimer benchmark while binding an **off-target site**, and repaired it by adding an on-target constraint — **≥80% of user-specified interface residues within 5 Å of the binder** by closest heavy atoms, or 100% for interfaces of three residues or fewer — calling the result the AF2M+ Benchmark and using it for all downstream analysis. This is the chapter's only ingested instance of a published filter being fixed *because* it was caught passing off-target binders, and it is the literature counterpart of the gate pathologies measured operationally in Ruijgrok 2026.

What the oracle still cannot see. No ingested source uses a predictor to assess binding kinetics, aggregation propensity, expression yield, or off-target cross-reactivity. Wicky 2022 is the proof that expression and solubility are invisible to it. And Gainza 2023 supplies the chapter's sharpest counterexample to oracle authority: for several of its experimentally confirmed binders the **AlphaFold predictions did not agree with the design models**, while competition and mutagenesis experiments nevertheless established that the designed interfaces were the ones in use — and the paper notes that AlphaFold agreement is "a predictive exercise" that "cannot be interpreted as absolute evidence that the designed binding mode is occurring, which will ultimately require an experimental structure."

Circularity is a newer failure, and it arrives with unified models. When the generator and the scorer share weights or training data, the score reports the generator's own objective back to it. Ruijgrok 2026 handles this explicitly for the BoltzGen arm, declaring its Boltz-2 scoring tier **circular** because Boltz-2 is a component of BoltzGen's own pipeline — with the asymmetric reading that "a BG advantage here is not evidence; a BG deficit would be" — and promoting AlphaFold2, which shares no weights or training pipeline, to primary binding tier for that arm. The same report records that several axes for that arm are **inflated** because the delivery's rank ordering is BoltzGen's own weighted rank, and that self-consistency RMSD is outright censored for it. Since Stark 2025 and Didi 2026 both make unification of design and prediction a selling point, the wiki's reading is that circularity is a structural consequence of the field's current direction rather than an implementation slip.

And the gate can select against the thing it is meant to select for. Ruijgrok 2026 reports that for the Complexa pipeline, placement and pass/fail are measured on *different structures* and agree at chance — Cohen's κ of +0.006 and −0.091 — because placement is read off

the search pose while passing is read off AlphaFold2's independent re-dock of the redesigned sequence, and that **the AlphaFold2 gate depletes epitope contact rather than enriching it**. For ProteinHunter the same report records that moving a promotion gate from 0.8 to 0.7 shifted promotion from 14.1% to 43.8% on byte-matched generation, meaning the stricter gate had been **discarding on-epitope designs**.

11. Compute, search, and a false dichotomy

By 2025 the methodological split in binder design was between **generative** models trained on binder–target complexes and **hallucination** methods that optimise a sequence against a structure predictor's confidence. Didi 2026 argues the split is false, and the argument is this chapter's most useful reframing.

Its analogy is to language and image modelling, where a pretrained base model is combined with adaptive inference-time compute in one framework; by comparison, generative binder design is **pure training-time optimisation** and hallucination is **pure inference-time optimisation with no generative prior** (Didi 2026). Proteína-Complexa supplies the missing combination in three parts: **Teddymer**, a large synthetic dataset built by treating the domain–domain interactions of computationally predicted monomers as surrogate binder–target complexes, used for pretraining before post-training on experimental multimers; a **partially latent** flow-matching architecture over a Cα coordinate variable plus per-residue latents decoded to all-atom structure and sequence; and inference-time optimisation adapted from the diffusion literature — **best-of-N, beam search, Feynman–Kac steering** (importance sampling of a tilted distribution) and **MCTS** — with interface pAE as the reward.

Without any test-time optimisation it reports 9.10 unique successes on self-generated sequences and 14.4 with full ProteinMPNN redesign, against 4.68 for RFdiffusion and 0.73 for Protpardelle-1c under MPNN redesign; with test-time search it reports beating the hallucination baselines **BindCraft** and **BoltzDesign** under **normalised compute budgets measured in test-time GPU-hours**, with simple best-of-N already sufficient on easy targets and the more elaborate searches paying off as targets harden (Didi 2026). It claims to be the first systematic exploration of test-time scaling for protein design.

Cho 2025 reaches the same unification from the opposite side and is the more surprising result. It is **fine-tuning-free**: start from an **all-X sequence**, let a diffusion-based structure predictor hallucinate a structure for it, then cycle sequence redesign with ProteinMPNN against structure re-prediction. There is no diffusion over coordinates and no training — the generative capacity is borrowed wholesale from a predictor's willingness to hallucinate. The X token is chosen empirically (X-initiated designs gave the highest ipTM and lowest loop fractions) and mechanistically (AF3-style models encode unknown PDB residues as X). Boltz-2 outperformed Chai-1 and AlphaFold3 as the hallucination substrate, which the paper attributes to Boltz-2 encoding each non-canonical residue as a single learned token, so X behaves as one residue, whereas atom-level tokenisation increases sampling complexity — and to AlphaFold3 having been fine-tuned on distilled AF2 predictions, which encourages spaghetti-like low-confidence regions and so suppresses the very hallucination the method depends on (Cho 2025).

Its results: higher ipTM than RFdiffusion on **9 of 11** protein targets with Boltz-2 and 7 of 11 with Chai-1; higher small-molecule success rates than RFdiffusionAA, BoltzDesign2 and its own single-pass mode on all four ligands tested, under criteria of backbone RMSD ≤ 1.5 Å, ligand RMSD ≤ 5 Å, interface PAE ≤ 1.5 and ipTM ≥ 0.8; plus cyclic peptides, multi-motif scaffolding, antibody CDR redesign and DNA/RNA binders. Its efficiency claim is that iterative cycling **reduces the need for extensive sequence sampling**, co-optimising structure and sequence across cycles. Its stated limitation is that it inherits the predictor's prior including its biases — structures from AF3-style models showed **lower diversity than RFdiffusion's**, with a preference for α-helical topologies (Cho 2025).

A third answer: don't combine them, just win. Lin 2026 declines the synthesis. Genie 3 is a pure diffusion model with no inference-time oracle optimisation in the loop, and on the ten AlphaProteo binder problems at a fixed budget of 200 generated structures per model per problem it **produces the most successful designs on 7 of 10, with BindCraft best on the remaining 3** — while RFdiffusion, BoltzGen and Proteína-Complexa are "substantially" behind both. Once success is normalised by per-sample runtime the gap over BindCraft widens, which is the paper's central claim: a generative model "can outperform a far more expensive hallucination-based model" (Lin 2026). Against Proteína-Complexa specifically, on inference-time scaling across the same ten problems, Complexa wins one, ties two and loses seven — and **in four cases exhibits no scaling behaviour at all** while Genie 3 improves robustly with compute, with the gains concentrated in multimeric targets (Lin 2026).

It also reverses the field's other recent architectural move. Where all-atom design has meant abandoning SE(3)-equivariance — "mirroring the evolution of AlphaFold2 to AlphaFold3" — Genie 3 keeps equivariance and reaches atomistic detail by treating the protein as a **branched polymer**: Frenet–Serret frames anchored at Cα for the backbone, then sidechain frames constructed from the backbone frame in `atom14` ordering, with **partial atomization** giving all-atom representation for known motif residues and backbone-only for scaffold residues (Lin 2026). Its efficiency gain is separable from its architecture and worth noting on its own: moderating the reverse process by a **direction scale on the mean** rather than a noise scale on the variance allows $\Delta t = 10$, cutting sampling from 1,000 steps to 100 — **10× faster with negligible quality loss**.

The wiki's reading. The hallucination lineage in this wiki runs Anishchenko 2021 → Wicky 2022 → Frank 2024, and its recurring failure was adversarial overfitting to the oracle (§10). Didi 2026 and Cho 2025 suggest the problem was never the optimisation but the **absence of a prior to optimise inside**: search constrained to a learned manifold cannot easily reach the adversarial region. The first builds such a prior; the second borrows one from a predictor. Lin 2026 complicates that story rather than confirming it — it shows that a sufficiently good prior with *no* search at all beats the hallucination state of the art, which suggests the prior was doing most of the work in the combined methods too. Taken together, the wiki's reading is that the field's 2024–25 consensus (drop equivariance, adopt hallucination) was adopted faster than it was tested. Whether that is the right account of the overfitting failure is recorded as open in §13.

What search costs in practice is measurable only from the first-party source, and the numbers are sobering. Over the whole history of one project, Ruijgrok 2026 measures **21,707 Slurm tasks and 10,376.1 GPU-hours** (7,418.0 generation, 2,956.8 scoring) plus 127,987.1 CPU-hours — of which **303.7 GPU-hours were charged to failed tasks**. Per design: Protpardelle 8.40 GPU-minutes, ProteinHunter 15.40, RFdiffusion3 20.00, Complexa **54.60**. The search-heavy pipeline is roughly 6.5× the cost per candidate of the cheapest, before any consideration of how many candidates each needs.

That report also supplies the counter-example to naive scaling. Protpardelle, which scales by embarrassingly parallel replication up to 2,048 independent draws rather than guided search, was found on pool clustering to gain nothing from breadth: **tripling n on one recipe added zero new folds** (Ruijgrok 2026). Independent draws and guided search are not interchangeable uses of the same compute, which is precisely Didi 2026's thesis arriving as an operational result.

12. What actually gets validated — and why the rates do not compare

12.1 The ladder

Across the sources, validation proceeds in a consistent order, and each rung proves something different.

Display for hit rate. Yeast surface display, with the operational bar **SC50 < 4 µM** in Bennett 2023 (suppl.), and deep-sequencing readout over the top 2,000 designs per target in Gainza 2023. **Single-point binding for triage.** Biolayer interferometry at 10 µM with success at ≥50% of the positive control's maximal response in Watson 2023. **Titration for affinity.** SPR and BLI in Gainza 2023 and Stark 2025; fluorescence polarisation in Vasquez-Torres 2023. **Behaviour.** Size-exclusion chromatography, monodispersity, thermostability and expression yield — the properties Wicky 2022 showed the oracle cannot see. **Structure.** Crystallography at 1.81–1.95 Å in Vasquez-Torres 2023 and 0.8 Å model agreement in Gainza 2023; cryo-EM at 2.9 Å local resolution in Gainza 2023 and of a designed binder in complex with influenza haemagglutinin in Watson 2023. **Specificity, as controls.** Knockout mutants, competition with clinical antibodies, and homologue non-binding in Gainza 2023; related-peptide non-binding in Vasquez-Torres 2023. **Function.** Viral neutralisation in Gainza 2023; biosensors and serum immunoaffinity enrichment in Vasquez-Torres 2023; in-cell binding and growth inhibition in Stark 2025.

12.2 Why the rates do not compare

Here are the chapter's headline success figures, each in its own units. They are deliberately **not** aggregated.

Source	Rate	Denominator	Assay and bar
Bennett 2023	~10× improvement	designs ordered, per campaign	yeast display, SC50 < 4 µM
Watson 2023	19%	95 designs/target × 5 targets, post-AF2 filter	BLI single point at 10 µM, ≥50% of positive control
Watson 2023 (p53/MDM2)	55/95	designs ordered	BLI, >50% max response
Vasquez-Torres 2023	44/192, then 25/96	designs screened	binding at 10 nM peptide, then FP titration
Gainza 2023	binders for 3/3 targets	top 2,000 designs/target	yeast display + deep sequencing
Stark 2025	66% of targets	15 designs/target, from 60,000 generated	SPR/BLI, nanomolar
Didi 2026	9.10–14.4 unique successes	samples per target	AF2-Multimer: pLDDT > 90, i-pAE < 7.0, RMSD < 1.5 Å
Cho 2025	9/11 targets	designs per target	higher ipTM than RFdiffusion (relative, not absolute)

Source	Rate	Denominator	Assay and bar
Lin 2026	best on 7/10 problems	200 structures/problem × 8 MPNN sequences	AF2M+: pTM > 0.8, min interaction pAE < 1.5 Å, complex RMSD < 2.5 Å, **plus ≥80% of specified hotspots within 5 Å**
Lin 2026 (Nipah)	1/8 = 12.5%	8 submitted, from 200 generated and 34 filtered	SPR K_D, blind third-party competition
Bennett 2026	**0–2% per target**	9,000 designs/target (display) or 95/target (SPR)	yeast display enrichment, or single-concentration SPR

One partial counterexample now exists, and it deserves naming before the incomparabilities. Lin 2026 runs **five methods — Genie 3, BindCraft, RFdiffusion, BoltzGen and Proteína-Complexa — on one shared ten-problem set, at a fixed budget of 200 generated structures per model per problem, through one evaluation pipeline**. That is the closest thing to a sound cross-method comparison the wiki holds. Its limits are equally clear: it is an *in silico* comparison, its oracle tops out at 12% precision and fails entirely on three targets, and the single wet-lab datapoint it carries comes from a blind competition whose comparator hit rates "encompass diverse generation and filtering procedures performed by groups with varying levels of expertise" (Lin 2026).

Four distinct incomparabilities are visible. The **assay** differs (display SC50, single-point BLI, titrated K_D, *in silico* confidence). The **bar** differs even within one assay family (19% at ≥50% of a positive control is not the same event as "nanomolar by SPR"). The **denominator** differs in kind: 95 post-filter designs in Watson 2023 versus 15 designs selected from 60,000 in Stark 2025 versus 2,000 in Gainza 2023. And the **target sets** differ, which matters most of all: Stark 2025's 66% is against targets with no >30%-identity bound-context homologue in the PDB, while its own 80% is against benchmark targets with known high-affinity binders — the same model, the same assay, a 14-point spread attributable entirely to target difficulty.

12.3 The first-party ledger

Ruijgrok 2026 is the only source in the wiki that attacks this problem directly rather than inheriting it, and it does so by refusing to rank: in place of a results table it carries a **comparability ledger** stating, per pipeline, what its rate is a fraction of, why it cannot sit beside another's, and what would have to be held fixed for a fair comparison. Its findings on six pipelines run against one target class:

- **Protpardelle-1c** — denominator is designs generated, with no pre-scoring gate, the cleanest in the set; but the metric is complex-level self-consistency from the model's own sequence, which asks whether a design is self-consistent, not whether it binds (Ruijgrok 2026).
- **Proteína-Complexa** — denominator is finalists surviving a filter capped at `filter_samples_limit: 1000`, so campaigns at n=16 and n=512 have **incomparable denominators**; and the rate is computed on redesigned rather than co-designed sequences, measuring the redesign pathway rather than the generator (Ruijgrok 2026).
- **ProteinHunter** — denominator is designs *promoted* by a generation-stage gate, with **79% of tasks lost before any validation runs** (Ruijgrok 2026).

- **RFdiffusion3** — clean denominator at n=64, but a different target set entirely, and on three of five benchmark targets the 0/64 result is **unmeasurable rather than undesignable**, because ESMFold cannot rebuild those targets (median target prediction RMSD 2.18, 9.54 and 17.68 Å); those rows carry binder-only self-consistency at 84–94% instead (Ruijgrok 2026).
- **DiffuseBio and BoltzGen** — pass rate is **banned by preregistration**, because both are post-filter deliveries with the generated pool withheld, so any rate would describe the supplier's selection rule rather than the method. Comparison is made instead as **paired deltas within each design across two target contexts**, which is immune to selection because each design is its own control (Ruijgrok 2026).

That last device is the chapter's most transferable methodological point. When a denominator cannot be recovered, a **within-design paired contrast** can still be sound where a rate cannot.

12.4 Representation as a validity condition

The same report contributes an axis absent from the published literature: an audit of **what each pipeline stage can actually see** — all-atom, backbone, Cα-only or sequence-only — on the argument that a stage seeing only backbone cannot evaluate packing, a stage that rebuilds sidechains is inferring them from a sequence rather than observing them, and reading a packing or interface number off a backbone-only stage is "the single most common way to over-read a result" (Ruijgrok 2026). Two concrete findings follow: in one pipeline the file from which every contact and interface-residue count is read is **Cα-only**, while the all-atom file goes to the sequence designer; in another the generator's sidechains are **discarded** before scoring. Given §2's result that interface energetics are a packing phenomenon correlating with occupied volume at $r^2 = 0.94$ (Halperin 2004), the wiki's reading is that atomic representation per stage is not a software detail but a **validity condition** on any interface number a pipeline reports.

13. Open questions

- **Why is the hit rate what it is?** No ingested source diagnoses *why* 19% of filtered RFdiffusion binders bind and 81% do not (Watson 2023), or whether the failures are dominated by interface quality, expression, aggregation or wrong binding mode. The chapter's two mechanistic failure accounts — surface hydrophobicity inherited from training on native oligomers (Anishchenko 2021) and adversarial overfitting to the oracle (Wicky 2022) — are both about *hallucination*, and neither transfers cleanly to a conditional diffusion model that never optimises against a predictor.
- **Is specificity ever designed, or only observed?** Most specificity results in the chapter are negative controls run after the fact (Vasquez-Torres 2023, Gainza 2023). Bennett 2026 goes furthest the other way — an epitope with no antibody in the PDB, chosen in advance, confirmed by competition against a structurally characterised binder to that site, with non-binding to a 70%-homologous toxin and functional neutralisation in a receptor-knockout line — so epitope *targeting* is now demonstrably designed. Whether broader off-target cross-reactivity is designed remains untested. The one source that treats specificity as a design objective with its own machinery does so for oligomers, not binders (Boyken 2016). No ingested source reports a binder campaign in which off-target panels were screened systematically, and the chapter cannot say what fraction of designed binders are cross-reactive.
- **Is the epitope a design variable?** Every method takes hotspots as input. Gainza 2023 gets closest to automating the choice by predicting high-propensity patches with MaSIF-site, and Stark 2025 generated against nine targets **without specifying any binding**

site at all — but neither reports how epitope choice affects outcome, and Ruijgrok 2026 shows that within one project epitope comparisons are confounded by per-arm promotion gates. Whether some epitopes are simply undesignable, and whether that is predictable in advance, is open.

- **Did the interface prior survive the transition from hallucination to diffusion?** Anishchenko 2021 attributes unintended oligomerisation to a network having learned native interface sequence signatures from the PDB. RFdiffusion is fine-tuned from a structure predictor trained on the same data, so the prior is presumably present — but whether it *helps* binder design (a useful bias toward realistic interfaces) or *hurts* it (a bias toward native-like rather than designed interfaces) is not addressed by any ingested source.
- **Was adversarial overfitting a property of the optimiser or of the missing prior?** Didi 2026 and Cho 2025 both imply the latter, by searching inside a prior and succeeding where Wicky 2022 failed. But neither runs the controlled experiment — the same objective, the same evaluation budget, with and without a prior — so the inference is architectural rather than empirical.
- **Why do the published and operated accounts of the same model disagree?** Didi 2026 concludes MPNN redesign is unnecessary for Complexa; Ruijgrok 2026 finds the co-designed sequence unusable on its target because of antigen copying. Candidate explanations — target hardness, the use of a search reward that copying can exploit, benchmark targets having bound homologues in training — are not distinguished by either source. This is the chapter's most consequential open question because it is the only place where a benchmark number and an operational measurement of one model can be compared at all.
- **Does success transfer off the targets the filters were tuned on?** The AF2-with-initial-guess protocol was validated on minibinder complexes including FGFR2 and IL-7Rα (Bennett 2023 (suppl.)), and the field's benchmark targets recur across papers (PD-L1 appears in Watson 2023, Gainza 2023 and Stark 2025). Stark 2025's 66%-versus-80% spread between novel and benchmark targets is the only quantification of the gap, from one model.
- **How far does a 256-residue training ceiling generalise?** Lin 2026 trains on nothing longer than 256 residues — AFDB monomers filtered to 20–256 residues, and Pinder complexes with the *whole complex* capped at 256 — then attributes its performance on 300–800-residue monomers and on multimeric design problems to out-of-domain generalisation, with an ablation showing its architectural changes "only impact designs longer than 700 residues". The claim is not new to Genie 3 — Lin 2023 trained its long models at the same 256 maximum and Lin 2024 states the computational reason and claims length generalisation past it on the same grounds, so this is the third iteration of a standing house argument. It is plausible on its own evidence and untested at the sizes that matter operationally: the PfLDH system in Ruijgrok 2026 is a 632-residue target in a 711-residue complex, roughly 2.8× the largest complex Genie 3 ever saw in training. Whether equivariance is what buys the extrapolation, as that paper argues, or whether the extrapolation simply has a limit nobody has looked for, is open.
- **Was abandoning SE(3)-equivariance a mistake?** Lin 2026 is the only ingested source that tests the question rather than assuming an answer, and finds equivariance competitive-to-better while being faster. One paper from one group is not a settled reversal, and no source runs the matched-architecture ablation that would isolate the equivariance contribution from the branched-polymer representation, the sampling change and the training recipe.
- **Is there a shared benchmark?** No. Butcher 2025 (suppl.) defines PPI, DNA-binder, small-molecule and atomic-motif-enzyme suites; Didi 2026 follows a different convention with different thresholds; Cho 2025 reports relative ipTM against RFdiffusion. This is the `benchmarks/` gap already noted in Introduction — the field at a glance, and binder design is where it bites hardest — three benchmark and affinity-dataset PDFs sit un-ingested in

`sources/` (see §14).

- **What does an interface cost in compute, per binder that works?** Ruijgrok 2026 gives cost per *candidate* (8.40 to 54.60 GPU-minutes) and is explicit that this is not cost per usable lead. Combining the two requires a comparable hit rate, which §12 argues does not exist. No source in the wiki can state the compute cost of one validated binder.

14. Flagged references — the history this chapter cannot yet cite

Written strictly from ingested sources, per the sourcing discipline in `CLAUDE.md`. Listed in rough order of how much they would improve the chapter; mirrored in index.

Present in `sources/` but as a supplement with no main text — the chapter's hardest block:

1. ~~**Bennett et al. —** antibodies with RFdiffusion~~ — **supplied and ingested 2026-10-21** as Bennett 2026 (+ Bennett 2026 (suppl.)), *Nature* 649 (2026). §8.2 is now written. The orphaned 2025 supplement copy remains in `sources/` and is superseded; flagged for the next lint sweep.
2. **Rocklin et al., *Science* 2017 — "Global analysis of protein folding using massively parallel design, synthesis, and testing"** (DOI 10.1126/science.aan0693). Again `sources/ProteinDesign/` holds only the supplementary materials (`RocklinBaker_Science_2027_sm.pdf` — the filename year is a typo for 2017). The yeast-display protease assay it introduced underlies the display readouts §12.1 describes.

Not in `sources/` at all, and needed for the chapter to be honest about the field:

3. **BindCraft (Pacesa et al. 2025).** The most widely used open binder-design pipeline in practice, hallucination against AlphaFold2 by gradient optimisation. It is named as a baseline by Didi 2026, Cho 2025 and Lin 2026, and is carried as a *planned* comparator at 27% field coverage in Ruijgrok 2026. Batch 6 improved the secondary picture materially — Lin 2026 supplies BindCraft's competitive position (best on 3 of 10 problems, beaten on 7), its cost disadvantage under runtime normalisation, a named weakness in hotspot fidelity on H1 and IL-17A, and a competition hit rate of 1/100 — but the chapter still cannot describe how BindCraft works from a primary source. **Highest-priority acquisition.** See BindCraft.
4. **Fleishman et al., *Science* 2011 — computational design of influenza haemagglutinin binders.** The founding de novo binder paper; §3 currently has no origin. *Not to be confused with* the on-disk `Fleischman_Baker_PloS_2011_RosettaScripts…`, a different paper (a scripting interface) which is itself un-ingested and already flagged in index as a Rosetta-baseline candidate.
5. **Chevalier et al., *Nature* 2017 — massively parallel de novo design and testing of minibinders.** Where the "minibinder" this chapter discusses throughout comes from, and the first success statistics worth quoting.
6. **Cao et al., *Nature* 2022 — "Design of protein-binding proteins from the target structure alone."** The direct precursor to Bennett 2023; §4 describes an improvement to a pipeline the wiki cannot cite. It is cited inside Stark 2025 as the source of nanomolar binders for one of its benchmark targets.
7. **AlphaProteo (DeepMind 2024).** The other major non-Baker pipeline, with reported hit rates on shared targets — one of the few remaining opportunities for a genuine cross-method comparison of the kind §12 says is currently impossible.
8. **Bogan & Thorn — the "O-ring" account of hot spots.** Cited inside Halperin 2004 for the observation that hot spots sit in regions buried in the interface, and the source of the alanine-scanning database that paper screens. §2 rests on a 53-hot-spot sample

read through one paper; the primary hot-spot literature would widen it.

9. **Chao et al., *Nature Protocols* 2006 — yeast surface display.** Every display-based rate in §12 is quoted in units of an assay no ingested source describes. (It is cited in Gainza 2023's reference list.)
10. **A source on binder specificity and off-target cross-reactivity.** §13's second open question has no literature behind it at all. This may be a genuine gap in the field rather than in the wiki.

Already in `sources/`, un-ingested, each closing a named gap:

11. **`Proteins/Hoare_2001_Analyzing Kinetic Binding Data`** — the kinetics half of §2. Every affinity in this chapter is an equilibrium constant; k_on and k_off appear nowhere.
12. **`Proteins/Vreden_JMolBiol_2015_…Docking Benchmark Version 5 and Affinity Benchmark Version 2`**, **`Proteins/LiuFan_ScientificData_2024_PPB-Affinity`**, **`Proteins/LiuWang_Bioinformatics_2014_PDBbind`** — three benchmark and affinity datasets that would answer §13's benchmark question and would be the natural first content for a `benchmarks/` page type, listed as an open format question in `CLAUDE.md`.
13. **`ProteinDesign/QianBaker_bioRxiv_2025_Accelerating protein design by scaling experimental characterization`** (+ supp) — §12's thesis that the bottleneck has moved from generation to characterisation currently has no source that *states* it.
14. **`Proteins/PassowFordayce_biorxiv_2026_Amplicon:Protein Bead Display…`** — quantitative high-throughput binding measurement; pairs with the above.
15. **`ProteinDesign/ShuarHie_BioRxiv_FAMPNN`** — full-atom sequence design, the successor to ProteinMPNN for §9.
16. **`ProteinDesign/ShuaiGrey_CellSystems_2023_IgLM`** — antibody sequence language modelling; would partly mitigate the §8.2 block.
17. **Scaffold and modality sources** for §1 and §8.3: `Proteins/ShaKoide_ProteinScience_2017_Monobodies`, `Proteins/Koide_Koide_JMB_1998_The fibronectin type III domain as a scaffold…`, `ProteinDesign/Missonmindrebo_Jardine_PNAS_2023…tetravalent bispecific nanobody…` (+supp), `Proteins/Kubala_Collins_2010_…GFP-nanobody complex` (a fully worked structural-plus-thermodynamic analysis of one binder), `Proteins/KeebleHowarth_2019_Approaching infinite affinity` and `Proteins/ZakeriHowarth_2012_SpyTa` (the covalent limit of the affinity axis).
18. **`ProteinDesign/SchneiderDuhovnik_NAR_2005_PatchDock and SymmDock…`** — would give the docking application in Halperin 2004 a primary method citation.

Missing supplements for pages this chapter leans on: Gainza 2023 (`supp: missing-requested`) — the SI holds Supplementary Table 4 with the per-target design and binder counts, without which the one-shot campaign's denominators cannot be stated beyond "top 2,000 per target". Halperin 2004 (`supp: missing-requested`) — the Supplemental Data identifies the 8 alanine-scanned complexes and carries the per-residue neighbour and volume data. Wang 2022 (`supp: missing-requested`) — already flagged in index, and §3's thinness is partly this.

A filing correction for the next lint sweep: `GainzaCorreia_NatureMethods_2023_…` is *Nature Methods* **17**, 184–192 (2020; online 9 December 2019). The filename year is wrong; Gainza 2020 follows the publication year.

15. Sources

Ingested sources feeding this chapter, roughly chronologically:

- Halperin 2004 — hot spots defined by ΔΔG; packing density correlates with binding energy at $r^2 = 0.94$ · *Structure* 2004

- Boyken 2016 — HBNet; buried polar satisfaction and specificity as a separate axis · *Science* 2016
- Gainza 2020 (+ Gainza 2020 (suppl.)) — molecular surfaces as a learnable, homology-free representation · *Nature Methods* 2020
- Jumper 2021 (+ Jumper 2021 (suppl.)) — the confidence oracle the whole filtering discipline rests on · *Nature* 2021
- Anishchenko 2021 (+ Anishchenko 2021 (suppl.)) — hallucination founded; the learned interface prior and the sticky-surface pathology · *Nature* 2021
- Dauparas 2022 (+ Dauparas 2022 (suppl.)) — the sequence-design step in every pipeline here · *Science* 2022
- Wang 2022 — deep-learning motif scaffolding; the grafting era generalised · *Science* 2022
- Wicky 2022 — adversarial overfitting when an oracle becomes an objective · *Science* 2022
- Ingraham 2023 (+ Ingraham 2023 (suppl.)) — programmable conditioning · *Nature* 2023
- Bennett 2023 (+ Bennett 2023 (suppl.)) — deep learning as filter; pLDDT and pAE_interaction; ~10× · *Nature Communications* 2023
- Watson 2023 (+ Watson 2023 (suppl.)) — hotspot-conditioned diffusion; 19% over five targets; 0.5 nM p53/MDM2 · *Nature* 2023
- Gainza 2023 — the independent lineage; nanomolar binders to spike, PD-1, PD-L1, CTLA-4 · *Nature* 2023
- Vasquez-Torres 2023 (+ Vasquez-Torres 2023 (suppl.)) — flexible peptide targets; partial diffusion; picomolar; biosensors · *Nature* 2023
- Krishna 2024 (+ Krishna 2024 (suppl.)) — RFdiffusionAA; ligand-conditioned binders · *Science* 2024
- Abramson 2024 (+ Abramson 2024 (suppl.)) — all-atom prediction; the newer oracle · *Nature* 2024
- Chu 2024 (+ Chu 2024 (suppl.)) — all-atom generation by superposition · *PNAS* 2024
- Frank 2024 (+ Frank 2024 (suppl.)) — hallucination by gradient descent · *Science* 2024
- Butcher 2025 (+ Butcher 2025 (suppl.)) — all-atom protein/DNA/small-molecule interfaces at ~10× less compute · bioRxiv 2025
- Cho 2025 — fine-tuning-free hallucination inside a diffusion predictor · bioRxiv 2025
- Stark 2025 — universal binder design; 26 targets; 66% against structurally novel targets; 60,000:15 · bioRxiv 2025
- Didi 2026 — generative prior plus test-time compute; the false dichotomy dissolved · ICLR 2026
- Lin 2023 — Genie; the SE(3)-equivariant frame asymmetry, at 4.1 M parameters · ICML 2023
- Lin 2024 — Genie 2; SE(3)-invariant motif conditioning, multi-motif scaffolding, AFDB augmentation · arXiv 2024
- Lin 2026 — Genie 3; atomistic SE(3)-equivariance; generation beats hallucination on 7 of 10 problems; a 92 nM Nipah binder · bioRxiv 2026
- Bennett 2026 (+ Bennett 2026 (suppl.)) — de novo epitope-specific VHHs, scFvs and IgGs; 0–2% per target; all six CDR loops confirmed by cryo-EM · *Nature* 2026
- Ruijgrok 2026 — first-party nine-pipeline comparison; measured compute; the comparability ledger · internal report 2026

Related pages: concepts Binder design, Motif scaffolding (functional-site design), Sequence design (fixed-backbone), Diffusion generative design, Structure prediction · topic Symmetric oligomer design · tools RFdiffusion, RFdiffusion3 (RFD3), ProteinMPNN, AlphaFold, RoseTTAFold, Chroma, Protpardelle, BindCraft, BoltzGen, ProteinHunter, Proteína-Complexa, MaSIF, Genie 3.

Catalysis

De novo enzyme design

A chapter on designing proteins that catalyse chemical reactions from scratch. It is a late chapter in this wiki for a reason: until 2026 the field's honest summary was that de novo enzymes worked on easy reactions and that the hard ones — the reactions biology actually cares about — remained out of reach. That changed within a single year, and the chapter's first job is to record how.

The defining difficulty is stated most clearly by Choi 2026: **amide bonds have half-lives of hundreds of years in neutral aqueous solution**, so cleaving one requires substantial activation, and de novo enzyme design "has been largely limited to low-energy-barrier model reactions". The closest prior achievement, designed serine hydrolases, acted on "small non-polar ester substrates that are much more activated than the amides of proteins and easier to bind than polar protein substrates" (Choi 2026). Chen 2026 puts the same barrier in chemical terms: amides are less reactive than esters because of stronger resonance stabilisation, and peptide cleavage requires departure of an intrinsically poor amine leaving group, with a pK_a above 35. Amide-bond hydrolysis is therefore the field's benchmark not by convention but by physics.

The chapter's argument. De novo enzyme design has largely solved the problem of *building* an active site and has largely not solved the problem of *preorganizing* one. Three independent campaigns cleared the amide barrier in 2026, by three different chemistries, and all three report the same pair of facts: the crystal structures match the design models to within roughly an ångström, and the catalytic efficiency is still one to four orders of magnitude below native enzymes. The gap is not that the designs were built wrong. It is that catalysis is a property of an **ensemble of conformations across a multi-step reaction coordinate**, while the design and filtering apparatus is built around **single predicted structures** — and the two are not the same object.

A second thread runs through §6 and is the chapter's most transferable finding: **what you model is what you get**. The single largest effect on hit rate anywhere in this chapter comes not from changing the generator but from changing what was present in the modelled active site during filtering.

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1. What counts as de novo enzyme design

The object is a protein that accelerates a chosen reaction, designed from a specification of the chemistry rather than evolved from a natural enzyme. In practice the specification is a **theozyme** — a geometric arrangement of catalytic groups around a substrate or transition state — and the design problem is to build a protein that holds that arrangement.

Where the theozyme comes from varies, and the variation matters. Chen 2026 took catalytic residue positions and orientations from a cocrystal structure of **aminopeptidase N with a transition-state analogue** and used an AlphaFold3 prediction of the native astacin–substrate complex as a guide for substrate placement. Choi 2026 took the catalytic triad, oxyanion hole and four substrate backbone residues from the yeast protease **Ulp1 (PDB 1EUV)**. Butcher 2026 used a mix: for its phosphotriesterase campaign the theozyme was **derived from a quantum chemistry calculation** (Butcher 2026 (suppl.)), while its transthyretin protease theozyme merged catalytic residues from aminopeptidase N with substrate backbone atoms and stabilising protease backbone atoms taken from an AF3 prediction. Elnesr 2026 uses no theozyme at all, which is §7.

Three capabilities distinguish enzyme design from the binder design of De novo binder design, and each is a reason the methods had to change:

- **Sub-ångström placement of several specific atoms**, not a complementary surface. Butcher 2026 frames the entire method around this: catalytic function "arises from the placement of many atoms in three-dimensional space", and residue-level representations "lack the resolution to specify enzyme active sites".
- **Unknown sequence positions.** The backbone coordinates and sequence positions of catalytic residues "are generally not specified in advance", so a method must **jointly infer the scaffold, the catalytic side-chain rotamers, and the sequence positions** from minimal atomic constraints (Butcher 2026).
- **A substrate that must be bound *and* oriented.** A binder needs affinity; an enzyme needs the scissile bond presented to the nucleophile in a particular geometry. Choi 2026 handles this by letting **both the conformation and the sequence of the substrate be freely sampled** alongside the enzyme, and Chen 2026 by modelling the enzyme–substrate complex as a separate filtered state.

2. What an active site has to do

A catalytic site is not a binding site with extra steps. The sources identify four requirements, and designs now routinely satisfy the first two and struggle with the last two.

Activate a nucleophile. The cysteine protease family uses a hydrogen-bonded **aspartate–histidine–cysteine catalytic triad** in which the histidine deprotonates the cysteine, and cysteine was chosen over serine in Choi 2026 partly because it "is more readily activated". The metalloprotease family instead uses a **metal ion to activate a water molecule**, with two histidines and a glutamate forming the Zn(II) centre and water as the fourth ligand (Chen 2026). That the designs achieve this is measurable: Dokki-15's cysteine pK_a is shifted from the free-thiol value of ~8–8.5 down to **6.0**, and its bell-shaped pH–rate profile with pK_a1 = 6.0 and pK_a2 = 8.0 matches ionisation of the catalytic cysteine and histidine (Choi 2026).

That the shift is incomplete is equally measurable: native enzymes reach cysteine pK_a values as low as **2.5–4.0**.

Stabilise high-energy intermediates. The **oxyanion hole** stabilises "multiple high-energy intermediates and transition states along the multistep reaction pathway" (Choi 2026). This is where the designs are most clearly partial. Mutating the designed Gln/Asn oxyanion side chain to alanine **did not uniformly inactivate** the designs, costing only 1.9- to 3.8-fold in the optimised variants, with the authors suggesting ordered waters may compensate (Choi 2026). A designed feature that can be deleted at a cost of twofold is not doing what the equivalent native feature does.

Bind and orient the substrate. Designed proteases bind **8–13-residue substrates, creating an extended recognition interface that can encode substantial sequence and conformational selectivity** (Chen 2026). The contrast with native enzymes is instructive: Choi 2026 notes that Mpro, TEV and HRV3C form well-defined substrate-binding interfaces with key hydrophobic contacts, while **Ulp1 engages the entire folded SUMO domain** over a large surface to position its scissile bond — and the designs, which copy Ulp1's catalytic geometry, have **no auxiliary folded domain** doing that work.

Hold all of this in the right geometry across the whole reaction. This is preorganization, it is §5, and it is where the field now is.

The biophysics the chapter can cite is thinner than the chemistry. Buried polar satisfaction — why an oxyanion hole is hard to build — is grounded in Boyken 2016, which establishes that a hydrogen bond is partially covalent and so restricts donors and acceptors to narrow ranges of orientation and distance, that nearly all buried polar atoms must find a partner or incur a desolvation penalty, and that such networks are inherently **many-body** and therefore badly handled by pairwise energy functions. For metals, Durr 2023 supplies the design-relevant framing: metals are attractive because they "tether to the protein using strong interactions and have favorable catalytic properties", **about 10% of all enzyme-catalysed reactions use zinc**, and computational design "is hampered by the complex electronic structure" of such metals — classical force fields handle them poorly and quantum treatment is too expensive for design-scale work.

3. The method lineage: from functional sites to free atoms

Scaffolding a functional site is the ancestral formulation, and the wiki's source for it is Wang 2022: given a site's geometry and residue identities — "from quantum chemistry, fragment docking, or extracted from a native protein" — build a protein that holds it. The concept page is Motif scaffolding (functional-site design). Anishchenko 2021 had already anticipated the application, noting its loss could be generalised to include binding motifs or catalytic sites, and arguing that because the network hallucinates plausible structures from scratch, no scaffold blueprint need be specified in advance.

Conditional diffusion generalised this. Watson 2023 reports state-of-the-art on metal-binding and enzyme active-site scaffolding among its tasks, achieved by fine-tuning on **minimal motifs** (Watson 2023 (suppl.)). Krishna 2024 made the generator ligand-aware.

RFdiffusion2 — the step that unlocked everything in §4. Ahern 2025 diagnoses the limiting constraint precisely. Existing motif-scaffolding methods represent motifs as **backbone frames**, parameterising each residue's N, Cα, C atoms in SE(3), and **each motif frame requires a prespecified sequence index** giving its position along the chain. That is adequate for scaffolding a contiguous domain and wrong for a theozyme, which is a handful of functional groups whose sequence positions are unknown and whose rotamers must be inferred. The prior workaround was **inverse rotamer sampling** — enumerating backbone placements consistent with each side-chain geometry — which the paper objects to as expert-intensive

and sampling-limiting. Before that, RosettaMatch searched existing scaffolds for placements, "restricted to theozyme geometries that could be matched to the input scaffold set".

RFD2's answer has two parts. Residues may be represented **either as a frame or as heavy atoms**, so the model can atomize a catalytic residue and learn the distribution of its side-chain poses — which is how it can "forgo inverse rotamer sampling". And modifying the index features yields **unindexed residues**, which — with **no auxiliary losses** — the network learns to superimpose onto residues of the generated protein, so conditioning on an **unindexed motif** samples proteins containing the motif without being told where it sits (Ahern 2025). The supplement shows this is a trained behaviour rather than an inference-time device: each residue is annotated unindexed with probability `P_IS_UNINDEXED` during training-example construction (Ahern 2025 (suppl.)). Training uses **flow matching** rather than denoising diffusion, over a data representation containing both atoms in $\mathbb{R}^3$ and frames in SE(3).

The measured effect is the clearest before/after in this chapter: **RFD2 solves all 41 AME benchmark cases where previous methods solved 16** (Ahern 2025).

Both 2026 Nature protease campaigns use **RFD2-MI**, a variant optimised for enzyme *and* protein–protein interface design, "as the de novo design of proteases requires solving both challenges" (Choi 2026, Chen 2026). RFD2-MI has no methods paper of its own; Choi 2026 cites a phospho-tyrosine peptide binder preprint for it and Butcher 2026 cites the metalloprotease preprint — see §11.

Free-atom diffusion. Butcher 2026 identifies RFD2's remaining limitation precisely: because diffusion remains at the residue level, it "is incapable of creating additional side chain interactions with ligands". RFD3 diffuses at the atomic level throughout. For catalytic sites, an extra token is appended per catalytic residue, and crucially **the atomic motif conditioning tokens correspond only to the atoms actually being fixed**, where RFD2 had to represent the full catalytic residue in order to locate a backbone position. The supplement makes the mechanism exact: standard residues are single tokens, while **modified residues, all ligands and unindexed conditions are tokenized per-atom**, and tokens are *indexed* or *unindexed* by whether residue index is revealed (Butcher 2026 (suppl.)) — so "scaffolding an unindexed atomic motif" is a precise statement about the representation.

The AME benchmark is itself a contribution of Ahern 2025, which built it after finding the pre-existing enzyme motif-scaffolding benchmark inadequate: PDB crystal structures **where all reactants and cofactors were present**, curated to **41 active sites** across diverse reactions, 100 structures sampled per case (Ahern 2025). The measured benefit of free-atom diffusion on it: on **atomic motif enzyme (AME)** problems drawn from the **Mechanism and Catalytic Site Atlas**, where difficulty rises with the number of catalytic motif islands, **RFD3 outperforms RFD2 on 37 of 41 (90%) tasks**, with improvement concentrated on the harder problems; on four six-residue active sites it recapitulated active-site residues to within 1.25 Å after refolding more often in all four cases; and rotamer quality, scored by a **Dunbrack backbone-dependent rotamer probability loss**, is consistently better (Butcher 2026). The metric's definition carries a caveat worth holding: it **excludes Gly, Ala and Pro** (Butcher 2026 (suppl.)).

Two further capabilities arrive with the atomic representation. **Symmetric enzyme complexes**: tested on 15 C2-symmetric AME cases, RFD3 produced designs with consistent catalytic-residue alignment across subunits "even in challenging cases involving up to seven catalytic residues per subunit" (Butcher 2026) — connecting this chapter to Symmetric oligomer design. And **mixed protein/RNA active sites**, demonstrated on the RNase P site which uses amino-acid side chains and RNA bases together.

4. 2026 — the amide barrier falls, three times

Three campaigns, three chemistries, one year. They are best read as independent replications of the claim that de novo amide hydrolysis is now possible, because they share a lab lineage but not a mechanism.

4.1 An activated cysteine nucleophile

Choi 2026 scaffolded the Ulp1 triad and oxyanion hole with RFD2-MI, letting substrate sequence and conformation sample freely, designing sequences with ProteinMPNN to optimise enzyme and substrate together, and filtering on AlphaFold3 **apo-state** self-consistency — a deliberate guard, since large substrate-binding sites "are susceptible to collapse in the absence of substrate".

13 of 69 designs were active (19%), from 3 of 12 unique backbones (Choi 2026) — the families Gawi, Kal and Dokki. LC–MS confirmed **all 13 cleaved precisely at the intended site**. Cys-to-Ala and His-to-Ala abolished activity. After ProteinMPNN redesign (21 Gawi and 89 Dokki variants, of which 7 and 24 improved by 1.2- to 31-fold), steady-state kinetics gave **k_cat/K_M of 0.24 ± 0.03 M^{-1} s^{-1} for Gawi-09 and 12 ± 2 M^{-1} s^{-1} for Dokki-15**, rising to **413 ± 50 M^{-1} s^{-1} for Dokki-v2**. Rate enhancements reach **3 × 10^7 before optimisation** and **1.8 × 10^9** for Dokki-v2 against the measured uncatalysed rate of 3.6 × 10^{-11} s^{-1} (Choi 2026).

Two results beyond the kinetics. **Six crystal structures within Cα r.m.s.d. < 1.2 Å of the design models**, on folds with **TM-score < 0.50 to natural proteases** — with the novelty computed on the experimental structures by Foldseek, not on models (Choi 2026 (suppl.)). And the designs **work in living mammalian cells**: in a nuclear-localised sfGFP–substrate–mScarlet reporter, cells expressing the Dokki series, "including the original Dokki designs straight out of the computer", relocalised mScarlet out of the nucleus, while Cys-to-Ala mutants did not (Choi 2026).

4.2 A water-activating zinc centre

Chen 2026 took the opposite chemical route — metal-activated water rather than an activated protein nucleophile — on the same generator. Each design was modelled in **three states**: apo, enzyme–substrate complex, and a **transition-state-analogue complex** using a phosphonamidate TSA, chosen because many crystal structures exist for it and it is therefore "better represented in ML training data than the actual transition state".

Of 135 designs tested, 36% cleave their cognate substrate at the targeted scissile bond (Chen 2026) — a figure §6 shows to be an average across two very different filters. **Zn45** on the ZnO36 substrate reached k_cat = 0.025 ± 0.002 s^{-1}, K_M = 26 ± 5 µM, **k_cat/K_M = 900 ± 200 M^{-1} s^{-1}**, a **>10^8-fold rate acceleration**; an optimised variant reached **k_cat/K_M = 31,000 ± 5,000 M^{-1} s^{-1}**, a 30-fold gain in both k_cat and k_cat/K_M, taking acceleration past **10^{10}-fold**. Crystal structures of Zn5–Zn(II) (3.41 Å, PDB 11CE) and the ZnO7–Zn(II)–substrate complex (2.37 Å, PDB 11CM) match design models at **0.82 Å and 1.35 Å Cα r.m.s.d.**, with zinc resolved at the intended site at full occupancy and the catalytic water resolved in ZnO7 (Chen 2026).

4.3 Free-atom diffusion, two zero-shot campaigns

Butcher 2026 ran two 96-construct campaigns with no experimental optimisation.

Phosphotriesterases. The active site is unusually demanding: six catalytic residues coordinating **two metal ions, including a post-translationally carbamylated lysine bridging**

them. From 96 designs, **4 were active, all from the same base scaffold** (Butcher 2026); **ZAPP-1** performs multiple-turnover catalysis at k_cat = 0.015 ± 0.003 s^{-1} and k_cat/K_M = 2.0 ± 0.6 M^{-1} s^{-1}. The designed modification is load-bearing and was tested: removing carbamylation gave an **11-fold decrease in k_cat and 4-fold in k_cat/K_M**. A second round gave 7 characterised hits across **4 distinct scaffolds**, with ZAPP-2 at a rate enhancement of **(5 ± 2) × 10^5** (Butcher 2026).

Transthyretin proteases. The substrate is **PAINVAVHVF**, the longest β-strand in the core of the TTR amyloid fibril (PDB 8GBR), chosen for disease relevance rather than assay convenience. RFD3 simultaneously scaffolded the catalytic residues, **determined the substrate conformation**, and generated a bespoke active-site cleft. **56 of 96 designs (58%) showed >10% cleavage in cis** (Butcher 2026); the best, TTRg11, cleaved the majority of substrate in an hour, with mass spectrometry confirming the designed site, and gave **k_cat 0.072 ± 0.008 s^{-1}, K_M 70 ± 10 µM, k_cat/K_M 1100 ± 300 M^{-1} s^{-1}** as a one-shot design — "within the range (although at the low end) of native proteases with well-characterized biological functions". The paper names the advance over RFD2-era protease design as two things: the very high in-cis success rate, and **activity on natural peptide-derived substrates** rather than synthetic substrates optimised for cleavability.

4.4 Reading the three together

The three campaigns are not directly comparable — different chemistries, substrates, assays and success definitions — and §8 says so in detail. What they jointly establish is narrower and more robust than any one of them: **designed proteins can hydrolyse peptide bonds, at designed positions, by at least two mechanisms, on folds unlike natural proteases, with crystal structures confirming the active site was built as specified.** That proposition had no support in this wiki a year ago.

4b. The ester precedent, and an independent lineage

Two papers published in *Nature* 649 on the same day frame everything in §4, and neither is about amides.

The ester precedent — how high designed catalysis had already reached

Kim 2026 is the direct companion to Ahern 2025, and it is the activity bar the protease papers are measured against. Target: hydrolysis of the fluorogenic ester **4MU-phenylacetate**. The theozyme came from **DFT** — the transition-state geometry for Zn(II)-hydroxide attack on the ester, with no native enzyme's geometry copied.

86 of 96 designs expressed; 5 had activity above background; the best gave k_cat/K_M = 16,000 ± 2,000 M^{-1} s^{-1} — against **3 to 60 M^{-1} s^{-1} for previously designed metallohydrolases** (Kim 2026). A second 96-design round, whose theozymes now **contained the catalytic base**, produced three more: **ZETA_2 at 53,000 ± 5,000 M^{-1} s^{-1} with k_cat = 1.5 ± 0.1 s^{-1}**, ZETA_3 at 19,000, ZETA_4 at 1,100. The four most active differ from known structures **and from each other**; ZETA_1 retains activity for **at least 1,000 turnovers** and its closest PDB and AFDB matches score TM 0.41 and 0.49 with no analogous catalytic arrangement. Crystallography: ZETA_2 apo at **3.5 Å, Cα r.m.s.d. 1.1 Å (PDB 9PYJ)**, and Zn-soaked at **2.1 Å, Cα r.m.s.d. 0.8 Å with zinc at 100% occupancy** at the designed location (PDB 9PYL).

Two details worth carrying. The **Zn(II) ion itself acts as the oxyanion hole**, stabilising developing negative charge with no nearby side-chain hydrogen-bond donors — which sidesteps the many-body buried-polar problem of Boyken 2016 that Choi 2026 struggles with. And the designed mechanism is only partly confirmed: the triple-histidine knockout inactivates, but **H130A costs only 13-fold and D67A — the proposed general base — barely affects k_cat/K_M and increases zinc affinity**, suggesting competing zinc-binding configurations (Kim 2026).

Why this matters for §4. 53,000 M^{-1} s^{-1} on an ester against **413 M^{-1} s^{-1}** on an amide in Choi 2026 is roughly two orders of magnitude — a measure of what the substrate change costs rather than of what the methods differ by. The amide barrier is real, and these numbers size it.

An independent lineage — Riff-Diff

Braun 2026 (Oberdorfer lab) is the only non-Baker-lab enzyme-design source in the wiki, and its complaint is about *rates* rather than scope: "an often underemphasized limitation of current approaches to enzyme design is the **low initial catalytic rates** of the designed biocatalysts… the current paradigm for generating efficient biocatalysts is to **compensate for low initial efficiencies with high-throughput screening and directed evolution**."

Riff-Diff builds *on* Watson 2023 rather than retraining it — the methodological contrast with Ahern 2025, which learned the inverse-rotamer problem away. Two contributions:

1. **Artificial motif libraries.** Each catalytic residue is embedded in a **helical fragment** with rotamers chosen for compatibility with that fragment's backbone; the pipeline then **inverts rotamers** to fix functional groups in space while varying backbone positions, searches all non-clashing fragment combinations, and ranks assemblies by **rotamer probability and PDB abundance** (Braun 2026). An enumerative solution to the same problem RFD2 solved by learning.
2. **The placeholder helix.** From **PDBbind**, natural enzymes bury substrates such that an average of **7.1 α-carbons lie within 8 Å**, normalised by ligand heavy atoms. RFdiffusion's substrate auxiliary potential "decreased the number of clashes… but **failed to promote the formation of well-defined substrate binding pockets**". The fix is to add an **α-helix as an entry-channel placeholder** at the pocket position, centre the denoising trajectory on it, constrain backbone distances to that centre — then **delete the helix after diffusion, leaving a vacant pocket** (Braun 2026).

Results that change the chapter's sense of the ceiling. From the **RA95.5-8F** retroaldolase tetrad — the same theozyme Ahern 2025 used — 36 designs ordered, 35 produced, **29 of 35 correctly folded**, and the best two cleaved rac-methodol at **k_cat 3.6 × 10^{-2} and 3.1 × 10^{-2} s^{-1}**, a **~5 × 10^{6}-fold acceleration**, faster than the evolved catalytic antibody 38C2. **RAD29's efficiency of 290 M^{-1} s^{-1} approaches the extensively evolved RA95.5-5 at 320 M^{-1} s^{-1}** (Braun 2026). For comparison, Ahern 2025's retroaldolase from the same theozyme reached **6.34 ± 0.92 M^{-1} s^{-1}** from 96 designs — roughly 45-fold lower, which is this chapter's only direct cross-method comparison on one reaction.

Stereoselectivity, reported nowhere else in the wiki. RAD35 showed an E-value over 200 and 99% enantiomeric excess for (R)-methodol, with HPLC determination of absolute configuration (Braun 2026, Braun 2026 (suppl.)). RAD29 was modestly selective (E-value 4, 60% ee).

And a non-biological reaction. The **Morita–Baylis–Hillman** reaction has no known natural enzyme. From two evolved catalytic arrays, **17 of 18 (94%)** and **42 of 45 (93.3%)** designs showed product formation, with MBH48 at k_cat 0.025 min^{-1}, 1.5× higher than its evolved comparator, and an unfolding onset near 85 °C (Braun 2026). Six high-resolution

structures across both campaigns, down to **1.13 Å (MBH2, Cα RMSD 0.53 Å)** and MBH48 at Cα RMSD 0.78 Å.

The honesty worth noting. The paper reports that **22 designs had k_cat below 10^{-3} s^{-1}, "within the range achievable by an isolated lysine in a hydrophobic pocket"** (Braun 2026) — most designs are not doing designed catalysis, and it says so. And it states that explaining the rate enhancement "will require more than the two crystal structures presented here".

The wiki's reading. This lineage changes what the chapter can claim about the ceiling. §5's diagnosis — that designs are 1–4 orders of magnitude below native because preorganization is imperfect — holds across the Baker-lab campaigns. But a method that reaches **90% of an extensively evolved enzyme's catalytic efficiency** on a well-studied reaction, with 99% ee, suggests the ceiling is a property of current pipelines rather than of de novo design. That is the strongest available argument that the gap is closable, and it comes from the one independent group in the chapter.

5. Preorganization — the diagnosis everyone converges on

Every campaign reports a gap to native enzymes, and every campaign reaches the same explanation. This convergence is the chapter's strongest synthesis.

The gap, quantified. Chen 2026 states that catalytic efficiencies "straight out of the computer are still **10^3–10^4-fold lower**" than native enzymes. Choi 2026 places Dokki-v2's k_cat "at the lower end of the native range, approximately **one to two orders of magnitude** below the most active native enzymes", benchmarked against Ulp1, Mpro, TEV and HRV3C with the comparison set given explicitly (Choi 2026 (suppl.)). Chen 2026 also puts a thermodynamic number on it: machine-learning force-field calculations place the activation barriers of Zn45 and ZnO7 **3.6 and 15.1 kcal mol^{-1} higher than native astacin** — though that paper's own supplement discusses the limitations of those calculations (Chen 2026 (suppl.)), so the chapter treats the figure as indicative.

The gap is not a construction error. This is what makes the diagnosis interesting. The crystal structures confirm the designs were built as specified — Cα r.m.s.d. < 1.2 Å across six structures in Choi 2026, 0.82 Å and 1.35 Å in Chen 2026, with catalytic residues at 0.9 Å and metal at full occupancy. The active sites are where they were designed to be and the enzymes are still slow.

The mechanistic account. Choi 2026 supplies the chapter's best evidence, using **PLACER** ensembles spanning the catalytic cycle — the first transition state, the acyl–enzyme and other states. In Dokki and Dokki-15, **His45 samples a non-productive interaction with Ser44**; the S44V substitution in Dokki-v2 suppresses that state and improves dyad preorganization, while P67V and T99F improve positioning of the oxyanion Gln65. The decisive observation is that the **fraction of productive conformations — defined by Cys–His < 4.2 Å — tracked activity across the series**, consistent with initial nucleophilic attack being rate-limiting. Molecular dynamics of the Michaelis complex found productive conformations sampled more frequently in Dokki-v2. Chen 2026 reaches the same place from the other side: four substitutions improving preorganization bought a 30-fold gain and took acceleration past 10^{10}-fold.

Substrate positioning is the second half. Choi 2026 separates the two explicitly: imperfection "both in the precise geometry of the catalytic machinery, and in positioning the peptide substrate relative to this machinery so that the peptide bond to be cleaved is properly oriented". The Ulp1 comparison makes the point concrete — the designs copy Ulp1's catalytic geometry without Ulp1's large SUMO-engaging interface — and the supplement's measurement of **Ulp1 on peptide substrates lacking SUMO** is what makes the comparison fair (Choi 2026 (suppl.)).

The wiki's reading. A designed enzyme is specified and filtered as a *structure*; catalysis is a property of a *distribution over structures* along a reaction coordinate. The PLACER result is the clearest statement of this in any ingested source: the quantity predicting activity was not the designed geometry but the **fraction of an ensemble occupying a productive geometry**. Every campaign's residual gap is consistent with the design apparatus optimising the mode of a distribution while catalysis depends on its tail. §10 records what would test this.

6. What you model is what you get

The single largest effect on outcome in this chapter is not a generator, an architecture or a sequence-design model. It is **what was present in the active site when the design was filtered**.

6.1 The catalytic water

Chen 2026 ran two filtering strategies differing only in whether the catalytic water was modelled in the enzyme–substrate complex, ordering genes for **95 designs from the Zn(II)-only strategy and 40 from the Zn(II)-water strategy**:

Filtering strategy	>10% cleavage at 1 h	>10% cleavage at 12 h
Zn(II)-only models	**3 of 89** (3.4%)	14 of 89 (16%)
Zn(II)-water models	**31 of 38** (82%)	35 of 38 (92%)

Roughly a **24-fold difference in one-hour hit rate**, from the same generator, the same designers and the same assay. The mechanism is shown on a matched pair: in Zn(II)-only models the inactive Zn56 and the active ZnO7 both place substrate in their clefts, "providing no basis for differentiating active and inactive designs"; in Zn(II)-water models the water in Zn56 **clashes with substrate binding, displacing the target amide bond by several ångströms**, while ZnO7's substrate binds productively. Zn(II)-water modelling "would have eliminated" the inactive design (Chen 2026).

What makes this more than an anecdote is that the **numerical thresholds did not change**. The supplement gives them per state — apo at pLDDT > 86 and catalytic side-chain RMSD < 1.5 Å; enzyme–substrate at pLDDT > 92, mean interface PAE < 5, minimum interface PAE < 1.2; TSA complex at pLDDT > 92 and mean interface PAE < 5 (Chen 2026 (suppl.)) — and both design sets passed them. The difference was in **what the model was asked to place**, not how confidently it had to do so.

6.2 A confident oracle endorsing an impossible site

Elnesr 2026 supplies the complementary failure, and it is sharper because it was published as a negative control. The design **desD1** was deliberately chosen far from the Pareto knee point; its site "contains only a single ligand within binding distance — a coordination environment insufficient for stable metal chelation". Experimentally it binds no zinc: null by MagFura-2 competition titration and by ICP-MS.

Post hoc, **AlphaFold3 predicted desD1 with excellent overall structural agreement and assigned the metal a pLDDT of 86.55**, placing it exactly where the design had (Elnesr 2026). The authors state the consequence without softening it: **"By standard computational metrics and filtering criteria, this design would have been classified as a success."**

The cause was traced to training data. Auditing the training sets of LigandMPNN and Durr 2023, they found zinc sites with **fewer than three coordinating residues within 3.0 Å**, which "may reflect crystallization additives or soaking agents rather than biologically" meaningful sites, and retrained **Metal3D-Clean** excluding them, observing the expected drop in confidence on exactly those sites (Elnesr 2026). Metal3D's own paper anticipates the mechanism, noting metals are often present in crystallization buffers because they help overcome the enthalpic cost of protein surface association (Durr 2023), and its supplement stratifies nearly every performance figure by coordination number and occupancy, reporting MAD separately for all sites and for sites with at least two protein ligands (Durr 2023 (suppl.)). The artefact was visible in the evaluation all along; it became consequential only when the model was used generatively.

PLACER as a prospective filter, validated both ways. Kim 2026 is the chapter's one case where an ensemble filter was used to *select* rather than to explain. PLACER — a network that, given a backbone with a substrate, **fully randomises the substrate and all side chains within a 600-atom sphere and regenerates their coordinates**, with repeated trajectories giving an ensemble — predicted the eventual best design, ZETA_1, to have a highly preorganized site with catalytic side chains fixed and substrate held in place, while in the inactive designs H7 and H8 "it fluctuated considerably". ZETA_1 **was the top-ranked design in the in silico ranking**. The confirming test is retrospective and decisive: **seven designs from the same ZETA_1 backbone family, filtered out during selection for suboptimal PLACER metrics, were later expressed and found to have very low or no activity** (Kim 2026). Same backbone family, same campaign, separated by a PLACER score. The supplement compares AlphaFold2, PLACER and Chai-1 metrics head to head on the ordered designs (Kim 2026 (suppl.)).

Taken with §5's finding that the productive-conformation fraction tracks activity (Choi 2026), the wiki's reading is that **the ensemble is the filterable quantity**, and that the field has now demonstrated this prospectively once.

Two further negative results belong here, both from Chen 2026 (suppl.): **"Introduction of substrate-stabilizing residues by MPNN reduces activity"** — sequence design aimed at better substrate binding made enzymes worse — and an explicit assessment of **the predictability of AF3 and MPNN in the mutagenesis space** around a designed active site.

The wiki's reading. De novo binder design §10 catalogues four ways an oracle misleads: adversarial overfitting, circularity, gates that deplete the signal they select for, and plain disagreement with experiment. This chapter adds a fifth and more basic one: **the oracle scores the model it is given, and an active site modelled without its catalytic machinery is a different object from the one that has to work.** Confidence thresholds cannot detect this, because the model is confidently right about the wrong thing.

7. The alternative premise — designing the coordination sphere

Every source in §3 and §4 begins from a native catalytic motif. Elnesr 2026 argues that premise is insufficient and tests the alternative.

Its objection: current methods extract or generate a catalytic motif and place it in a designed backbone, and "while this workflow is intended to preserve the motif geometry to hopefully recapitulate catalytic function, **optimizing for the motif alone does not guarantee catalytic competence**". The reason is second-shell — residues around the motif "play an important role in mediating affinity, specificity, and reactivity", well documented in natural enzymes and largely absent from motif-centric design.

dEVA (design by EVolutionary Algorithm) replaces motif scaffolding with multi-objective optimisation on **NSGA-II**, returning solutions on a **Pareto front** and selecting the knee point rather than maximising any single score. For metal sites it pairs **p(sequence | metal, structure)** from LigandMPNN with **p(metal | sequence, structure)** from Durr 2023, alternating until the population converges. It is zero-shot in a stronger sense than the campaigns of §4: "it imposes no requirement for structure prediction, natural templates, or evolutionary information". Scaffolds came from Protpardelle and sequences were initialised with ensemble Caliby (Elnesr 2026).

Metal sites without templates (Elnesr 2026). Eight designs, all well expressed as soluble monomers. **desH2C2** — an immunoglobulin-like fold with a **His_2Cys_2 site designed without a motif template** — binds zinc at **K_d = 37 ± 4 nM** at 1:1 stoichiometry, with **EXAFS establishing 2 Zn–N and 2 Zn–S** coordination and a Zn K-edge distinguishable from canonical zinc fingers. Three crystal structures confirm backbone fidelity on demanding folds: a de novo asymmetric β-propeller at **1.48 Å (PDB 12WO)** with zinc resolved **within 0.90 Å of the designed location**, a half-open TIM barrel at **1.55 Å (PDB 12WN)**, and desDEH (PDB 12WM) (Elnesr 2026).

The enzyme. To obtain catalysis rather than binding, Metal3D was retrained as **Metal3D-Cat** on annotated catalytic zincs from MAHOMES-II, approximating p(catalytic metal | sequence, structure). dEVA returned **only three Pareto-optimal solutions** (Elnesr 2026); of these, **desB** emerged active — and carries a **binuclear zinc site**, "the same bi-zinc site that underlies the exceptional efficiency of many naturally occurring metallohydrolases", which it was not told to build. Its novelty is established four ways: **Folddisco and RCSB Structure Motif Search returned no structural motif neighbours, BLAST no sequence homologues, and Foldseek structural neighbours contain no bi-zinc sites** (Elnesr 2026).

The catalysis is the headline. Phosphomonoester hydrolysis — uncatalysed half-life **>500,000 years** — at **k_cat/K_M = 1300 ± 400 M^{-1} s^{-1}** (4-MUP), a rate enhancement of **3 × 10^{13}**, described as **the highest reported for any de novo designed hydrolase**, and within the range of natural phosphomonoesterases. desB is also promiscuous, hydrolysing **phosphodiesters** — uncatalysed half-lives **>13 million years** — at enhancements of 3 × 10^{11} and 6 × 10^{12} (Elnesr 2026). Mechanistic support: ICP-MS confirms two zincs per monomer; alanine scanning of **first- and second-shell** residues reduces activity; alkaline pH optimum is consistent with two proximal zincs lowering metal-bound water's pK_a to ~7; XAS with orthophosphate gives **Zn–Zn ~4.3 Å** in solution, within the range of competent binuclear zinc hydrolases, with EXAFS fitting 4 Zn–N, 1 Zn–P and 1 Zn–Zn. The crystal structure at **2.29 Å (PDB 36BS)** shows **Zn–Zn 5.9 Å** against a design model's 5.8 Å, contrasting with the compressed 4.3 Å on phosphate binding — suggesting an open resting state that closes on substrate.

How to weigh it. The result rests on **three Pareto-optimal designs, one of which worked**, and the substrate class was chosen because it dominates Metal3D-Cat's training data, so the demonstration is in-distribution for the objective model even where it is out-of-distribution for the PDB (Elnesr 2026). Against that: the site is novel by four independent searches, the rate enhancement is the highest recorded, and nothing about a binuclear zinc centre was specified. The wiki's reading is that this is a genuine second hypothesis about where catalytic competence comes from — **from optimising the coordination environment and its second shell, rather than from transplanting a geometry** — and that §5's preorganization diagnosis is the reason to take it seriously, since second-shell residues are precisely what preorganize a site.

Two senses of "zero-shot" now circulate and the chapter keeps them apart: **no experimental optimisation** (Butcher 2026) and **no template, predictor or evolutionary information** (Elnesr 2026).

8. What actually gets validated

The validation ladder here is deeper than in De novo binder design, because catalysis admits mechanistic tests that binding does not.

Activity, in two formats. A **cis** assay fuses substrate to the design so cleavage is intramolecular and reads out as a molecular-weight shift — used with mScarlet and SDS–PAGE in Chen 2026, Choi 2026 and Butcher 2026. A **trans** assay uses separate substrate, typically FRET-based with GFP/mScarlet or EDANS/Dabcyl reporters. The two do not give the same number: in Butcher 2026, 56 of 96 designs passed in cis and three were selected as best in trans.

Mechanism, by knockout. Catalytic residue to alanine, with the expectation of abolition: Cys and His knockouts abolish activity in Choi 2026, while the oxyanion Gln knockout costs only 1.9- to 3.8-fold — a partial dissociation between designed and operative mechanism. Metal dependence is tested by chelation or omission (Chen 2026, Elnesr 2026), and cofactor modification by removing it (Butcher 2026's carbamylation test).

Site identity, by mass spectrometry. LC–MS confirming cleavage at the intended residue, in all three protease campaigns.

Kinetics, as k_cat, K_M and k_cat/K_M, with rate enhancement quoted against a measured uncatalysed rate — **k_uncat = 3.6 × 10^{-11} s^{-1} at 25 °C, pH 6.8** in Choi 2026.

Structure, by crystallography, with the designed feature resolved: zinc at full occupancy (Chen 2026), the catalytic water (Chen 2026), the binuclear site and its Zn–Zn separation (Elnesr 2026).

Coordination environment in solution, by **XAS and EXAFS** — unique to Elnesr 2026 in this wiki, and the only technique here that reports on the metal site in the state the enzyme actually works in rather than in a crystal.

Ensembles, by PLACER and MD (Choi 2026), and **energetics**, by machine-learning force fields (Chen 2026).

Function in context: mammalian cells (Choi 2026), full-length human TDP-43 and masked-cytokine activation (Chen 2026).

Why these success rates do not compare either

The same caution De novo binder design §12 raises applies here, with enzyme-specific twists:

Source	Rate	Denominator	What counts as success
Choi 2026	13/69 = 19%	designs ordered	cleavage detected in cis by SDS–PAGE/SEC
Chen 2026	36% of 135	designs ordered, **two filters pooled**	>10% cleavage, 1 h or 12 h, ± Zn
Chen 2026 (Zn-water arm)	31/38 = 82%	one filter's designs	>10% cleavage at 1 h
Butcher 2026 (TTR)	56/96 = 58%	designs ordered	>10% cleavage in cis at 18 h
Butcher 2026 (ZAPP)	4/96 = 4%	designs ordered	activity detected, round 1
Elnesr 2026	1 of 3	Pareto-optimal solutions	activity above background

The denominators differ in kind, the incubation times differ by a factor of 18, and the **4% versus 58% within a single paper** shows that the dominant variable is reaction difficulty rather than method quality. Pooling across two filters, as the 36% figure does, is the enzyme analogue of the denominator problems catalogued in Ruijgrok 2026.

9. Specificity, and designing against disease

Specificity arrives in this chapter in a way it never did in De novo binder design, where §13 records as open whether specificity is ever designed or only observed.

It emerges unrequested, and is predicted. Chen 2026 tested cross-reactivity across five scaffold families and found **four of five selective for their cognate substrates even though selectivity was not explicitly enforced**. The fifth, Zn45, was promiscuous — and "this experimentally observed cross-reactivity is consistent with the AF3 screening metrics", i.e. it was visible in advance. Quantified, Zn45 turned over roughly **200 molecules** of a non-cognate substrate per enzyme overnight.

It is testable against decoys. Butcher 2026 measured TTRg11 against **six further β-amyloidogenic sequences all with substantial β-strand propensity** and found much higher activity on the cognate substrate — a harder specificity test than a random panel, because the decoys share the structural propensity the enzyme exploits.

The structural basis is stated. Designs bind **8–13-residue substrates**, "creating an extended recognition interface that can encode substantial sequence and conformational selectivity" (Chen 2026).

And the targets are real. Chen 2026 designed proteases against **TDP-43** (residues 320–350), **serum amyloid A** and **amyloid-β**, with rate enhancements to 10^9-fold; TDPn3 cleaved **full-length human TDP-43** in monomeric and oligomeric forms, though more slowly than the peptide, attributed to liquid–liquid phase separation or precipitation. Butcher 2026 targeted the TTR fibril core strand, for amyloidosis. Both papers then convert proteases into control elements: Chen 2026 built **masked IL-2 mimics and a masked EGFR antagonist** selectively activated by the designed proteases, achieving "bioorthogonal control of cell state" — the enzyme counterpart of the biosensor applications in Vasquez-Torres 2023.

10. Open questions

- **Is the residual gap an ensemble problem?** §5's synthesis — that designs optimise a single structure while catalysis depends on a distribution — is supported by the PLACER correlation in Choi 2026 and, more strongly, by Kim 2026, which filtered on PLACER ensembles prospectively, got its best design ranked first, and found same-backbone designs rejected by that filter to be inactive. What is still missing is the controlled comparison: one campaign, one generator, split between an ensemble filter and a single-structure filter, with both arms ordered. No ingested source reports that.
- **Does better preorganization require better *modelling* or better *objectives*?** Chen 2026 names "incorporation of energetic alongside geometric design criteria" as one of four deficits, but every filter in this chapter is geometric plus confidence. Whether an energy term would help, or whether the available energy functions are too inaccurate at these barriers to use, is unaddressed.
- **Motif transplant or coordination-sphere optimisation?** Elnesr 2026 reports the highest rate enhancement in the chapter from a method that uses no motif, on n = 3 designs; the motif-scaffolding campaigns report lower enhancements on much larger n. No source runs both approaches on one target, so the comparison is unavailable.

- **Why does Riff-Diff reach 45× higher efficiency than RFD2 on the same theozyme?** Braun 2026 and Ahern 2025 both scaffold the RA95.5-8F retroaldolase tetrad; the former reaches 290 M^{-1} s^{-1} from 36 designs, the latter 6.34 M^{-1} s^{-1} from 96. Candidate explanations are all plausible and none is isolated: the enumerative artificial-motif library versus learned atomization, the enforced binding pocket via placeholder helix, the backbone refinement cycles, or differing assay and purification conditions. This is the chapter's single most informative unresolved comparison, because it is the only place two independent pipelines attack one identical theozyme.
- **Is the activity ceiling a property of de novo design or of current pipelines?** §5's diagnosis of a 1–4 order-of-magnitude gap holds across the Baker-lab campaigns, but Braun 2026 reaches ~90% of an extensively evolved enzyme's catalytic efficiency with 99% enantiomeric excess. Whether that generalises beyond well-studied retro-aldol chemistry, where the catalytic array has been evolution-optimised and is being transplanted rather than invented, is open.
- **How much of each hit rate is the filter?** §6 shows a 24-fold swing from one modelling choice (Chen 2026). No campaign in the chapter reports an unfiltered baseline, so the generator's own contribution is unmeasured everywhere.
- **What is the oracle's precision for enzymes?** Lin 2026 put a 12% ceiling on binder evaluation pipelines. Nothing equivalent exists for enzymes, and Elnesr 2026's desD1 result suggests it could be worse for metal sites, where a confident prediction may reflect training-set crystallisation artefacts (Durr 2023 (suppl.)).
- **Does free-atom diffusion beat equivariant representation on enzymes?** Butcher 2026 and Lin 2026 appeared within days, both arguing that representation gates functional design, and reaching opposite conclusions about SE(3)-equivariance. Neither benchmarks against the other on enzyme tasks, though both report on AME-style problems.
- **Why did RFD2-MI's two campaigns differ so much in hit rate?** Choi 2026 got 19% and Chen 2026 got 82% in its better arm, on the same generator and overlapping authors. Candidate explanations — chemistry, filter design, substrate class — are not separated by either paper.
- **Is promiscuity a defect or a finding?** desB hydrolyses two substrate classes separated by 7.5 orders of magnitude in uncatalysed half-life (Elnesr 2026), and Zn45 is broadly cross-reactive (Chen 2026). Whether designed active sites are intrinsically more promiscuous than evolved ones — and why — is unexamined.

11. Flagged references — the history this chapter cannot yet cite

Per the sourcing discipline in `CLAUDE.md`; mirrored in index.

The method the two Nature campaigns are built on:

1. **~~RFdiffusion2~~ — supplied and ingested 2026-10-21** as Ahern 2025 (+ Ahern 2025 (suppl.)), *Nature Methods* 23, 96–105 (2026). §3 was rewritten around it. **RFD2-MI remains unsourced and appears to have no methods paper of its own**: Choi 2026 cites Bauer et al., *De novo design of phospho-tyrosine peptide binders*, bioRxiv 2025.09.29.678898, while Butcher 2026 cites the metalloprotease preprint (now Chen 2026) for the same variant. Recorded as a finding rather than a gap: the variant is described only in passing by its application papers.

The missing prior art the chapter's framing depends on:

2. **Designed serine hydrolases** — Lauko, A., Pellock, S. J., Sumida, K. H., Anishchenko, I., Juergens, D., Ahern, W. et al. *Computational design of serine hydrolases.* **Science 388,**

eadu2454 (2025). Choi 2026 positions itself as the step past these, and the comparison is the basis of the chapter's claim that amide hydrolysis is the real barrier. Citation now identified from Butcher 2026's reference list; the paper itself is not in `sources/`.

3. **The classical Rosetta enzyme-design era** (Kemp eliminases, retro-aldolases, and the directed-evolution campaigns that followed). Elnesr 2026 alludes to it — "early attempts to design biocatalysts resulted [in] low initial catalytic rates, requiring subsequent directed evolution" — and nothing primary supports it. Without this, §1 has no account of what the field tried for twenty years.
4. **PLACER** — Anishchenko, I. et al. *Modeling protein–small molecule conformational ensembles with PLACER*. **PNAS 122, e2427161122 (2025)**. Carries the preorganization analysis in Choi 2026 and the prospective filtering in Kim 2026, so it underwrites both §5 and §6. Citation identified; paper not in `sources/`. **Now the highest-priority acquisition in this chapter.**
5. **MAHOMES-II.** The catalytic-metal annotation database that Metal3D-Cat is trained on, and thus the thing that distinguishes a catalytic zinc from a structural one in Elnesr 2026.
6. **LigandMPNN.** Used by Elnesr 2026 as one of dEVA's two objectives and by Cho 2025 for ligand tasks; the wiki holds Dauparas 2022 but not its ligand-aware successor.
7. **The Mechanism and Catalytic Site Atlas (M-CSA)** — Ribeiro, A. J. M., Holliday, G. L., Furnham, N., Tyzack, J. D., Ferris, K. & Thornton, J. M. *Nucleic Acids Res.* **46**, D618–D623 (2018). The source of the AME benchmark curated in Ahern 2025, and the nearest thing enzyme design has to a shared benchmark. Also **Dunbrack, R. L. Jr., *Rotamer libraries in the 21st century*, Curr. Opin. Struct. Biol. 12, 431–440 (2002)**, behind the rotamer probability loss in Butcher 2026 (suppl.).
8. **Caliby** (ensemble Potts-based sequence design), used to initialise sequences in Elnesr 2026 and appearing as an optional scorer in Ruijgrok 2026.

Supplements and files:

9. Elnesr 2026 is `supp: no` — its Figs. S1–S24 and Tables S1–S10 are referenced throughout but not supplied, so the eight-design characterisation set, the Metal3D-Clean comparison and the kinetics fits cannot be inspected.
10. **The RFD3 preprint/published pair.** Butcher 2025 and Butcher 2026 are both retained, with different titles, framings and content. A deliberate reconciliation pass — what the published version dropped, added or revised — has not been done.

12. Sources

Ingested sources feeding this chapter, roughly chronologically:

- Boyken 2016 — buried polar satisfaction and the many-body problem · *Science* 2016
- Anishchenko 2021 (+ Anishchenko 2021 (suppl.)) — hallucination; anticipates catalytic-site scaffolding · *Nature* 2021
- Dauparas 2022 (+ Dauparas 2022 (suppl.)) — the sequence-design step throughout · *Science* 2022
- Wang 2022 — functional-site scaffolding, the ancestral formulation · *Science* 2022
- Watson 2023 (+ Watson 2023 (suppl.)) — conditional diffusion; enzyme active-site scaffolding by minimal-motif fine-tuning · *Nature* 2023
- Durr 2023 (+ Durr 2023 (suppl.)) — predicting metal ion location; the p(metal | sequence, structure) objective · *Nat. Commun.* 2023
- Krishna 2024 (+ Krishna 2024 (suppl.)) — ligand-aware all-atom design · *Science* 2024
- Abramson 2024 (+ Abramson 2024 (suppl.)) — the oracle used for apo, complex and TSA states · *Nature* 2024
- Chu 2024 (+ Chu 2024 (suppl.)) — all-atom generation; the scaffold source for dEVA ·

PNAS 2024

- Ahern 2025 (+ Ahern 2025 (suppl.)) — RFD2; atomization and unindexed motifs; all 41 AME cases against 16 for prior methods · *Nature Methods* 2025/26
- Kim 2026 (+ Kim 2026 (suppl.)) — DFT theozymes; 53,000 M^{-1} s^{-1} zinc esterases; PLACER validated as a prospective filter · *Nature* 2026
- Braun 2026 (+ Braun 2026 (suppl.)) — **independent lineage**; artificial motif libraries and the placeholder helix; evolved-enzyme-rivalling retro-aldolases at 99% ee; MBHases for a non-biological reaction · *Nature* 2026
- Butcher 2025 (+ Butcher 2025 (suppl.)) — the RFD3 preprint; the AME benchmark · bioRxiv 2025
- Elnesr 2026 — dEVA; coordination-sphere optimisation; 3×10^{13} rate enhancement; the desD1 oracle failure · bioRxiv 2026
- Choi 2026 (+ Choi 2026 (suppl.)) — de novo cysteine proteases; 13/69; preorganization diagnosed by PLACER · *Nature* 2026
- Chen 2026 (+ Chen 2026 (suppl.)) — de novo zinc proteases; the catalytic-water filtering result; disease targets and caged biologics · *Nature* 2026
- Butcher 2026 (+ Butcher 2026 (suppl.)) — free-atom diffusion; zero-shot phosphotriesterases and TTR proteases · *Science* 2026
- Lin 2026 — the competing representational thesis; enzyme-design benchmark · bioRxiv 2026

Related pages: concepts Catalytic site design, Metalloprotein design, Motif scaffolding (functional-site design), Diffusion generative design, Sequence design (fixed-backbone) · topics De novo binder design, Symmetric oligomer design · tools RFdiffusion, RFdiffusion2, RFdiffusion3 (RFD3), ProteinMPNN, AlphaFold, Protpardelle, Genie 3.

Assemblies, Scaffolds and Validation

Symmetric oligomer design

A chapter on the design of protein assemblies in which several copies of one designed chain are arranged under a **point-group symmetry** — cyclic (C*n*) rings, dihedral (D*n*) stacks, and the cubic groups (tetrahedral, octahedral, icosahedral). The interesting thing about symmetric homo-oligomers, from a design standpoint, is that they let one small design decision — a single protomer and a single interface — determine a much larger object. The interesting thing historically is that this problem has been solved three separate times, by three genuinely different mechanisms, in roughly thirty years: symmetry as **geometry** (imposed exactly, by construction), symmetry as an **objective** (rewarded, by optimization against a structure predictor), and symmetry as a **constraint on a sampling process** (enforced at every step of denoising). A fourth, hybrid mechanism has since appeared — symmetry **tied** in the sequence representation while the interface is rewarded — and is treated in §7. That progression, and what each step cost and bought, is the argument of this chapter.

Contents

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1. Why symmetry is a design primitive

Symmetric assemblies are not a niche of the natural protein world. Cyclic protein oligomers alone constitute nearly **30% of all deposited PDB structures**, and they participate in almost every biological process (Wicky 2022). The reason usually offered is energetic: symmetric interfaces can be favourable under evolutionary pressure, and natural proteins accordingly oligomerize into linear, cyclic, dihedral, cubic, planar and spatial symmetries (Luo 2016).

For a designer the appeal is combinatorial. If *n* copies of one chain are arranged under a point group, then designing one protomer and one protomer–protomer interface specifies the entire assembly; the relative orientation of two adjacent subunits together with the comple-

mentarity of their contact interface determines the multiplicity of the whole complex (Luo 2016). A designed ring is therefore cheap in design effort relative to its size — the property that makes symmetric oligomers attractive as vaccine platforms, delivery vehicles and catalysts (Watson 2023), as building blocks for nanocage assemblies and as substrates for small-molecule binding and catalysis (Wicky 2022).

The costs are equally structural. An error in the single designed interface is repeated *n* times, so failure modes are amplified rather than averaged out. And the geometry is unforgiving: for fusion-based approaches, a specific arrangement between the symmetry axes of two different oligomeric proteins is required for the design to be accurate at all (Luo 2016). Symmetry makes the design problem smaller and the tolerance for error smaller at the same time.

2. The pre-learning era — borrow, parameterize, or dock

Before deep learning there were three ways to obtain a symmetric assembly, and it is worth being precise about them because the later methods are best understood as removing specific constraints that each imposed.

Borrow the symmetry. The oldest computational route takes symmetry from nature rather than creating it. The founding strategy is **genetic oligomeric fusion**, introduced by Padilla, Colovos and Yeates and described first-hand in Yeates 2002: fuse two distinct, naturally oligomeric proteins into a single chain, so that the fusion is driven to self-assemble through the natural oligomeric interfaces of its components. What results depends entirely on the **geometric relationship between the components once fused** — the disposition of the symmetry axes each one carries. A dimer joined to a trimer through an **α-helical linker** gives different geometries as the linker length varies, and different symmetry-axis geometries yield different regular complexes: the same strategy produces a two-dimensional hexagonal layer or a tetrahedrally symmetric cage. Demonstrated experimentally: a **tetrahedral cage of twelve 49 kDa subunits**, a fusion of trimeric bromoperoxidase with dimeric influenza matrix protein M1, confirmed by molecular model and transmission electron micrograph, alongside a self-assembling filament (Yeates 2002).

The approach kept going for a decade after that. A later refinement gave a homogeneous **12-subunit, 16-nm cage with a 5-nm central opening**, resolved crystallographically; it was extended to porous cubic cages — a KDPGal–FkpA fusion with **225 Å outer and 132 Å inner diameter** under octahedral symmetry — and to symmetry-matching fusions producing 1D filaments, 2D layers and 3D crystals (Luo 2016).

Fusion is powerful precisely because it inherits well-characterized, stable natural interfaces. Its limitation is the same fact: the designer chooses among the symmetries nature already provides, and the conformational flexibility of the borrowed building blocks can divert the assembly. The KDPGal–FkpA cube is the clean illustration — native mass spectrometry found **12-subunit tetrahedral and 18-subunit prism assemblies alongside the intended 24-subunit cube** (Luo 2016).

Parameterize the backbone. The second route generates the symmetric backbone from a closed-form geometric description — see Parametric design for the idea in general. It is worth being clear why it works at all, because the reason constrains where it applies: reducing a 3D fold to parametric equations that still capture its salient features is possible only for a **subset of proteins, namely repeat proteins**, of which the α-helical coiled coil — an estimated **3–5% of all known protein-encoding regions of DNA** — is the exemplar (Wood 2018). A shape built by applying one rigid-body transform repeatedly is describable by the few parameters of that transform; see Repeat protein design.

The conceptual prerequisite is the demonstration that oligomeric state is *encodable*: Harbury 1993 systematically mutated the hydrophobic **a** and **d** core positions of the dimeric GCN4-p1

leucine zipper in concert and obtained peptides forming **two-, three- and four-stranded** structures, showing that the geometry of buried apolar side chains — not merely the hydrophobic/polar pattern — selects the number of chains.

Two decades later this became a generative procedure. Huang 2014 built backbones directly from the **Crick coiled-coil equations**, sampling the supercoil twist ω0, helical twist ω1, supercoil radius R0, per-helix phases and axial offsets on a grid, running Rosetta sequence design at each grid point, and filtering with Rosetta *ab initio* prediction before synthesis. The resulting designs — including a **pentameric parallel left-handed five-helix bundle** — were extremely stable (**extrapolated ΔG_fold > 60 kcal/mol**) with crystal structures showing nearly identical core packing to the models. The crucial mechanical detail is in the supplement: helices are arrayed symmetrically about the z axis, so for an n-helix bundle the superhelical phase takes values $\{0, 2\pi/n, \ldots 2(n-1)\pi/n\}$ (Huang 2014 (suppl.)). **Cyclic symmetry is exact by construction** — a property of the coordinate generator, never optimized for and never checked afterwards.

The parametric family is broader than closed-form generators, though. Brunette 2015 reaches the same place by a different route: Rosetta fragment assembly over a helix–loop–helix–loop repeat unit, with **every Monte Carlo move made at the equivalent position in each repeat** and every sequence substitution applied identically across units, so symmetry constrains the search rather than emerging from it. Because the transform relating adjacent repeats is identical by construction, the designs are globally helical and collapse onto three coupled parameters — radius, twist and axial rise. From 5,776 helix- and loop-length combinations, 761 designs passed in silico filters and 83 were tested: **53 monomeric and stable at 95 °C**, 43 SAXS-consistent, and **15 crystal structures at 0.7–2.5 Å RMSD** (Brunette 2015). That library matters for this chapter because the designed helical repeat proteins it produced became the standard component set that later symmetric assemblies were built from.

Published alongside Huang 2014, Thomson 2014 applied a related geometric logic to the sparse region above four helices. Classical coiled coils of two to four helices account for **more than 98% of known coiled-coil structures**; α-helical barrels, in which each helix contacts two neighbours through *two* independent hydrophobic seams, are the remainder, and most of their possible structures had never been observed. Enumerating the ways two seams can be accommodated in a heptad repeat, scoring with a knowledge-based function and building atomistic models produced discrete water-soluble **C5, C6 and C7** assemblies whose crystal structures matched the models — CC-Pent being the first crystal structure of a designed pentameric barrel and CC-Hept the first heptamer. That paper also introduced a theme worth carrying forward: the **channels are chemically defined and their diameters track oligomer state**, so a ring's lumen is itself a designable feature.

Dock and design. The third route specifies a protomer in advance, rigid-body docks copies of it into a higher-order symmetric arrangement, then optimizes the interface to make the assembled state low-energy. This was the mainstream approach, and Wicky 2022 characterizes the state of the art it inherited in exactly these terms: current approaches required the structure of the monomer to be specified in advance and, **with the exception of parametrically designed helical bundles**, involved rigid-body docking of previously characterized monomers followed by interface optimization.

That single sentence names the binding constraint of the whole era. Because the protomer had to be specified first, the accessible space of oligomers was bounded by the space of already-solved monomers — which in particular excluded assemblies with **more intertwined chains**, where protomer and interface cannot be separated (Wicky 2022).

3. Specificity as a second axis

Getting the geometry right is not the same as getting the *addressing* right. In DNA, specificity comes from modular hydrogen bonds in the core of the double helix; in proteins it comes largely from buried hydrophobic packing complemented by irregular peripheral polar interactions (Boyken 2016). A set of designed coiled-coil oligomers with hydrophobic cores may each fold correctly and still cross-assemble promiscuously.

Boyken 2016 added the missing axis with **HBNet**. Buried polar interactions are hard to design for two compounding reasons: the partially covalent hydrogen bond restricts donors and acceptors to narrow ranges of orientation and distance, and nearly all buried polar atoms must find a partner or incur a considerable desolvation penalty. Conventional design algorithms make this worse by expressing system energy as a sum of *pairwise* residue interactions, whereas a hydrogen-bond network is inherently many-body. HBNet instead precomputes hydrogen bonds and traverses the resulting graph to enumerate all networks of **three or more residues**, rejecting any network leaving a buried donor or acceptor unsatisfied. Hosting those networks in parametrically generatable coiled coils — which have repeating geometric cross sections, so a network can be reused at equivalent positions around the ring — produced **dimers, trimers and tetramers** built as two concentric rings of helices, in triangular, square and supercoiled topologies not previously seen. **Ten crystal structures** confirmed the designs, with left-handed trimers matching at sub-ångström RMSD and dimer networks as close as **0.39 Å**, and the networks conferred **interaction specificity in vivo**.

This is the point at which symmetric-oligomer design stops being “can we build a ring” and becomes “can we build a set of rings that do not talk to each other” — the prerequisite for anything multi-component.

4. The ceiling of the hand-built approach

Courbet 2022 is the best available measure of how far the parametric-plus-Rosetta toolchain could be pushed, and it is instructive that the paper is not really about making symmetric oligomers at all — it is about making symmetric oligomers *move*.

The premise, as Courbet 2022 states it, is that designed nanomaterials to that point — the paper names icosahedral nanocages and two-dimensional arrays — are assemblies whose components have **fixed orientations** relative to one another; the goal was a kinematic pair with internal degrees of freedom. The design proceeds hierarchically: rigid components first, directed self-assembly into hetero-oligomeric complexes second, shaping of the multistate energy landscape along the mechanical degrees of freedom third. The rigid components are largely **designed helical repeat proteins** rigidly fused to cyclic homo-oligomers (Courbet 2022 (suppl.)) — that is, the output of Brunette 2015 used as a parts bin, which is the clearest illustration in the chapter of why Repeat protein design is upstream of point-group assembly design. The component library is substantial — **C3, C4, C5 and C8 cyclic rotors** and **D2, D3, D4, D5 and D8 dihedral axles** — confirmed by cryo-EM at 6.0 Å (C3 rotor, 28 Å internal diameter), 7.9 Å (C4 rotor, 57 Å internal diameter), 4.2 Å (D3 axle) and 7.4 Å (D8 axle) (Courbet 2022). The payoff is the **symmetry-mismatched** assemblies: **D3–C3** and **D3–C5** axle-rotor systems that assemble in vitro and in vivo and, by cryo-EM, populate conformationally variable relative orientations reflecting the symmetry of the coupled components and the designed interface energy landscape — the D3–C3 system largely constrained, the mismatched D3–C5 rotor free to rotate and translate along the main symmetry axis (Courbet 2022).

The result is a genuine conceptual advance: **symmetry as a mechanical variable**, where matching symmetries rigidify and mismatching them leaves a shallow periodic landscape along a rotational coordinate. But the supplement shows what it cost. Reaching a D3 axle required sampling **two different starting superhelical radii per helix (6 Å and 12 Å)**

over a ~5 Å range in 0.5 Å bins, with helical phase swept 0°–90° in 10° steps, *z*-offsets from −1.51 Å to +1.51 Å in 0.1 Å steps, and supercoil phases fixed by hand at 0° for D3 and 30° for D2 — followed by HBNet interface design, RosettaRemodel helical fusion to cyclic partners, a second design round across the fusion, and SymDofMover for the D8 case (Courbet 2022 (suppl.)). Every symmetry group is a separate engineering project.

5. Hallucination — symmetry as an objective (and, first, as an accident)

5.1 The method, and the oligomers nobody asked for

Hallucination as a design method was established by Anishchenko 2021, and the mechanism is worth stating in its original, symmetry-free form, because everything the symmetric version does is a modification of this. Start from a **random amino acid sequence**, whose predicted inter-residue distance and orientation maps are diffuse and featureless. Run **Monte Carlo simulated annealing in sequence space**, at each step substituting a random residue at a random position, re-predicting with **trRosetta**, and accepting by the Metropolis criterion on the change in **Kullback–Leibler divergence** between the predicted distributions and generic length- and separation-dependent background distributions. The objective is pure **contrast** — make the network's prediction sharp — with no target structure specified anywhere; the authors' stated analogy is DeepDream, inverting a network trained to recognize patterns so that it generates them (Anishchenko 2021).

It worked, and the numbers set the baseline against which the symmetric variants should be read. After around **20,000 Monte Carlo steps** the maps were as featured as those for natural sequences; **2,000** proteins were generated; outputs were novel (median BLAST E-value **0.17** to UniRef90, average TM-score **0.78** to the closest PDB match) and spanned all-α, all-β and mixed α/β with **95 sub-folds**. Of **129** designs expressed, **27** gave SEC peaks for monomeric or small oligomeric species with CD spectra consistent with the targets, **21** were thermostable above 70 °C, and three structures were solved — NMR at **1.82 Å** backbone RMSD (design 0515), X-ray at **2.53 Å** (0217) and X-ray at **3.68 Å** overall but **1.32 Å** and **2.17 Å** across the N- and C-terminal halves (0738_mod) (Anishchenko 2021). A separate, telling detail: the closest PDB structure to 0738_mod is **Top7**, the canonical Rosetta de novo design (Anishchenko 2021 (suppl.)).

And here is why this paper belongs in a chapter about symmetric oligomers. Every sequence was modelled as a **monomer**, and yet **many of the hallucinated proteins formed oligomers in solution**. Design 0217 crystallized as a **dimer**, corroborated by SEC-MALS and NMR rotational correlation time; design 0417 self-associated transiently, which ultimately prevented NMR structure determination (Anishchenko 2021). The diagnosis is specific and, for this chapter, important: **trRosetta was trained on native structures including many homo- and hetero-oligomers, so it never learned to disfavour surface hydrophobic residues to the degree required for soluble monomers** — the network had absorbed the sequence signatures of native protein–protein interfaces, and β-sandwich designs suffered worst, plausibly reflecting immunoglobulin-fold interdomain contacts in the training data (Anishchenko 2021). The rescue was manual and effective: substituting exposed hydrophobics with polar residues converted several oligomers to monomers, demonstrated directly by paired before/after SEC traces (Anishchenko 2021 (suppl.)).

So the first thing a predictor-inverted generator did, unprompted, was **make homo-oligomers** — because interface-forming sequence patterns are what it had learned from the PDB. The wiki's reading is that this reframes what Wicky 2022 achieved a year later. The hard part was never eliciting an oligomer from the network; it was **specifying which one**, converting an uncontrolled propensity into an addressable input.

5.2 Wicky — making valency an input

Wicky 2022 removes the constraint that the protomer be known in advance, and it does so by changing what kind of object symmetry is. Instead of building symmetry into the coordinates, it **rewards** symmetry in a loss function — adding a symmetry term to the confidence objective of Anishchenko 2021 and swapping trRosetta for AlphaFold2.

The method takes only two inputs: a chain length L and an oligomer valency N. Starting from a random sequence, a Monte Carlo search in sequence space feeds N copies of the current sequence into **AlphaFold2** (Jumper 2021) and combines the network's confidence metrics — pLDDT and pTM — with a measure of cyclic symmetry, namely the **standard deviation of the distances between the centres of mass of adjacent protomers** in the predicted structure (Wicky 2022). Trajectories for monomers through heptamers with 65–130 residue chains typically coalesce within a few hundred steps, at a cost of roughly **1 to 7 CPU-days** per design (Wicky 2022). The outputs — called HALs — span all-α, mixed α/β and all-β topologies and differ from the cyclic de novo designs already in the PDB; median TM-scores to the closest PDB relative are **0.67 for protomers and 0.57 for oligomers**, with **29% of structures below the 0.5 fold-assignment cutoff** in CATH/SCOP, indicating substantial generalization beyond the training set (Wicky 2022).

The failure that makes this paper worth reading closely. 150 designs with pLDDT > 0.7 and pTM > 0.7 were selected for experimental testing, and virtually none showed appreciable soluble expression in *E. coli* — median soluble yield **9 mg/L** — with none of the marginally soluble ones showing both the expected SEC oligomerization state and a consistent circular-dichroism profile (Wicky 2022). The authors' diagnosis is **overfitting to the oracle**: hundreds of MCMC steps against AF2 generate adversarial sequences that the network confidently predicts as structured but which have aberrant biophysical properties, analogous to activation maximization producing unrealistic outputs in image classifiers (Wicky 2022).

This is not a new pathology so much as a much more severe instance of a known one. Anishchenko 2021 had already reported that hallucinated sequences carry **exposed surface hydrophobics** inherited from the network's training on native oligomeric structures, and had fixed it by hand (Anishchenko 2021 (suppl.)). The wiki's reading is that adding a symmetry term makes the same failure much worse rather than introducing a different one: a cyclic-symmetry objective *rewards* inter-protomer contact, so the optimizer is now being paid to produce exactly the sticky surfaces that hallucination was already prone to. What changed between the two papers is not the diagnosis but the remedy — manual polar substitution on a handful of designs in 2021 (Anishchenko 2021 (suppl.)), versus wholesale sequence replacement by a dedicated inverse-folding network in 2022 (Wicky 2022).

The fix is instructive. Discard the hallucinated *sequences*, keep the hallucinated *backbones*, and redesign the sequences with Dauparas 2022 — 24 to 48 sequences per backbone, validated with AF2. Because dozens rather than hundreds of evaluations are performed, overfitting is far less likely; independently, an updated RoseTTAFold confidently predicted almost all ProteinMPNN sequences while failing on most of the original hallucinated ones. The redesigned set behaved completely differently: of 96 tested, **71 (74%) expressed at high levels** (median yield **247 mg/L**, a ~27-fold increase), **50 (52%) had SEC retention volumes consistent with the target oligomer** (30 of those monodisperse), **at least 21 (22%) had the correct oligomeric state by SEC-MALS**, and **67 of 71 (94%)** had circular-dichroism secondary-structure content consistent with the design. **54 of 71** retained secondary structure to 95 °C.

Structural validation was strong where it was attempted. Crystallization was tried for 19 designs and succeeded for **seven** (three C2, two C3, two C4), all with the correct oligomerization state and a **median Cα RMSD of 0.6 Å** at 1.8–3.4 Å resolution, with side-chain rotamers also matching (Wicky 2022). Six large designs with C6–C42 internal repeat symmetry gave negative-stain 3D reconstructions, and three were solved by cryo-EM at 4.38, 6.51 and 6.32

Å: a C15 homopentamer, a C18 homohexamer and a **C33 homotrimer**, **87, 99 and 100 Å in diameter** and 40–50 Å high, each with a continuous parallel β sheet lining the lumen and outer helices enforcing ring closure, with backbone RMSDs to the design models of **0.81, 1.69 and 2.30 Å** and TM-scores of 0.41–0.49 to anything in the PDB (Wicky 2022). For scale: the highest cyclic symmetry recorded in the PDB for a naturally occurring protein is **C39**, the vault proteins (Wicky 2022). Aggregated over the whole campaign, **27 of 30** designs evaluated by SEC-MALS, nsEM, cryo-EM or crystallography had the intended oligomeric state (Wicky 2022).

What it could not do. Hallucination-based cyclic design **fails for higher-order dihedral, tetrahedral, octahedral and icosahedral symmetries**, probably in part because such structures are far less represented in the PDB and therefore in the oracle's learned prior (Watson 2023). The symmetry objective — a standard deviation over *adjacent* protomer centre-of-mass distances — is intrinsically a statement about a ring. It has no natural generalization to a point group with multiple distinct rotation axes.

That limitation, plus the cost of random search, is what handed the field to diffusion in 2023. It is worth flagging now that the story does not end there: hallucination returns in §10 with gradients instead of MCMC, and beats diffusion on large designs (Frank 2024).

6. RFdiffusion — symmetry as a constraint on sampling

Watson 2023 is the pivot of this chapter, and symmetric oligomer design is one of the capabilities it was built to unify — alongside unconditional and topology-constrained monomer design, motif and enzyme active-site scaffolding, and binder design. The model fine-tunes the RoseTTAFold structure-prediction network on a denoising task: starting from noised residue frames, it iteratively denoises to a coherent backbone, with conditioning inputs steering generation. For assemblies, the conditioning input is a **point-group symmetry operator**.

The mechanism is worth stating precisely, because it is the third distinct answer to "how do you make a design symmetric." Given a point group and a monomer chain length, random starting residue frames are generated for a **single** subunit as in the unconditional case, and $n-1$ copies are placed according to the point group. Because RFdiffusion inherits **equivariance** with respect to rotation and chain relabelling from RoseTTAFold, symmetry is *largely maintained* through the denoising predictions on its own; the model nonetheless **explicitly resymmetrizes at each step**, which changes the structures only slightly. For octahedral and icosahedral architectures only the smallest generating subset of monomers is modelled explicitly — for icosahedra, the subunits at the five-, three- and twofold axes — to reduce compute and memory. Design diversity is tuned by **scaling the noise added at inference**, and an **auxiliary inter- and intra-chain contact potential** guides symmetric sampling; external potentials are the same machinery used for ligand and metal pocket design (Watson 2023 (suppl.)).

The striking result is that **RFdiffusion was not trained on symmetric inputs** and nonetheless achieves high in silico success rates on symmetric oligomers, particularly with the contact potential engaged (Watson 2023). Symmetry here is neither built into a coordinate generator nor rewarded in a loss — it is **projected onto the state at every denoising step**, and the model's equivariance is what makes that projection nearly free.

Coverage. Cyclic designs included topologies not seen in the PDB — two-layer β barrels at **C10**, and mixed α/β topologies at **C8** whose closest TM-aligns in the PDB are **0.47 and 0.43** (Watson 2023). Of **608 designs** selected for experimental characterization, **at least 87** had SEC oligomerization states within the 95% confidence interval of the calibration (**126 within the 99% interval**), and negative-stain EM of a **C3 design with 350-residue subunits (1,050 residues total)** matched the design both per-subunit and in overall architecture,

including a distinctive pinwheel shape; C5 and C6 designs of more than 750 residues were also consistent with their models (Watson 2023). A separate family of α/β barrel oligomers effectively extends the TIM barrel: **HE0626**, a C6 with 18 strands and 18 helices, and **HE0675**, a C8 with an inner ring of 16 strands and an outer ring of 16 helices in a 1:1 helix:strand pattern locally very similar to a TIM barrel, both confirmed by nsEM 3D reconstruction (Watson 2023). The framing the authors give this is worth keeping: natural evolution extensively explored variations *within* the eight-strand TIM barrel, while a generative model more readily explores **global changes in barrel curvature** (Watson 2023).

Beyond cyclic. This is where RFdiffusion separates from hallucination. SEC indicated that **38 D2, seven D3 and three D4** designs had the expected molecular weights (four, six and eight chains respectively); D3 and D4 2D class averages and 3D reconstructions matched the design topologies, and cryo-EM of the D4 design **HE0537** recapitulated the ~45° offset between tetrameric subunits (Watson 2023). A **12-chain tetrahedron** (HE0964) gave consistent 2D class averages, and of **48 icosahedra** selected for validation, **HE0902** — a **15 nm**, highly porous assembly — formed homogeneous particles whose 2D class averages and 3D reconstruction closely match the design model, with triangular hubs arrayed around empty C5 axes and outward-facing N and C termini suited to antigen display (Watson 2023).

Two things should be read together here. The **coverage** is unprecedented: C, D, T and I in one framework, from noise, with no per-symmetry engineering of the kind Courbet 2022 (suppl.) documents. The **yield** is sobering: 87 of 608 by SEC, one validated icosahedron out of 48 attempts. Generality and success rate are different quantities, and this chapter's reading is that RFdiffusion's contribution to symmetric design is overwhelmingly the former.

7. Four ways to impose symmetry, compared

The mechanisms are easy to conflate because they produce superficially similar objects. Stated side by side:

Exact by construction. In parametric design the symmetry is a property of the coordinate generator — the superhelical phases are *set* to {0, 2π/n, …} and no other value is representable (Huang 2014 (suppl.)). Symmetry error is identically zero. The price is expressivity: only folds the parameterization can write down, in practice helical bundles and barrels (Huang 2014, Thomson 2014).

Rewarded. In hallucination the symmetry is a **term in a loss** — the standard deviation of adjacent-protomer centre-of-mass distances — added to the pure confidence objective that Anishchenko 2021 established and optimized jointly with it (Wicky 2022). Expressivity is enormous: protomer and assembly are discovered together, so intertwined topologies inaccessible to dock-and-design become reachable, and the outputs generalize well past the training set. Three prices are paid. The optimization is expensive (1–7 CPU-days per design). Optimizing *against* a predictor for hundreds or thousands of steps invites adversarial solutions — the 150-design expression failure of Wicky 2022, foreshadowed by the surface-hydrophobic problem already documented in Anishchenko 2021. And the objective is ring-shaped: a statement about *adjacent* protomers does not extend to point groups with several distinct axes, which is why the approach fails for dihedral and cubic symmetries (Watson 2023).

Projected. In diffusion the symmetry is applied to the **state**, at initialization and again at every denoising step, and the model's equivariance means the denoiser's own predictions nearly respect it already (Watson 2023, Watson 2023 (suppl.)). Symmetry error is driven to near zero without ever entering the objective. Expressivity extends to any point group, including cubic ones, and the asymmetric-unit trick keeps icosahedral design tractable. The price is that success is now dominated by whether the *interface* the model invents is physically real — hence the reliance on an auxiliary contact potential, and the 87-of-608 SEC yield.

Tied — a hybrid, added by the gradient era. Frank 2024 splits the two things the other mechanisms bundle together. Homo-oligomers are built by **copying the input logits** across protomers with a residue-index shift of 50 at each breakpoint, then adding **interface pAE and contact losses** (Frank 2024 (suppl.)). So *sequence* symmetry is exact by construction — the protomers are literally the same optimization variable — while *interface geometry* is rewarded through the loss. That is a genuinely different point in the design space from the three above, and it is closest in spirit to the tied-position decoding of Dauparas 2022 (suppl.), which imposes sequence identity around a ring without saying anything about the structure.

The through-line the wiki draws from this: **as symmetry moved from the coordinates into the sampling process, the binding constraint moved from geometry to interface quality.** The parametric era's hard problem was writing down a backbone; the diffusion era's hard problem is that writing down backbones is easy and most of them do not assemble.

8. Sequence design is not a detail

It would be easy to treat sequence design as a downstream step. The symmetric-oligomer literature is the strongest evidence in this wiki that it is not.

Dauparas 2022 predicts sequence autoregressively from backbone geometry alone and explicitly handles cyclic oligomers, nanoparticles and interfaces, including **tied/symmetric positions**. The mechanism enabling tying is the **random-order autoregressive decoding** scheme — a detail stated in the main text but derived only in the supplement (Dauparas 2022 (suppl.)) — and the same tied-decoding machinery underlies symmetric design in Watson 2023 (suppl.). Tying matters for a specific reason: under C*n* symmetry, the *n* copies of a residue position are the same residue by definition, so sequence design must either respect that constraint or be corrected afterwards.

The empirical case is the Wicky rescue described above. The same backbones, sequenced two different ways, gave a median soluble yield of **9 mg/L** versus **247 mg/L**, and went from zero validated designs to 27 of 30 with the intended oligomeric state (Wicky 2022). The generator was never the problem; the sequences were.

And the diagnosis has since sharpened. Frank 2024 ran the controlled version of this experiment: taking the sequences its own optimizer produced and using them directly instead of handing the backbone to ProteinMPNN. Argmax discretization gave structures deviating strongly from target; simulated annealing from the relaxed sequences improved *in silico* success but **still gave poor experimental success rates**. The stated reason is specific and mechanistic — **AF2 predictions are highly tolerant of mutations, which limits their ability to distinguish valid from adversarial sequences**. That is a better account than "overfitting": the oracle is not being fooled so much as it is insensitive in the direction that matters, so a search which only ever asks the oracle has no gradient toward biophysical realism. Including a component built specifically to produce valid sequences for a given backbone is what fixes it (Frank 2024). Three papers spanning three years and two optimization schemes reach the same conclusion. That result also justifies the pipeline shape adopted everywhere afterward — **generate backbone → design sequence with ProteinMPNN → filter with AlphaFold2** (Watson 2023) — and it is a large part of why Dauparas 2022 shows up as a dependency of essentially every later symmetric-design result.

9. What actually gets validated

Symmetric assemblies are unusually well served by orthogonal experimental methods, and the literature has settled into a fairly consistent ladder. It is worth being explicit about it, because "success rate" means different things at different rungs.

In silico, success is defined by AlphaFold2 self-consistency: **pAE < 5**, **pLDDT > 90** (relaxed to > 80 for motif cases) and backbone RMSD to the design under ~1.0–1.5 Å; these thresholds, not wet-lab data, define the success rates quoted throughout Watson 2023 (Watson 2023 (suppl.)). Hallucination used a looser in-house bar — pLDDT > 0.75 and RMSD to the original backbone < 1.5 Å for the ProteinMPNN redesigns (Wicky 2022).

Experimentally the rungs are: **soluble expression**, then **SEC** retention volume against a calibration curve (cheap, high-throughput, and the metric behind both the 50/96 of Wicky 2022 and the 87/608 of Watson 2023), then **SEC-MALS** for an absolute mass rather than a hydrodynamic proxy, then **circular dichroism** for secondary-structure content and thermal stability, then **negative-stain EM** for shape and 2D class averages, then **cryo-EM** for a 3D reconstruction, and finally **crystallography** for atomic accuracy. The gap between the rungs is real: in Wicky 2022, 52% passed SEC but only ~22% passed SEC-MALS on the same set.

Two observations follow. First, symmetric assemblies are the one design class where **EM is routinely decisive**, because the assemblies are large enough — the giant HAL rings exceeded 100 kDa, which is what made them well suited to EM characterization (Wicky 2022) — and because a ring's symmetry is directly legible in a 2D class average. Second, the crystallization rate itself carries information: seven of 19 attempted HALs formed diffracting crystals, which the authors note is considerably higher than is typical for Rosetta de novo designs (Wicky 2022).

10. After RFdiffusion

The symmetric-design capability has propagated rather than plateaued. Ingraham 2023, developed independently and contemporaneously at Generate Biomedicines, frames design as **Bayesian inference under external constraints**, with **symmetry** among the first-class conditioners alongside substructure, shape, semantic class and natural-language prompts; conditioners enter as likelihood or constraint terms on the diffusion posterior, giving one uniform mechanism for imposing symmetry (Ingraham 2023 (suppl.)). That two lineages arrived at conditional diffusion with symmetry conditioning separately is, on this wiki's reading, evidence that the formulation is natural rather than incidental — the same argument made at the level of the whole field in Introduction — the field at a glance.

Butcher 2025 carries the capability into the all-atom, token-plus-atom regime, with a **dedicated inference loop for symmetric design** — token initializers plus symmetry handling — distinct from the main inference loop, explicitly carrying forward the symmetry capability of Watson 2023 (suppl.) (Butcher 2025 (suppl.)). Given that RFdiffusion3 matches or beats its predecessors at roughly **one-tenth the compute** (Butcher 2025), and that symmetric-oligomer success rates are dominated by yield rather than coverage, cheaper sampling is a directly relevant improvement: the 87-of-608 problem is partly a compute-budget problem.

And the hallucination line came back. Frank 2024 revisits the approach that diffusion appeared to supersede, fixing the two things that had held it back: slow random search, and the difficulty of gradient descent in a *discrete* sequence space. The fix is to let the sequence be **continuous during optimization** — backpropagate the loss through AlphaFold2 to the input sequence and follow the gradient, accepting that the result is a logit-like or PSSM representation in which every position is a weighted superposition of all 20 amino acids, and declining to force it back to a real sequence at each step. Only the converged *backbone* is handed to ProteinMPNN. This is exactly the improvement Anishchenko 2021 proposed in its closing paragraphs, delivered by a group including one of its authors.

The result matters for this chapter because of **scale**. RSO produced viable designs up to **1000 amino acids** with a median TM score of **0.89** at that length, while Watson 2023 "faced challenges with producing viable backbones beyond 600 amino acids" in the same head-to-

head, at comparable GPU cost per successful design (Frank 2024). Symmetric assemblies are exactly where chain length runs large — the giant rings of Wicky 2022 reached 1550 residues, and RFdiffusion's own symmetric successes include subunits over 750 residues (Watson 2023) — so a generator that holds up at 1000 residues is not a peripheral improvement. Validation was substantial: over 100 proteins characterized, three crystal structures and two cryo-EM maps of single chains up to 1000 residues, one at **1.91 Å** Cα-RMSD (Frank 2024).

For multi-chain work it handles homo-oligomers and heterodimers through AF2's residue-index-gap trick, with designed heterodimers binding at **K_d 560 nM (MST) / 480 nM (fluorescence polarization)** and a second set spanning **<4.0 to 790 nM** (Frank 2024). Point-group assemblies beyond simple copies are not demonstrated, so on *coverage* of the symmetry groups this does not displace Watson 2023 — the comparison the chapter draws is on scale and on sequence-space mechanism, not on breadth of point groups.

A third sighting of the same bias. The heterodimer campaign included a set designed with the **in silico homo-oligomer filter deliberately omitted**, and those proteins **independently formed homo-oligomers** at elution concentrations, converting to the intended heterodimer only on mixing (Frank 2024). Set beside the unintended dimers of Anishchenko 2021 and the sticky adversarial surfaces of Wicky 2022, this is the third independent observation in the chapter that **predictor-derived generators drift toward homo-oligomers unless actively pushed away from them** — and the first where the push is an explicit, quantified negative-design filter (normalized homodimer I_PAE above 0.8, per Frank 2024 (suppl.)) rather than a manual fix. The wiki's reading: self-association is the default output of a model trained on the PDB, and every symmetric-design method in this chapter is, in one way or another, a way of *aiming* a propensity that is already there.

One further thread connects symmetry to function. Watson 2023 scaffolds **symmetric functional motifs** — C3-symmetric trimers rigidly holding three binding domains, and C4-symmetric histidine square-planar metal sites — noting that the symmetry of an assembly can be matched to the symmetry of a coordination geometry or a viral glycoprotein. Symmetric oligomer design and Motif scaffolding (functional-site design) are not separate problems at the level of the model; they are two conditioning inputs to the same generator.

11. Open questions

- **Why is the yield what it is?** No ingested source diagnoses *why* only 87 of 608 symmetric RFdiffusion designs pass SEC (Watson 2023), or whether failures are dominated by interface quality, kinetic trapping in the wrong oligomeric state, or solubility. The chapter's two mechanistic failure accounts — surface hydrophobics inherited from training on native oligomers (Anishchenko 2021) and adversarial overfitting to the oracle (Wicky 2022) — both concern *hallucination*, and neither transfers cleanly to a diffusion model that never optimizes against a predictor. Whether diffusion's symmetric failures share a cause with hallucination's is open.
- **Did the interface prior survive the transition?** Anishchenko 2021 attributes unintended oligomerization to the network having learned the sequence signatures of native protein–protein interfaces from the PDB. RFdiffusion is fine-tuned from a structure predictor trained on the same kind of data (Watson 2023), so the same prior is presumably present — but whether it *helps* symmetric design (a useful bias toward realistic interfaces) or *hurts* it (a bias toward native-like rather than designed interfaces) is not addressed by any ingested source.
- **Does success rate decay with symmetry order, and how?** The numbers are suggestive — 38 D2 versus seven D3 versus three D4 (Watson 2023) — but the denominators per symmetry group are not stated in the main text, so no rate can be computed. Watson 2023 (suppl.) indicates per-symmetry SEC success rates are characterized in the SI;

extracting them would settle this.

- **Is exact resymmetrization necessary?** Watson 2023 reports that explicit resymmetrization at each step changes structures only slightly, since equivariance largely preserves symmetry anyway. Whether the residual asymmetry is noise or a useful relaxation is not addressed.
- **Can hallucination's expressivity be recovered inside diffusion?** Hallucination discovered intertwined protomer/assembly topologies precisely because it never separated them (Wicky 2022). Whether symmetric diffusion samples that same region, or has quietly reintroduced a bias toward separable protomers, is untested in the ingested sources.
- **Two-component and heteromeric assemblies.** Courbet 2022 achieves hetero-oligomeric coupling by design, and the specificity machinery of Boyken 2016 exists to make orthogonal components possible, but the wiki has no ingested source on designed **two-component** symmetric nanomaterials — see the flagged references below.
- **Benchmarks.** There is no shared benchmark for symmetric-oligomer design comparable to the PPI or motif-scaffolding suites in Butcher 2025 (suppl.), so cross-paper numbers in this chapter are not strictly comparable. This is an instance of the `benchmarks/` gap already noted in Introduction — the field at a glance.
- **Does the gradient route reach the cubic groups?** Frank 2024 beats Watson 2023 on single-chain size and handles homo-oligomers via tied logits, but demonstrates no dihedral, tetrahedral, octahedral or icosahedral assemblies. Whether the ring-shaped-objective limitation that defeated Wicky 2022 is inherent to hallucination or was an artefact of MCMC search is therefore still open — and it is the single question that would decide whether the two lineages are converging or dividing the field by task.
- **Would RSO's size advantage transfer to symmetric assemblies?** The 1000-residue result is for *single chains* (Frank 2024). Symmetric assemblies reach comparable total sizes through many small protomers instead — 1550 residues across 33 chains in Wicky 2022 — and no ingested source establishes that a generator scaling well in chain length also scales well in chain count.

12. Flagged references — the history this chapter cannot yet cite

The sections above are written strictly from ingested sources, per the sourcing discipline in `CLAUDE.md`. Several load-bearing pieces of the historical record are therefore either cited **second-hand through a review** or absent entirely. Supplying these PDFs into `sources/` would let the chapter stand on primary literature. Listed in rough order of how much they would improve the chapter; these are also mirrored in index.

Cited through a review rather than the primary paper:

1. **Padilla, Colovos & Yeates, *PNAS* 2001 — "Nanohedra: using symmetry to design self-assembling protein cages, layers, crystals and filaments."** Still the primary source §2 rests on. *Substantially upgraded 2026-08-24*: Yeates 2002 is now ingested, and because it is a review by the *same authors* describing their own design, §2 no longer depends on a third party's paraphrase. The geometric reasoning, the tetrahedral cage of twelve 49 kDa subunits, and the bromoperoxidase/M1 fusion are all now first-party. The 2001 paper itself would still be better — a review does not give the linker-design details or the full geometric derivation — but this is no longer the chapter's weakest citation.
2. **The Yeates-lab follow-ups on the crystallographically resolved 12-subunit cage** — the refinement producing the homogeneous 16-nm cage with a 5-nm central opening, still reachable only through Luo 2016. This is now the most valuable single addition on the list.

Named as prior art by ingested sources, but with no primary paper in `sources/`:

3. **King et al., *Science* 2012 — "Computational design of self-assembling protein nanomaterials with atomic level accuracy."** Appears in the reference list of Luo 2016; the icosahedral-nanocage line it founded is what Courbet 2022 refers to as the fixed-orientation nanomaterials it sets out to move beyond. The chapter currently cannot describe how those cages were designed.
4. **King et al., *Nature* 2014 — "Accurate design of co-assembling multi-component protein nanomaterials."** The two-component branch; needed for the open question in §11.
5. **Bale et al., *Science* 2016 — "Accurate design of megadalton-scale two-component icosahedral protein complexes."** The scale demonstration; also the natural bridge to vaccine-platform applications, which Watson 2023 invokes without this wiki being able to support it.

Closed 2026-08-14: the previous entry here requested Anishchenko et al., *Nature* 2021 as the missing foundation of the hallucination lineage. It has since been supplied and ingested as Anishchenko 2021 (+ Anishchenko 2021 (suppl.)), and §5.1 was rewritten around it. All remaining gaps in this list are in the *pre-deep-learning* history.

Already present in `sources/` but not yet ingested — adjacent branches this chapter deliberately excludes:

6. **`Proteins/Shen_Baker_Science_2018_...De novo design of self-assembling helical protein filaments (+supp).`** Helical rather than point-group symmetry; would support a companion section on unbounded assemblies.
7. **`Proteins/Brunette_Baker_Nature_2015_...Exploring the repeat protein universe through computational protein design.`** Repeat symmetry; relevant because the giant HAL rings have LRR-like protomers (Wicky 2022).
8. **`ProteinDesign/SchneiderDuhovnik_NAR_2005_PatchDock and SymmDock servers for rigid and symmetric docking.`** Would let the "dock-and-design" paragraph in §2 cite a primary symmetric-docking method rather than Wicky 2022's retrospective characterization of the era.

Supplements missing for pages this chapter leans on: Thomson 2014 and Boyken 2016 are both *Science* papers whose supplementary materials contain the actual method descriptions (the barrel scoring function; the HBNet algorithm and scripts). Both are marked `supp: missing-requested`.

13. Sources

Ingested sources feeding this chapter, roughly chronologically:

- Harbury 1993 — oligomeric state is encoded in core residue geometry · *Science* 1993
- Yeates 2002 — review, first-party account of the symmetric-fusion strategy and the nanohedra cage · *Curr. Opin. Struct. Biol.* 2002
- Luo 2016 — review; the later symmetric-fusion literature · *Chem. Rev.* 2016
- Huang 2014 (+ Huang 2014 (suppl.)) — parametric backbones from the Crick equations · *Science* 2014
- Thomson 2014 — designed C5/C6/C7 barrels with defined channels · *Science* 2014
- Brunette 2015 — 83 de novo repeat proteins; symmetry imposed on the search · *Nature* 2015
- Boyken 2016 — HBNet; specificity as a separate design axis · *Science* 2016
- Wood 2018 — CCBuilder 2.0; why parameterization works for repeat proteins · *Protein Sci.* 2018
- Anishchenko 2021 (+ Anishchenko 2021 (suppl.)) — hallucination founded; oligomers as an unintended byproduct · *Nature* 2021

- Dauparas 2022 (+ Dauparas 2022 (suppl.)) — sequence design with tied symmetric positions · *Science* 2022
- Courbet 2022 (+ Courbet 2022 (suppl.)) — symmetry-mismatched machines; the hand-built ceiling · *Science* 2022
- Wicky 2022 — hallucinating cyclic oligomers; giant rings to C33 · *Science* 2022
- Watson 2023 (+ Watson 2023 (suppl.)) — point-group-conditioned diffusion; C, D, T, I · *Nature* 2023
- Ingraham 2023 (+ Ingraham 2023 (suppl.)) — symmetry as a Bayesian conditioner · *Nature* 2023
- Frank 2024 (+ Frank 2024 (suppl.)) — hallucination by gradient descent in relaxed sequence space; 1000-residue chains · *Science* 2024
- Butcher 2025 (+ Butcher 2025 (suppl.)) — dedicated symmetric inference loop, ~10× cheaper · bioRxiv 2025
- Jumper 2021 — the confidence oracle used by both the hallucination objective and the diffusion filter · *Nature* 2021

Related pages: concept Symmetric assembly design · concepts Parametric design, Repeat protein design, Diffusion generative design, Motif scaffolding (functional-site design), Sequence design (fixed-backbone) · tools RFdiffusion, RFdiffusion3 (RFD3), Chroma, ProteinMPNN, AlphaFold, CCBuilder 2.0.

Concepts — a lookup section

Binder design

De novo design of a protein that binds a chosen target with high affinity and specificity, starting from the target's structure — or, latterly, its sequence alone — rather than from immunization or library selection (Bennett 2023, Watson 2023).

Promoted 2026-10-12. This page is now the short lookup entry; the full treatment is the chapter De novo binder design, which covers the biophysics of interfaces and hot spots, the four methodological lineages, the oracle and its failure modes, the compute and search axis, and the comparability of success rates.

The lineages, in one line each.

1. **Grafting** — transplant a binding mode that already exists and scaffold it (Wang 2022); see Motif scaffolding (functional-site design). Still the highest-affinity route when a natural mode is available: scaffolding the p53 helix against MDM2 gave 0.5 and 0.7 nM binders against a 600 nM native interaction (Watson 2023).
2. **Filter-first** — keep the pipeline, add a structure predictor in front of the ordering decision. Monomer pLDDT plus interface pAE raised experimental success roughly tenfold with the generator unchanged (Bennett 2023, Bennett 2023 (suppl.)).
3. **Hotspot-conditioned generation** — diffuse a binder backbone against named target residues (Watson 2023), then all-atom and ligand-conditioned (Krishna 2024, Butcher 2025), then modality-specified (Stark 2025), then **antibody-format** with a fixed framework supplied global-frame-invariantly so the rigid-body placement is designed (Bennett 2026).
4. **Surface fingerprints** — the independent lineage: match learned descriptors of molecular surfaces to find a complementary binding seed, then graft it (Gainza 2020, Gainza 2023).

Hallucination, which optimises a sequence against a predictor's confidence, is treated in the chapter as the inference-time half of lineage 3 rather than a fifth family, following Didi 2026 and Cho 2025. Lin 2026 complicates that: a pure generative model with no oracle optimisation in the loop beats the hallucination state of the art on 7 of 10 benchmark problems, and by a wider margin once runtime is normalised.

What the designer still supplies. The epitope. Every method in the chapter takes hotspot residues as input, and the physical reason that is a coherent instruction is that interface binding energy concentrates in densely packed regions — occupied volume around a residue correlates with its alanine-scanning $\Delta\Delta G$ at $r^2 = 0.94$ (Halperin 2004).

The standing caution. Reported success rates across this literature are fractions of different denominators measured by different assays against different bars — a yeast-display SC50 < 4 µM hit rate (Bennett 2023 (suppl.)), a single-point BLI response at 10 µM (Watson 2023), nanomolar by SPR from 15 designs out of 60,000 generated (Stark 2025), an AlphaFold2 self-consistency pass rate (Didi 2026) — and they should not be placed side by side. The nearest thing to a sound comparison is the five-method, one-protocol, fixed-budget benchmark in

Lin 2026, which is *in silico* and whose own oracle tops out at 12% precision. Ruijgrok 2026 documents the same problem within a single project and six pipelines.

Sources: see §15 of De novo binder design for the full annotated list.

Catalytic site design

Specifying and building the arrangement of atoms that performs a chemical transformation — as distinct from scaffolding it, which is Motif scaffolding (functional-site design), and from binding a substrate, which is Binder design. The full treatment is the chapter De novo enzyme design.

The specification is a theozyme: a geometric arrangement of catalytic groups around a substrate or transition state. Its provenance varies and the variation is consequential — extracted from a cocrystal with a **transition-state analogue** (Chen 2026), from a native enzyme structure (Choi 2026, using Ulp1 PDB 1EUV), derived from **quantum chemistry** (Butcher 2026 (suppl.)), or dispensed with entirely in favour of optimising the coordination environment directly (Elnesr 2026).

What a catalytic site must do, with the sources that establish each:

1. **Activate a nucleophile** — by a hydrogen-bonded Asp–His–Cys triad (Choi 2026) or by a metal activating water (Chen 2026). Designs achieve this partially: a designed cysteine pK_a shifts from ~8–8.5 to 6.0, against 2.5–4.0 in native enzymes (Choi 2026).
2. **Stabilise high-energy intermediates** via an **oxyanion hole** — the hardest part to design, because buried polar networks are many-body and badly served by pairwise energy functions (Boyken 2016). Evidence it is incompletely designed: knocking out the oxyanion side chain costs only 1.9- to 3.8-fold (Choi 2026).
3. **Bind and orient the substrate** so the reacting bond is presented correctly — designed proteases use 8–13-residue substrates and an extended recognition interface (Chen 2026).
4. **Preorganize** all of the above across a multi-step reaction coordinate.

Preorganization is where the field currently is. Crystal structures confirm designed active sites are built as specified — Cα r.m.s.d. < 1.2 Å across six structures in Choi 2026, 0.82–1.35 Å in Chen 2026 — while catalytic efficiencies remain **10^3–10^4-fold below native** (Chen 2026). The mechanistic account comes from PLACER ensembles across the catalytic cycle: a non-productive His rotamer suppressed by a single substitution, with the **fraction of productive conformations tracking activity across a design series** (Choi 2026). The wiki's reading is that designs optimise a single structure while catalysis depends on a distribution.

A design-time corollary with a large measured effect. What is modelled in the active site during filtering determines the hit rate more than the generator does: including the **catalytic water** in the enzyme–substrate model moved a one-hour hit rate from 3/89 to 31/38, at unchanged numerical thresholds (Chen 2026, Chen 2026 (suppl.)). The converse failure is equally sharp — AlphaFold3 assigned pLDDT 86.55 to a metal in a site with a single coordinating ligand that binds no metal experimentally (Elnesr 2026).

Preorganization is filterable, and has been filtered once prospectively. Kim 2026 selected designs on **PLACER** ensembles — a network that randomises the substrate and all side chains within a 600-atom sphere and regenerates them, repeated to give an ensemble

— ranked its eventual best design first, and then found that **seven designs from the same backbone family, rejected for suboptimal PLACER metrics, had very low or no activity**. That is the chapter's one prospective demonstration that the ensemble, not the single structure, is the quantity to filter on.

The ceiling may be a pipeline property, not a method property. Across the Baker-lab campaigns, efficiencies sit 1–4 orders of magnitude below native. But Braun 2026 reaches **290 M^{-1} s^{-1} against an extensively evolved enzyme's 320**, with **99% enantiomeric excess**, by enumerating artificial motif libraries and enforcing a binding pocket with a deletable placeholder helix. On the one reaction where two independent pipelines attack the same theozyme, the gap between them is ~45-fold (Braun 2026 vs Ahern 2025).

Sources: see §12 of De novo enzyme design.

Diffusion generative design

Generating novel protein backbones (and complexes) by learning to reverse a noising process — the paradigm that now dominates de novo structure generation.

Two independent demonstrations (2023). Watson 2023 fine-tunes the RoseTTAFold predictor into a backbone denoiser, excelling at monomers, symmetric assemblies, motif scaffolding and binders. Ingraham 2023 builds a from-scratch diffusion model with a polymer-statistics-aware noise prior and programmable (including natural-language) conditioning. Different lineages, same conclusion: diffusion works for proteins.

Convergence with prediction. Abramson 2024 adopts a diffusion coordinate head for *prediction*, so prediction and design increasingly share machinery.

Going all-atom. Krishna 2024's denoising fine-tune (RFdiffusionAA) diffuses protein backbones *around* a small-molecule ligand — extending diffusion design from protein-only to ligand-conditioned Binder design (validated for digoxigenin, heme, bilin).

The non-diffusion alternative did not go away. Frank 2024 revives predictor inversion by running gradient descent through AlphaFold2 in a **relaxed** (continuous, PSSM-like) sequence space rather than MCMC over discrete sequences, handing only the converged backbone to ProteinMPNN. It matches or beats Watson 2023 on designability and reaches **1000-residue** single chains with median TM 0.89, where RFdiffusion "faced challenges with producing viable backbones beyond 600 amino acids", at comparable GPU cost per success (Frank 2024). Diffusion still has the broader demonstrated coverage of point-group symmetries; the gradient route currently has the size advantage.

Conditioning is the point. The value is steerability — hotspots for binders, symmetry operators, fixed functional motifs, shape/semantic constraints. Backbones are then sequenced by Sequence design (fixed-backbone) (ProteinMPNN) and filtered by Structure prediction.

All-atom generation — three routes. The field is now pushing past backbone-only generation: Krishna 2024 atomizes ligands inside a predictor (RFdiffusionAA); Chu 2024 (Huang lab) makes side-chain generation intrinsic via a "superposition + collapse" scheme, codesigning sequence; and Butcher 2025 uses a token+atom sparse architecture to hit prior quality at ~10× less compute across ligand/nucleic-acid/enzyme tasks. Its published successor Butcher 2026 reframes the same model around **free-atom diffusion for atomic functional site scaffolding**, and backs it with zero-shot experimental enzymes — phosphotriesterases carrying a carbamylated lysine bridging two metals, and transthyretin-cleaving zinc proteases at 56/96 in-cis success.

Open contrast to file: fine-tuned-predictor (RFdiffusion) vs. bespoke correlated-noise prior (Chroma) vs. superposition all-atom (Protpardelle) vs. token+atom sparse (RFD3) — tradeoffs not yet resolved in these sources.

Sources: Watson 2023, Ingraham 2023, Abramson 2024, Krishna 2024, Chu 2024, Butcher 2025.

A fourth independent arrival, and the only fully SE(3)-equivariant one. Lin 2023 diffuses Cα coordinates as a point cloud in the forward process and reasons over a cloud of **oriented Frenet–Serret frames** in the reverse process, an asymmetry chosen so that IPA's expressivity is available during generation without violating the Gaussian assumption of DDPMs. It reached 0.790 designability with **4.1 M parameters** against RFdiffusion's 0.951 at 59.8 M (Lin 2023). Lin 2024 then added SE(3)-invariant motif conditioning via Cα pairwise distance matrices — insensitive to initial motif placement, unlike fixed-coordinate conditioning — and was the first protein diffusion model trained on the AlphaFold Database, matching RFdiffusion's designability at 0.96 while far more diverse (0.91 vs 0.63) and novel (0.41 vs 0.26 PDB novelty). Lin 2026 extends the frames to sidechains. See Genie 3.

Metalloprotein design

Designing proteins that bind a metal ion in a specified coordination geometry — structurally, or to do chemistry. Treated in depth in De novo enzyme design §7.

Why metals. They are "biologically abundant, tether to the protein using strong interactions, and have favorable catalytic properties"; zinc in particular is an excellent Lewis acid, serves structural roles (zinc fingers) and catalytic ones in sites up to trinuclear, and **about 10% of all enzyme-catalysed reactions use zinc** (Durr 2023).

Why it is hard. Computational design "is hampered by the complex electronic structure of many biologically relevant metals": classical force fields treat metals poorly, and quantum-mechanical treatment is too expensive for design-scale work, usable only to verify selected candidates (Durr 2023).

Two routes, with different premises.

1. **Transplant a native metal site, or derive one from first principles.** Chen 2026 takes a Zn(II) centre of two histidines and a glutamate from aminopeptidase N, with water as the fourth ligand, and scaffolds it. Kim 2026 instead derives the theozyme from **DFT** — the transition-state geometry for Zn(II)-hydroxide attack on an ester — copying no native enzyme, and reaches **53,000 $\mathbf{M^{-1}\ s^{-1}}$**, against 3–60 $M^{-1}\ s^{-1}$ for previously designed metallohydrolases. In that design the **Zn(II) ion itself serves as the oxyanion hole**, with no nearby side-chain hydrogen-bond donors, sidestepping the many-body buried-polar problem entirely. Butcher 2026 goes further, scaffolding a di-metal site bridged by a **post-translationally carbamylated lysine** — and shows the modification matters, since removing it costs 11-fold in k_cat. Wang 2022 and Watson 2023 established metal-binding sites as a scaffoldable motif class.
2. **Design the coordination sphere directly.** Elnesr 2026 pairs LigandMPNN with Durr 2023 in a multi-objective evolutionary search, producing a **$\mathbf{His_2Cys_2}$ site with no motif template** binding zinc at K_d 37 ± 4 nM, and — after retraining the metal predictor on catalytic zincs only — a **binuclear zinc hydrolase** with no structural-motif, sequence or fold neighbour containing a bi-zinc site.

Prediction as a design objective. Durr 2023 predicts zinc location from a voxelized local environment to **0.70 ± 0.64 Å**, template-independently, and emits a per-residue density usable inside a design loop — which is what makes route 2 possible at all.

The training-data caveat, which is now a documented failure. Metal3D is trained on zinc alone, and its supplement stratifies performance by coordination number and occupancy throughout (Durr 2023 (suppl.)), including a named false positive where a calcium site is called as zinc at p = 0.9. Elnesr 2026 audited the training corpus, found zinc sites with fewer than three coordinating residues within 3.0 Å that "may reflect crystallization additives or soaking agents", retrained **Metal3D-Clean** excluding them, and separately **Metal3D-Cat** on catalytic zincs from MAHOMES-II. The lesson: a model trained on *zinc appears here in a crystal* learns something different from *zinc is functionally bound here*, and the difference only became consequential once the model was used generatively.

Validation specific to metals. Beyond crystallography: **ICP-MS** for stoichiometry, **chelation** for activity dependence, competition titration for affinity, and **XAS/EXAFS** for the solution-state coordination shell — the last unique in this wiki to Elnesr 2026, and the only method here that reports on the metal site in the state the protein actually works in.

Zinc affinity is a persistent weak point. Kim 2026 measures **K_D = 41 ± 5 nM** for its best design, similar to previous designed zinc enzymes but weaker than native zinc hydrolases, which are "typically below 10 nM"; Elnesr 2026 reports 37 ± 4 nM for a template-free His_2Cys_2 site and could not determine K_d for its bi-zinc enzyme because chelation destabilises the protein. Designed metal sites bind, and bind more weakly than evolved ones.

Mechanism is often only partly confirmed. In Kim 2026, the triple-histidine knockout inactivates as expected, but **H130A costs only 13-fold and the proposed general base D67A barely affects k_cat/K_M while increasing zinc affinity** — consistent with competing zinc-binding configurations rather than the designed one. Designed metalloenzymes work; the residue-by-residue account of why is frequently incomplete.

Sources: see §12 of De novo enzyme design.

Motif scaffolding (functional-site design)

Building a full protein *around* a fixed functional site — an enzyme active site, a metal/ligand pocket, a binding or epitope motif — so the site is held in its active geometry. The second half of functional de novo design (the first half being: obtain the motif geometry).

Two generations of method.

1. **Hallucination & inpainting** — Wang 2022 introduced both: *constrained hallucination* (optimize a sequence until the network folds it to contain the motif) and *inpainting/RFjoint* (one forward pass fills sequence+structure around the motif). Flexible vs. fast. The unconstrained hallucination engine both build on comes from Anishchenko 2021, which anticipated this application — noting that because the network hallucinates plausible structures from scratch, no scaffold blueprint need be specified in advance, and the loss can be generalized to include binding motifs or catalytic sites.
2. **Conditional diffusion** — Watson 2023 generalizes these: conditional training scaffolds motifs by denoising (see Watson 2023 (suppl.) §3.2), with higher success and diversity. Butcher 2025 extends to **atomic** motifs (tip-atom conditioning; the AME enzyme benchmark), and Chu 2024 scaffolds motifs **backbone- and rotamer-free** in all-atom space.

Common pattern. Motif in → scaffold out → sequence with ProteinMPNN → filter with AlphaFold2/self-consistency.

Sources: Wang 2022, Watson 2023, Butcher 2025 (and all-atom variant Chu 2024).

Conditioning without fixing coordinates. Lin 2024 encodes a motif as one-hot sequence features plus the **pairwise distance matrix of its Cα atoms**, which is SE(3)-invariant and therefore — unlike the fixed-coordinate conditioning of Watson 2023 and FrameFlow — insensitive to the motif's initial placement. Specifying intra-motif but not inter-motif distances yields **multi-motif scaffolding** as an inference-time choice: Genie 2 solves 4 of 6 curated multi-motif tasks without ever training on one. On the 24-task single-motif benchmark it equals or beats RFdiffusion in unique successes on 22 tasks. That paper also argues *success rate* is the wrong metric, because mode collapse inflates it, and reports **unique successes** after TM-score 0.6 clustering instead (Lin 2024).

Atomic motifs, and the sequence-position problem. The step that made de novo proteases possible is conditioning on **active-site atoms without specifying where along the sequence the catalytic residues sit** — the contribution of Ahern 2025, which does it two ways: residues may be represented as a frame *or* as atomized heavy atoms (so rotamers are inferred rather than enumerated), and **unindexed residues**, whose sequence position is withheld, are learned with no auxiliary loss to superimpose onto generated residues. It solves **all 41 cases of the AME benchmark it introduces, against 16 for prior methods**. Used via its molecular-interfaces variant RFD2-MI in Choi 2026 and Chen 2026.

The enumerative alternative. Braun 2026 solves the same problem without retraining a

generator: embed each catalytic residue in a helical fragment, **invert rotamers** to fix functional groups while varying backbone positions, search all non-clashing fragment combinations, and rank by rotamer probability and PDB abundance — then scaffold the resulting artificial motif with plain RFdiffusion. It also adds a **placeholder helix** at the pocket position, which the denoising trajectory is centred on and which is deleted afterwards, because RFdiffusion's substrate potential "failed to promote the formation of well-defined substrate binding pockets" (Braun 2026). On the one shared theozyme, the enumerative route reached ~45× the catalytic efficiency. Butcher 2026 then moved to **free-atom diffusion**, where conditioning tokens correspond only to the atoms actually fixed rather than to whole catalytic residues, and reports outperforming RFD2 on **37 of 41 (90%)** atomic motif enzyme benchmark tasks from the M-CSA, with better Dunbrack rotamer quality. Full treatment in De novo enzyme design §3.

A standing objection. Elnesr 2026 argues that scaffolding a motif is not sufficient for function: "optimizing for the motif alone does not guarantee catalytic competence", because second-shell residues mediate affinity, specificity and reactivity. Its alternative — optimising the coordination environment directly — produced the highest rate enhancement reported for a de novo hydrolase. See De novo enzyme design §7.

Parametric design

Generating protein backbones from a **closed-form geometric description** — a handful of numbers fed to a set of equations — rather than by sampling fragments, docking known structures, or denoising noise.

Why it works only for some folds. The challenge is reducing a 3D fold to parametric equations that still capture its salient features, and that is currently possible for a **subset of proteins: repeat proteins** (Wood 2018). The reason is structural rather than technical — a shape built by applying one rigid-body transform repeatedly is fully described by the few parameters of that transform. This is the conceptual link between Parametric design and Repeat protein design, and it is why the α-helical coiled coil, a repeat class making up an estimated 3–5% of protein-encoding DNA, has been the fold where the approach has paid off (Wood 2018).

The parameters. For a coiled coil the description is **Crick's, from 1953**: supercoil radius R0 and twist ω0, superhelical phase φ0, helical radius R1, twist ω1, phase φ1, axial offset Δz, and pitch angle α = arcsin(R0ω0/*d*). In practice *d* and R1 are **fixed at ideal α-helix values (2.26 and 1.51)** and the rest are sampled, which is what keeps the search low-dimensional (Huang 2014 (suppl.)). Supercoil and helical twist are coupled: successive Cα atoms rotate by ≈ (ω0 + ω1), and the backbone is strained when that departs from the ~100° of an ideal helix (Huang 2014).

Symmetry is free. For an *n*-helix bundle the superhelical phase simply takes the values {0, 2π/n, 4π/n, … 2(n−1)π/n}. **Cyclic symmetry is therefore exact by construction** — a property of the coordinate generator, never optimized for and never verified afterwards (Huang 2014 (suppl.)). This is the sharpest contrast in the wiki with symmetry as a rewarded objective (Wicky 2022) or as a projection applied during sampling (Watson 2023); see Symmetric oligomer design §7.

The workflow. Sample the parameters on a grid, generate backbones, run Rosetta sequence design at each grid point, refine the grid near the lowest-energy parameter sets, connect helices with loop modelling for monomeric targets, and filter with structure prediction before synthesis (Huang 2014). Interfaces can be given specificity independently, by installing buried hydrogen-bond networks into the repeating cross-section (Boyken 2016).

What it delivered. Hyperstable designs with **extrapolated ΔG_fold > 60 kcal/mol** and crystal structures matching in core packing, including a C5 bundle (Huang 2014); α-helical barrels at **C5, C6 and C7** with chemically defined channels whose diameters track oligomer state (Thomson 2014); and, pushed to its limit, the D2–D8 axles and C3–C8 rotors of Courbet 2022.

Its two limits. Expressivity — only folds the parameterization can write down, in practice helical bundles and barrels (Wicky 2022 notes parametric bundles were the sole exception to the rule that protomer structure had to be specified in advance). And labour — reaching a D3 axle required hand-chosen dual superhelical radii (6 Å and 12 Å) over a ~5 Å range in

0.5 Å bins, phase swept 0°–90° in 10° steps, z-offsets in 0.1 Å steps, and per-symmetry linker design (Courbet 2022 (suppl.)).

It is not obsolete. Wood 2018 makes the approach browser-accessible with force-field optimization, extending to α-helical barrels and collagen triple helices, and notes the parameterization also runs *backwards* — fitting Crick parameters to solved structures. Parametric generation remains the only route here where the symmetry error is identically zero.

Related: Repeat protein design (why parameterization is possible at all) · Symmetric assembly design · tool CCBuilder 2.0 · chapter Symmetric oligomer design.

Sources: Harbury 1993, Huang 2014, Thomson 2014, Boyken 2016, Wood 2018, Courbet 2022, Brunette 2015.

Repeat protein design

Designing proteins built from **multiple tandem copies of one modular structural unit**. The repeat is the design object; the overall shape is whatever repeated application of a single rigid-body transform produces.

Why the fold class matters. Repeat proteins are widespread in nature and carry out molecular recognition, signalling and other essential functions; natural ones have been re-engineered as recognition and scaffolding platforms (Brunette 2015). Coiled coils are themselves a repeat class, and account for an estimated **3–5% of all known protein-encoding regions of DNA** (Wood 2018).

Repetition implies global helical symmetry. If the rigid-body transform relating adjacent units is identical throughout — which it is, by construction, in a designed repeat protein — then repeated application of that transform generates a **helical array**. Every such design is therefore globally helical and can be classified by three coupled parameters: radius r, twist between adjacent repeats ω, and axial translation z. The coupling is geometric: a repeat spans arc length $\approx r\omega$ and has total unit length $\approx \sqrt{((r\omega)^2 + z^2)}$, so the radius–twist distribution is hyperbolic, with highly twisted structures having smaller radii (Brunette 2015).

Symmetry is imposed on the search, not discovered. In Brunette 2015 every Rosetta Monte Carlo move is made at the equivalent position in each repeat, and every sequence substitution is applied identically across units. This is the same move that defines Parametric design — reduce the search to one unit and let symmetry replicate it — implemented through fragment assembly rather than a closed-form generator.

Nature has barely explored the space. From a single helix–loop–helix–loop unit with helix lengths 10–28 and turn lengths 1–4 (5,776 combinations), 761 designs passed in silico filters and 83 were tested: **53 monomeric and stable at 95 °C**, 43 SAXS-consistent, and **15 crystal structures at 0.7–2.5 Å RMSD** across a broad range of curvatures. The conclusion is that existing repeat proteins occupy only a small fraction of the accessible sequence and structure space (Brunette 2015).

Why this matters downstream. The resulting designed helical repeat proteins (DHRs) became a standard component library — the cyclic rotors and dihedral axles of Courbet 2022 are built by fusing DHRs. And repeat topology recurs on the generative side: the giant hallucinated rings of Wicky 2022 have protomers with LRR-like repeat topology, though their closed cyclic assemblies differ considerably from the open horseshoe folds of natural leucine-rich repeats.

Boundary with Symmetric assembly design. Repeat proteins have *helical* symmetry within a single chain; symmetric assemblies have *point-group* symmetry across several chains. The two meet when a repeat protein closes into a ring — which is what makes DHRs natural building blocks for Cn components.

Sources: Brunette 2015, Wood 2018, Courbet 2022, Wicky 2022.

Sequence design (fixed-backbone)

Given a target backbone, find an amino-acid sequence that folds to it — the "inverse folding" problem, and the second stage of the dominant de novo recipe.

Shift from physics to learning. Classic approach: Rosetta energy optimization over rotamers — accurate but slow, with limited success on hard targets. Dauparas 2022 replaced it with an autoregressive message-passing network conditioned on backbone geometry, raising native sequence recovery to ~52% (vs. ~33% for Rosetta) in seconds, and generalizing to interfaces, oligomers and nanoparticles. Independently, Anand 2022 (Huang lab) reached the same conclusion from the "learned energy" direction — a network learning the design potential directly from crystal structures with no human priors, generalizing to unseen topologies (incl. a de novo TIM-barrel) with crystal-validated designs.

Toward codesign. The next step folds sequence into structure generation: Chu 2024 codesigns all-atom structure and sequence (miniMPNN in the loop), blurring the line between this concept and Diffusion generative design.

Role in pipelines. Sequence design is the step *after* backbone generation: Watson 2023 (or Ingraham 2023) produces a backbone, then ProteinMPNN assigns sequence, then a predictor filters. Bennett 2023 swapped Rosetta for ProteinMPNN specifically for this efficiency.

Key model: ProteinMPNN.

Sources: Dauparas 2022, Anand 2022, Bennett 2023, Chu 2024.

Structure prediction

Predicting 3D structure from sequence — historically the "protein folding problem." Now largely solved for single chains and increasingly for complexes, and the enabling substrate for modern design.

State of the art. Jumper 2021 reached atomic accuracy at CASP14 using MSA + pair reasoning (Evoformer) and an explicit structure module. Abramson 2024 extends this to protein–ligand, protein–nucleic-acid and antibody–antigen complexes with a diffusion coordinate head. RoseTTAFold (Baker lab) is the parallel lineage whose network is repurposed for generation in Watson 2023; its all-atom extension Krishna 2024 brings small molecules, nucleic acids and metals into a single model — the design-side counterpart to AlphaFold3.

Why it matters for design. Prediction models double as **design oracles**: their confidence scores (pLDDT, PAE, pTM) filter designs *in silico* before synthesis — see Bennett 2023 and the RFdiffusion→MPNN→AF2 loop. Prediction and design are converging on shared diffusion machinery.

Key models: AlphaFold · RoseTTAFold.

Sources: Jumper 2021, Abramson 2024, Krishna 2024, Bennett 2023.

Symmetric assembly design

Designing assemblies in which several copies of one chain are arranged under a **point-group symmetry** — cyclic (C*n*) rings, dihedral (D*n*) stacks, and the cubic groups — as building blocks for nanomachines, biomaterials and multivalent scaffolds.

> **This concept has been promoted to a chapter.** See Symmetric oligomer design for the full treatment: the historical arc, the four mechanisms for imposing symmetry, validation practice, and open questions. This page stays as the short canonical definition to cite from elsewhere.

The one-line definition. Symmetry lets a single designed protomer and a single designed interface specify an assembly of *n* chains, because the relative orientation of adjacent subunits plus the complementarity of their contact interface determines the multiplicity of the complex (Luo 2016).

Four ways to impose it. The distinction matters because the methods are not variations on one idea.

- **Exact by construction** — parametric generation, where symmetry is a property of the coordinate generator: for an *n*-helix bundle the superhelical phase simply takes the values {0, 2π/n, … 2(n−1)π/n} (Huang 2014 (suppl.)). See Huang 2014, Thomson 2014.
- **Rewarded** — deep-network hallucination, where symmetry is a **term in a loss**: the standard deviation of centre-of-mass distances between adjacent protomers in an AlphaFold2 prediction, added to the pure-confidence objective of Anishchenko 2021 and optimized jointly with it, from inputs of only chain length and valency (Wicky 2022).
- **Projected** — conditional diffusion, where point-group operators are applied to the noised state at initialization and the structure is **explicitly resymmetrized at each denoising step**; equivariance means this changes the predictions only slightly (Watson 2023, Watson 2023 (suppl.) §3.1).
- **Tied** — gradient-based hallucination, where the protomers share one set of **input logits** (copied, with a residue-index shift of 50 between chains) so sequence symmetry is exact by construction, while interface geometry is rewarded through interface pAE and contact losses (Frank 2024 (suppl.)).

Why the third one won on coverage. Hallucination-based cyclic design **fails for higher-order dihedral, tetrahedral, octahedral and icosahedral symmetries**, in part because those are much less represented in the PDB (Watson 2023); diffusion reaches all of them in one framework, using only the smallest generating subset of subunits for the cubic groups. Butcher 2025 carries a dedicated symmetric-design inference loop (Butcher 2025 (suppl.)), and Ingraham 2023 arrives at symmetry conditioning independently, as one Bayesian conditioner among several.

Specificity is a separate axis from geometry. Getting the right shape does not give orthogonal assemblies; buried hydrogen-bond networks enumerated by HBNet confer interaction specificity in vivo and were validated across ten crystal structures (Boyken 2016).

Sequence design is load-bearing, not downstream. The same hallucinated backbones went from a median soluble yield of 9 mg/L to 247 mg/L when the hallucinated sequences were discarded and redesigned with Dauparas 2022 (Wicky 2022); tied/symmetric decoding is what makes this possible (Dauparas 2022 (suppl.)).

Symmetry was a byproduct before it was a target. Unconstrained hallucination, with every sequence modelled as a monomer, nonetheless produced many designs that formed homo-oligomers in solution — design 0217 crystallized as a dimer — because the network had learned the sequence signatures of native protein–protein interfaces and did not disfavour surface hydrophobics enough for soluble monomers (Anishchenko 2021). Substituting exposed hydrophobics with polar residues reversed it (Anishchenko 2021 (suppl.)). Eliciting an oligomer was never the hard part; specifying *which* one was.

Chapter: Symmetric oligomer design · **Sources:** Harbury 1993, Huang 2014, Thomson 2014, Boyken 2016, Luo 2016, Anishchenko 2021, Courbet 2022, Wicky 2022, Watson 2023, Frank 2024, Butcher 2025.

Reference

Glossary

Concise definitions of the recurring terms in this book. Where a term has its own chapter or concept entry, follow its cross-references for the full treatment.

AF2 / AF3 AlphaFold 2 and AlphaFold 3, DeepMind's structure predictors. AF2 predicts protein monomers and complexes from an MSA; AF3 replaces the structure module with a diffusion decoder and generalises to nucleic acids, ligands and modified residues.

All-atom design Generative design that places side-chain (and often ligand) atoms jointly with the backbone, rather than generating a backbone first and assigning sequence afterwards. Contrast *two-stage design*.

Backbone The repeating N–Cα–C(=O) chain of a protein, ignoring side chains. Most first-generation generative models produce backbones only.

Binder A designed protein that binds a specified target with useful affinity and specificity. See the binder-design chapter.

Chroma Generate Biomedicines' programmable generative model, which diffuses under a polymer-statistics prior and steers sampling with modular conditioners.

Codesign Generating structure and sequence simultaneously in one model, as opposed to designing sequence onto a fixed backbone.

Conditioner A module that biases a diffusion model's sampling toward a constraint (symmetry, shape, substructure, a natural-language or property target) without retraining the base model.

CASP Critical Assessment of Structure Prediction — the biennial blind community benchmark on which AlphaFold2's accuracy was first established.

Denoising / reverse diffusion The generative process in a diffusion model: start from noise and iteratively remove it, each step conditioned on the model's current estimate of the clean structure.

De novo design Designing a protein from scratch rather than modifying a natural one.

Diffusion model A generative model trained to reverse a gradual noising process; the dominant architecture for structure generation (RFdiffusion, Chroma, Protpardelle, RFdiffusion3).

Evoformer AlphaFold2's trunk: paired sequence and residue-pair representations updated by attention with triangle operations that enforce geometric consistency.

FAPE (frame-aligned point error) AlphaFold2's training loss, comparing predicted and true atom positions in every residue's local frame — rotation- and translation-invariant, and sensitive to local as well as global error.

Foldseek A fast structural-alignment search tool; used in design pipelines to test whether a generated backbone resembles anything in the PDB (i.e. how novel it is).

Fold switching The ability of one sequence to adopt more than one distinct fold — a known failure mode for single-conformation structure prediction.

Hallucination Design by optimising a sequence so that a structure predictor confidently folds it into something — the input is an objective, not a target backbone. The route used for early motif scaffolding and symmetric assemblies.

Hotspot residues Target-surface residues specified as the intended contact patch when designing a binder.

Inpainting Filling in missing structure or sequence around a fixed region, the complement of hallucination for motif scaffolding.

Inverse folding Predicting a sequence that will fold into a given backbone — the problem ProteinMPNN solves. Synonym in practice for *fixed-backbone sequence design*.

IPA (invariant point attention) The attention mechanism in AlphaFold2's structure module that operates on 3D points in residue frames and is invariant to global rotation and translation.

lDDT / pLDDT The local distance difference test, a superposition-free measure of local structural accuracy; **pLDDT** is a model's per-residue self-estimate of it, widely reused as a design filter.

Logit An unnormalised score, one per amino acid per position; a softmax over the 20 logits at a position turns them into a probability distribution. Logits matter for design because a sequence is *discrete* and therefore cannot be differentiated — an L × 20 array of logits is the continuous stand-in that gradient-based methods actually optimise. See *relaxed sequence space*.

MaSIF Molecular surface interaction fingerprints — geometric deep learning on protein surfaces, used to find and design interaction patches.

Motif scaffolding Building a new protein around a fixed functional site (a binding motif, an active site) so the motif's geometry is preserved.

MPNN Message-passing neural network; in this field usually shorthand for ProteinMPNN.

MSA (multiple sequence alignment) A stack of evolutionarily related sequences whose covariation encodes contact information; the input signal that made AlphaFold2 accurate, and a dependency later models try to reduce.

Oracle A predictor used to score designs *in silico* before synthesis — typically AlphaFold2, which both filters designs and (via hallucination) can be optimised against.

pAE / PAE (predicted aligned error) A predicted pairwise error matrix; `pAE_interaction` across a designed interface is a standard in silico discriminator of binders that will work experimentally.

Parametric design Generating a backbone from a closed-form geometric description — for coiled coils, the Crick equations — rather than by sampling fragments, docking, or denoising. Symmetry comes out exact because the coordinate generator cannot express anything else. Works only for folds a parameterisation can capture, in practice repeat proteins.

Partial diffusion Noising an existing design only part-way and re-denoising it, to sample local variants around a promising solution rather than starting from scratch.

PDB The Protein Data Bank — the archive of experimentally determined structures that trains and benchmarks essentially every model in this book.

ProteinMPNN The Baker-lab graph neural network for fixed-backbone sequence design; fast, high sequence-recovery, and the default sequence step in most design pipelines.

Protpardelle The Huang-lab all-atom generative model, which handles side chains via a superposition over side-chain states.

Relaxed sequence space The continuous sequence representation that gradient-based hallucination optimises in. Instead of a one-hot L × 20 matrix (one amino acid per position), each row is a softmax over *logits* — a weighted superposition of all 20 amino acids, e.g. 60% Leu / 15% Ile / 8% Val at one position. Such a residue is physically meaningless, but AlphaFold2's sequence input is consumed by a matrix multiply against an embedding table, which accepts a soft row as readily as a hard one; the network therefore runs end to end and the loss can be backpropagated all the way to the logits. That is the whole point: a discrete sequence admits no gradient, a relaxed one does, so design becomes gradient descent rather than the random search of earlier hallucination. Two consequences recur in this book. **Copies of the same logit tensor** give exact sequence identity between protomers of a homo-oligomer — not as an enforced constraint but because there is only one variable, gradients from every copy accumulating into it. And the relaxed sequence is **scaffolding, not product**: discretising it by argmax yields structures that deviate badly from the converged backbone, so the backbone is kept and the sequence thrown away and redesigned with ProteinMPNN.

Repeat protein A protein built from multiple tandem copies of one modular structural unit. Because the rigid-body transform between adjacent units is identical, the whole chain is globally helical and describable by three parameters (radius, twist, rise) — the property that makes parametric design possible.

RFAA (RoseTTAFold All-Atom) The Baker-lab generalised predictor covering proteins with small molecules, nucleic acids and covalent modifications; the basis for RFdiffusionAA.

RFdiffusion The Baker-lab diffusion model that generates protein backbones, fine-tuned from RoseTTAFold; the workhorse for de novo backbones, motif scaffolding, symmetric oligomers and binders.

RFdiffusion3 The all-atom successor generating tokens and atoms directly, substantially cheaper to sample and extended to nucleic-acid and small-molecule interactions.

RMSD (root-mean-square deviation) The standard measure of structural agreement between two aligned structures; design success is usually defined by an RMSD threshold between the design model and the predictor's refolded prediction.

Rosetta The physics- and knowledge-based macromolecular modelling suite that defined computational protein design before deep learning, and still supplies energy functions and relax/repack protocols.

Scaffold (1) The designed protein framework that presents a functional motif; (2) a reusable natural framework (nanobody, monobody, fibronectin III domain) engineered for new binding.

Self-conditioning Feeding a diffusion model its own previous-step structure prediction as extra input; a large empirical accuracy gain in RFdiffusion.

Sequence recovery The fraction of native residues a sequence-design model reproduces when given a native backbone — the standard in silico metric for inverse folding.

Structure module AlphaFold2's final block, which converts the Evoformer representations into explicit 3D coordinates via residue frames and IPA.

Symmetric assembly A designed homo-oligomer with cyclic (C*n*), dihedral (D*n*) or point-group symmetry; symmetry is imposed during generation rather than discovered.

Test-time compute Spending more sampling or search effort per design at inference (more seeds, more denoising steps, re-ranking) to buy success rate without retraining.

Tokenization How a model discretises a biomolecular system into units — residues, atoms, or a mixture — which determines what kinds of molecules it can represent.

Two-stage design The dominant pipeline: generate a backbone (RFdiffusion), then design a sequence onto it (ProteinMPNN), then verify by refolding (AlphaFold2). Contrast *co-design* and *all-atom design*.

Yeast display A high-throughput experimental screen presenting designed proteins on the yeast surface, used to measure binding across large design libraries.

References

Every source ingested into the wiki, with a one-line description. In-text citations throughout the book link here.

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Abramson 2024 (suppl.). Accurate structure prediction of biomolecular interactions with AlphaFold 3 — Supplementary Information. — Methods/architecture companion to Abramson 2024. 45 pages of pseudocode-level specs. Best read against the Jumper 2021 (suppl.) to see exactly what changed from AF2.

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Ahern 2025 (suppl.). Atom-level enzyme active site scaffolding using RFdiffusion2 — Supplementary Information. — 67-page methods companion to Ahern 2025. Its value is the **training-example construction algorithm**, which is where the indexed/unindexed and frame/atom choices are actually made and sampled.

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Anishchenko 2021. Anishchenko, Pellock, Chidyausiku et al. (Baker lab). De novo protein design by deep network hallucination. Nature 2021. — The paper that **inverted a structure predictor into a generator**. Random sequences are optimized until the network's own predictions become confident, and the resulting sequences fold — establishing hallucination as a design method and founding the lineage that runs through Wang 2022 and Wicky 2022 into conditional diffusion.

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Bennett 2026 (suppl.). Atomically accurate de novo design of antibodies with RFdiffusion — Supplementary Information. — 129-page companion to Bennett 2026. It is the wiki's most detailed account of **how a generative design model is fine-tuned for one modality**, and of the filtering thresholds that modality needs.

Boyken 2016. Boyken, Chen, Groves et al. (Baker lab). De novo design of protein homo-oligomers with modular hydrogen-bond network–mediated specificity. Science 2016. — Introduces **HBNet** and uses it to design cyclic homo-oligomers whose **interaction specificity** is set by modular arrays of buried hydrogen-bond networks — the protein analogue of Watson–Crick base pairing — rather than by hydrophobic packing alone.

Braun 2026. Braun, Tripp, Chakatok, Kaltenbrunner, Fischer et al.; Hall & Oberdorfer (TU Graz / Univ. Graz). Computational enzyme design by catalytic motif scaffolding. Nature 2026 (649, 237–245; online 3 December 2025; doi 10.1038/s41586-025-09747-9). — The **independent lineage** the enzyme chapter lacked: a non-Baker-lab enzyme-design method, from the Oberdorfer lab, published back-to-back with Kim 2026 in the same issue of *Nature*. Its headline is not a new reaction but a new activity bar — **de novo catalysts "rivalling those optimized by in vitro evolution"**, with **exquisite stereoselectivity** and six high-resolution structures.

Braun 2026 (suppl.). Computational enzyme design by catalytic motif scaffolding — Supplementary information. — 49-page companion to Braun 2026, carrying the Supplementary Methods for the pipeline plus the biophysical characterisation the main text summarises.

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Butcher 2025 (suppl.). De novo Design of All-atom Biomolecular Interactions with RFdiffusion3 — Supplementary Information. — Methods/benchmark companion to Butcher 2025. 48 pages. Specifies the training regimen, tokenization, the DiffusionModule, and the four benchmarks behind the efficiency and capability claims.

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Choi 2026. Choi, Bauer, Coventry, Venkatesh, Chen et al. (Baker lab, with Hilvert). De novo design of cysteine proteases. Nature 2026 (doi 10.1038/s41586-026-11143-w; received 14 Oct 2025, accepted 9 Sep 2026). — De novo enzymes that hydrolyse the **peptide backbone itself** using a Cys–His–Asp catalytic triad, with sequence specificity, on folds unlike any natural protease — and that work inside living mammalian cells.

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symmetries of an axle and a rotor so that the coupled system retains an internal degree of freedom rather than locking into one rigid structure.

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Krishna 2024. Krishna, Wang, Ahern et al. (Baker lab). Generalized biomolecular modeling and design with RoseTTAFold All-Atom. Science 2024. — Extends structure prediction and design beyond protein-only systems to full biomolecular assemblies — proteins with nucleic acids, small molecules, metals and covalent modifications — and, via a diffusion variant, designs proteins around small-molecule ligands.

Krishna 2024 (suppl.). Generalized biomolecular modeling and design with RoseTTAFold All-Atom — Supplementary Information. — Methods/benchmark companion to Krishna 2024. 107 pages (Figs S1–S34). This is the clean, text-based counterpart to the OCR-recovered main PDF — the authoritative source for RFAA's featurization, training, benchmarks, and the RFdiffusionAA binder-design protocols.

Lin 2023. Lin & AlQuraishi (Columbia). Generating Novel, Designable, and Diverse Protein Structures by Equivariantly Diffusing Oriented Residue Clouds. ICML 2023 (PMLR 202); arXiv:2301.12485v3, 6 June 2023. — The founding paper of the Genie lineage. Its contribution is a deliberate **asymmetry**: represent a protein as a plain point cloud of Cα coordinates in the *forward* (noising) process, and as a cloud of **oriented reference frames** in the *reverse* (generation) process.

Lin 2024. Lin, Lee, Zhang & AlQuraishi (Columbia / Rutgers). Out of Many, One: Designing and Scaffolding Proteins at the Scale of the Structural Universe with Genie 2. arXiv:2405.15489v1, 24 May 2024. — The middle paper of the lineage, and the one the wiki most needed: it supplies the **training dataset and the motif-conditioning algorithm that Lin 2026 reuses rather than restates**. Two contributions — a multi-motif scaffolding framework that leaves inter-motif geometry unspecified, and massive data augmentation from the AlphaFold Database.

Lin 2026. Lin, Lee, Vermani, Jiang, De Cooman, Špeťko & AlQuraishi (Columbia). Fast and Ultra-Capable Protein Design: Advancing the Frontier Through Atomistic SE(3)-Equivariance with Genie 3. bioRxiv 2026 (posted 2026-05-05; doi 10.64898/2026.05.01.722168). — A diffusion model that **reverses two of the field's recent consensus moves at once** — it keeps SE(3)-equivariance where others abandoned it, and it beats hallucination with generation — then validates the claim with a 92 nM binder designed in a blind competition against a target with almost no structural characterisation.

Luo 2016. Luo, Hou, Bai, Wang, Liu. Protein Assembly: Versatile Approaches to Construct Highly Ordered Nanostructures. Chemical Reviews 2016. — A 60-page *Chemical Reviews* survey of how ordered protein nanostructures are built. Ingested for the wiki mainly as a **historical reference for the pre-deep-learning symmetry literature**, which the rest of `sources/` does not otherwise cover.

Ruijgrok 2026. Ruijgrok (dx_binder_design project, internal report). Binder design pipelines — a comparison of computational approaches. Internal technical report, 2026-10-12. — A first-party, machine-generated comparison of nine binder-design pipelines as operated in the `dx_binder_design` project, covering generative formulation, atomic representation, target and epitope conditioning, test-time search, scoring, measured compute, and — its distinctive contribution — a **comparability ledger** stating for each pipeline why its headline pass rate cannot be placed beside another's.

Stark 2025. Stark, Faltings, Choi, Xie et al. (MIT / Boltz, with DeGrado, Liu, Sabatini labs). BoltzGen: Toward Universal Binder Design. bioRxiv 2025 (preprint, 2025-11-20). — A single all-atom generative model for designing binders **across modalities and target classes**, validated in eight wet-lab campaigns over 26 targets — the broadest experimental validation of any source in the wiki, and the strongest available evidence on how binder design behaves against targets with no structural precedent.

Thomson 2014. Thomson, Wood, Burton et al. (Woolfson lab). Computational design of water-soluble α-helical barrels. Science 2014. — Designs **cyclic homo-oligomers of five, six and seven parallel identical helices** with open central channels, by combining coiled-coil geometry, knowledge-based scoring and atomistic modeling — the Woolfson-lab counterpart to the Baker-lab parametric work published in the same issue.

Vasquez-Torres 2023. Vázquez Torres, Leung, Venkatesh et al. (Baker lab). De novo design of high-affinity binders of bioactive helical peptides. Nature 2023. — Applies the generative design stack to a hard, high-value target class: **flexible α-helical peptide hormones** (parathyroid hormone, glucagon, secretin, neuropeptide Y, GLP1, GIP, BH3 peptides), producing high-affinity binders and turning them into biosensors.

Vasquez-Torres 2023 (suppl.). De novo design of high-affinity binders of bioactive helical peptides — Supplementary Information. — Methods companion to Vasquez-Torres 2023. 32 pages, heavily **experimental** — the wet-lab protocols behind the designs, which the main text compresses.

Wang 2022. Wang, Lisanza, Juergens et al. (Baker lab). Scaffolding protein functional sites using deep learning. Science 2022. — Introduces the two deep-learning routes to **motif scaffolding** — building a protein around a fixed functional site — that RFdiffusion later generalized. The conceptual bridge from prediction networks to functional de novo design.

Watson 2023. Watson, Juergens, Bennett et al. (Baker lab). De novo design of protein structure and function with RFdiffusion. Nature 2023. — The first broadly successful diffusion generative model for protein backbones, and the general-purpose engine behind much current de novo design.

Watson 2023 (suppl.). De novo design of protein structure and function with RFdiffusion — Supplementary Information. — Methods/data companion to Watson 2023. 129 pages. This page maps the SI and extracts the concrete details the main summary omits — architecture, diffusion math, training/conditioning recipes, and the *in silico* success criteria that define "success" throughout the paper.

Wicky 2022. Wicky, Milles, Courbet et al. (Baker lab). Hallucinating symmetric protein assemblies. Science 2022. — Uses deep-network **hallucination** to design symmetric homo-oligomers from almost nothing — just the number of protomers and their length — producing assemblies unlike anything in the PDB.

Wood 2018. Wood & Woolfson. CCBuilder 2.0: Powerful and accessible coiled-coil modeling. Protein Science 2018. — A web application that turns Parametric design of coiled coils into something anyone can do in a browser — and, in its framing of *why* parametric modeling works at all, the clearest statement of the link between coiled coils and repeat proteins.

Yeates 2002. Yeates & Padilla. Designing supramolecular protein assemblies. Current Opinion in Structural Biology 2002. — **A review, not original work** — but a review written by the authors of the nanohedra design it describes, which makes it a first-party account of the founding symmetric-fusion result rather than a third-party summary.

AlphaFold. DeepMind's protein structure predictor; the field-defining model and, in practice, the standard design oracle.

BindCraft. A binder-design pipeline that works by **hallucination against AlphaFold2** — optimising a binder sequence against the predictor's confidence and alignment scores by gradient feedback, rather than sampling from a trained generative model.

BoltzGen. An all-atom generative model for binder design **across modalities and target classes**, unifying design and structure prediction in one network. Introduced in Stark 2025; weights, data, training and inference code released under the MIT License.

CCBuilder 2.0. Woolfson-lab **web application for parametric modelling and optimization of coiled coils**, and the most accessible entry point to Parametric design (Wood 2018).

Chroma. Programmable diffusion generative model for proteins and complexes from Generate Biomedicines. Introduced in Ingraham 2023.

Proteína-Complexa. A fully atomistic binder generator built to **unify conditional generation with inference-time optimisation**, on the argument that the field's generative-versus-hallucination split is a false dichotomy. Introduced in Didi 2026.

Genie 3. The AlQuraishi-lab lineage of SE(3)-equivariant protein diffusion models, now fully ingested across three generations.

MaSIF. Molecular Surface Interaction Fingerprinting — a geometric-deep-learning framework that learns descriptors of protein *molecular surfaces* predictive of what a surface patch will bind, without explicit use of sequence or fold. Introduced in Gainza 2020; extended to de novo binder design in Gainza 2023.

ProteinHunter. A **fine-tuning-free** design framework that exploits the willingness of diffusion-based structure-prediction models to hallucinate plausible structures from an uninformative sequence. Introduced in Cho 2025.

ProteinMPNN. Baker-lab message-passing network for fixed-backbone Sequence design (fixed-backbone). Introduced in Dauparas 2022.

Protpardelle. Huang-lab all-atom protein diffusion model that generates backbone and side chains together. Introduced in Chu 2024.

RFdiffusion. Baker-lab diffusion generative model for protein backbones, made by fine-tuning the RoseTTAFold predictor on a structure-denoising task. Introduced in Watson 2023.

RFdiffusion2. Baker-lab generative model for **atom-level functional site scaffolding**, introduced in Ahern 2025. It is the generator behind every de novo protease and metallohydrolase in De novo enzyme design, and — fine-tuned — behind the de novo antibodies in Bennett 2026.

RFdiffusion3 (RFD3). Baker-lab all-atom diffusion generator for proteins in the context of ligands, nucleic acids and metals — the efficiency-focused successor to RFdiffusion and RFdiffusionAA. Introduced in Butcher 2025.

RoseTTAFold. Baker-lab protein structure predictor — the parallel lineage to DeepMind's AlphaFold, and the backbone repurposed for generative design.